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Task-specific models vs. large vision-language models in medical visual question answering: A survey

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ABSTRACT

Medical Visual Question Answering (Med-VQA) is an emerging interdisciplinary task that integrates computer vision and natural language processing to enable automated clinical decision support based on medical images. Recent advances in deep learning have led to two prominent model paradigms: task-specific models (TSMs), which are tailored to specific datasets and clinical tasks, and large vision-language models (VLMs), which leverage large-scale pretraining to achieve strong generalization across modalities. However, a comprehensive and comparative understanding of these two approaches remains lacking. This survey presents the first systematic comparison between TSMs and VLMs in the Med-VQA domain. An unified taxonomy is introduced covering architectural strategies, knowledge integration techniques, pretraining methods, and interpretability mechanisms. Over 50 representative models are reviewed and benchmark their performance across multiple datasets. Furthermore, this paper analyses the trade-offs between accuracy, generalizability, interpretability, and computational efficiency, and examine common challenges such as hallucination, modality bias, and data scarcity. Finally, the paper outlines emerging trends and future directions, including hybrid TSM-VLM systems, instruction tuning, and trustworthy VQA frameworks, to guide the development of clinically viable Med-VQA solutions.

1. Introduction

Medical visual question answering (Med-VQA) is a multimodal task where a model answers clinical questions about a medical image (Dong et al., 2025; Lin et al., 2023). It bridges computer vision and natural language processing to enable clinically grounded reasoning and communication in healthcare settings (Hartsock & Rasool, 2024). Specifically, Med-VQA involves developing computational models capable of accurately responding to clinical questions using various medical imaging modalities, including radiographs, computed tomography (CT), magnetic resonance imaging (MRI), ultrasound (US), and pathology images. Distinct from traditional medical image analysis tasks, which primarily focus on disease detection or classification, Med-VQA necessitates advanced capabilities for comprehending nuanced clinical contexts, reasoning about complex medical scenarios, and generating precise natural language responses. The need for Med-VQA systems arises from several pressing challenges in real-world medical settings. First, radiological and pathological interpretation often demands extensive expertise and time, especially in under-resourced or high-volume hospitals. Second, the increasing volume of imaging examinations exacerbates diagnostic delays and cognitive load on clinicians. Third, in regions with limited access to specialists, automated image-question-answering tools can assist in triage, patient education, and telemedicine. These demands underscore the clinical value of intelligent systems capable of providing fast, explainable, and accurate image-grounded answers.

To meet these challenges, two technical paradigms have emerged. **Task-specific models (TSMs)**, often designed for individual datasets and imaging modalities, prioritize accuracy and efficiency. They are carefully engineered with tailored fusion strategies, domain knowledge integration, and interpretability components, making them well-suited for deployment in constrained or safety-critical settings. In contrast, **Large Vision-Language Models (VLMs)**, inspired by general-purpose multimodal foundation models, leverage massive pretraining on diverse image-text corpora and exhibit strong zero-shot and transfer capabilities, albeit at the cost of high computational demands and potential hallucinations.

Despite promising progress in both directions, existing literature lacks a systematic comparison between these two families of methods. How do specialized and large multi-task models differ in terms of generalization, explainability, clinical reliability, and deployment feasibility? What are their respective trade-offs when applied to Med-VQA tasks? To answer these questions, this survey offers a comprehensive review

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任务特定模型与大规模视觉-语言模型在医学视觉问答中的比较：一项综述

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文章信息

关键词：视觉问答 医学影像 任务特定模型 视觉-语言模型

抽象

医学视觉问答（Med-VQA）是一项新兴的跨学科任务，将计算机视觉和自然语言处理结合起来，以实现基于医学影像的自动化临床决策支持。深度学习的最新进展催生了两种主要的模型范式：特定任务模型（TSMs），针对特定数据集和临床任务进行定制；以及大规模视觉语言模型（VLMs），利用大规模预训练实现跨模态的强泛化能力。然而，对于这两种方法的全面比较和理解仍然不足。本综述首次在Med-VQA领域对TSMs和VLMs进行了系统的比较。引入了统一的分类法，涵盖架构策略、知识整合技术、预训练方法和可解释性机制。回顾了50多个具有代表性的模型，并在多个数据集上评估其性能。此外，本文分析了...

1. 引言

医学视觉问答（Med-VQA）是一种多模态任务，模型需要对医疗图像相关的临床问题进行回答（Dong 等，2025；Lin 等，2023）。它桥接了计算机视觉与自然语言处理，使临床场景中的推理和交流成为可能（Hartsock & Rasool，2024）。具体而言，Med-VQA 涉及开发能够利用各种医学影像模态（包括X光片、计算机断层扫描（CT）、磁共振成像（MRI）、超声（US）以及病理图像）准确回答临床问题的计算模型。不同于传统的医学图像分析任务主要关注疾病的检测或分类，Med-VQA 需要具备理解微妙临床背景、推理复杂医学情境以及生成精确自然语言回答的高级能力。对 Med-VQA 系统的需求源于几个紧迫的挑战。

由于对专家的访问有限，自动化的图像问答工具可以协助分诊、患者教育和远程医疗。这些需求突出了能够提供快速、可解释和准确的基于图像的答案的智能系统的临床价值。

为应对这些挑战，出现了两种技术范式。任务特定模型（TSM），通常为单个数据集和成像模态设计，优先考虑精度和效率。它们经过精心设计，具有定制的融合策略、领域知识整合和可解释性组件，使其非常适合在受限或安全关键的环境中部署。相比之下，大型视觉-语言模型（VLM），受到通用多模态基础模型的启发，利用在多样化图文语料库上的大规模预训练，并展现出强大的零样本和迁移能力，尽管代价是高计算需求和潜在的幻觉问题。

尽管在这两个方向上都取得了令人鼓舞的进展，但现有文献缺乏对这两类方法的系统比较。专用模型和大型多任务模型在泛化能力、可解释性、临床可靠性以及部署可行性方面有何不同？在应用于医学视觉问答（Med-VQA）任务时，它们各自的权衡是什么？为回答这些问题，本综述提供了全面的回顾。

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2026年3月12日在线发布 0957-4174/© 2026 爱思唯尔有限公司。版权所有，包括文本和数据挖掘、人工智能训练及类似技术的权利。

Table 1
Comparison of current survey, where ✓ indicates the topic is covered, and × indicates it is not.

Paper	Title	Specialized model	VLMs	Comparison	Hallucination	Interpretable
Al-Sadi et al. (2021)	Visual question answering in the medical domain based on deep learning approaches: A comprehensive study	✓	×	×	×	×
Lin et al. (2023)	Medical visual question answering: A survey	✓	×	×	×	×
Al-Hadhrami et al. (2023)	A Critical Analysis of Benchmarks, Techniques, and Models in Medical Visual Question Answering	✓	✓	×	×	×
Chen et al. (2024a)	A Survey of Medical Vision-and-Language Applications and Their Techniques	✓	×	×	×	✓
Li et al. (2024)	Developing ChatGPT for biology and medicine: a complete review. of biomedical question answering	×	✓	×	×	×
Hartsock and Rasool (2024)	Vision-language models for medical report generation and visual question answering: a review	×	✓	×	✓	×
Dong et al. (2025)	Generative Models in Medical Visual Question Answering: A Survey	✓	✓	×	✓	✓

of both TSMs and large VLMs in Med-VQA, including their architectures, training strategies, evaluation results, and application scenarios. Currently, the distinction between TSMs and VLMs in Med-VQA lacks definitive temporal or technological demarcation. Therefore, for clarity in organization and comparison, this survey distinguishes task-specific models from vision-language models based on model architecture, data volume requirements, computational scale, and, most importantly, multi-task capability.

At present, there is no survey systematically comparing TSMs and large VLMs within the Med-VQA domain, a critical gap this survey aims to address. Table 1 provides a comparative summary of existing Med-VQA surveys. Al-Sadi et al. (2021), as the earliest survey, presents an early Med-VQA study centered on the ImageCLEF VQA-Med 2019 benchmark, proposing a hierarchical, question-type-routed pipeline whose core components are CNN-based classifiers. Lin et al. (2023) predates the widespread adoption of large vision-language models, focusing solely on specialized, task-specific models. Similarly, Chen et al. (2024a) discusses exclusively VLMs. Al-Hadhrami et al. (2023) includes both TSMs and large VLMs but lacks discussion on hallucination, a known limitation of large VLM-based approaches. Conversely, Li et al. (2024) focuses primarily on large vision-language models, neglecting contributions from those specialized models, and while Hartsock and Rasool (2024) addresses the hallucination issue, it similarly overlooks the potential of specialized models. Dong et al. (2025) provides a more balanced discussion encompassing both TSMs and large VLMs and addresses critical issues like hallucination and interpretability; however, it lacks a formal comparison of TSMs and large VLMs in terms of performance, computational resources, and reliability.

This survey systematically reviews recent developments, strengths, and limitations of both task-specific models and large vision-language models in the Med-VQA context. Additionally, this survey discusses open challenges and future research directions, providing practical insights for researchers, clinicians, and developers on advances and real-world deployment of medical visual question answering systems. The survey is organized as follows: Section 2 reviews existing TSM methods, including approaches such as fusion algorithms, external knowledge integration, counterfactual training, and other notable strategies. Section 3 covers large VLMs, discussing zero-shot approaches, fine-tuning strategies (full-parameter tuning and parameter-efficient tuning). Section 4 introduces commonly used datasets and essential evaluation metrics for Med-VQA tasks. Section 5 offers a detailed comparison between TSMs and large VLMs concerning performance, computational resources, and their respective advantages and disadvantages. Section 6 outlines existing challenges in Med-VQA, addressing issues such as modality bias, object grounding, interpretability, hallucination, and generalizability. Finally, Section 7 concludes the paper.

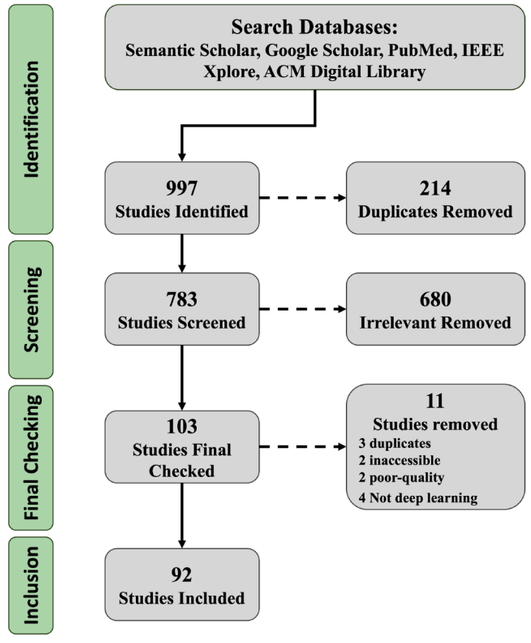


Fig. 1. Flowchart of the literature search and selection process for this survey.

2. Literature search and selection strategy

To ensure comprehensive coverage and reproducibility, this paper conducts a systematic literature search and screening procedure across five major scholarly databases: Semantic Scholar, Google Scholar, PubMed, IEEE Xplore, and the ACM Digital Library. The search is designed to capture both Med-VQA task-oriented studies and closely related medical vision-language research that explicitly involves Med-VQA-style evaluation or problem settings. The search queries were constructed around several keyword groups with synonym expansion: (i) “medical visual question answering” and its synonyms “Med-VQA”, “MedVQA”, “VQA-Med”, “VQA Med”, and “medical VQA”; (ii) “radiology visual question answering”; (iii) “clinical visual question answering”; and (iv) “medical vision-language model” together with

表1 当前调查的比较，其中✓ 表示涵盖该主题，× 表示未涵盖。

Paper	Title	Specia化	model	VLM的比较	幻觉输入	可解释的
Al-Sadi 等人（2021）	医学中的视觉问答 基于深度学习的领域 rning ap- 方法： A 综合研究	✓	×	×	×	×
林等 l. (2023)	医学病毒 g: A sur- vey	✓	×	×	×	×
Al-Hadhrami 等（2023）	A 基准测试与技术的批判性分析 技术和模型 在医学视觉中 问答	✓	✓	×	×	×
陈等 . (2024a) 李等人 (2024)	A 调查 医疗视觉 -and-语言 应用程序和 Their Techniques Developing ChatGPT 用于生物学和 医学：完整综述 生物医学的 ical问题回答 ring	✓	×	×	×	✓
Hartsock 和 Rasool (2024) Visio	n-l医学报告的语言模型 生成和 视觉问答： a 评论	×	✓	×	✓	×
董等 l. (2025)	生成模型 医学视觉提示 问题回答：A in M 调查	✓	✓	×	✓	✓

在医学视觉问答（Med-VQA）中，涵盖了TSM（任务特定模型）和大型VLM（视觉语言模型）的体系结构、训练策略、评估结果及应用场景。目前，Med-VQA中TSM和VLM之间尚无明确的时间或技术界限。因此，为了在组织和比较上更加清晰，本综述基于模型架构、数据量需求、计算规模，最重要的是多任务能力，将任务特定模型与视觉语言模型区分开来。

目前，还没有对Med-VQA领域中的TSM和大型VLM进行系统比较的调查，这是本调查旨在填补的关键空白。表1提供了现有Med-VQA调查的比较摘要。Al-Sadi等人（2021）作为最早的一项调查，提出了一项以ImageCLEF VQA-Med 2019基准为中心的早期Med-VQA研究，提出了一个分层、按问题类型路由的流程，其核心组件是基于CNN的分类器。Lin等人（2023）早于大型视觉语言模型的广泛应用，仅关注专门的、任务特定的模型。同样，Chen等人（2024a）仅讨论VLM。
Al-Hadhrami等人（2023）同时包括TSM和大型VLM，但缺乏对“大型VLM方法已知局限——幻觉”问题的讨论。相反，Li等人（2024）主要关注大型视觉语言模型，忽略了那些专门模型的贡献，而Hartsock和Rasool（2024）则涉及幻觉问题

本综述系统地回顾了任务特定模型（TSM）和大型视觉语言模型（VLM）在医学视觉问答（Med-VQA）背景下的最新进展、优势和局限性。此外，本综述还讨论了公开挑战和未来研究方向，为研究人员、临床医生和开发者在医学视觉问答系统的进展和实际部署方面提供了实用见解。本文综述结构如下：第2节回顾了现有的TSM方法，包括融合算法、外部知识整合、反事实训练及其他值得关注的策略。第3节介绍大型VLM，讨论零样本方法、微调策略（全参数微调与参数高效微调）。第4节介绍了Med-VQA任务中常用的数据集和必要的评估指标。第5节对TSM与大型VLM在性能、计算资源及各自优势方面进行了详细比较。

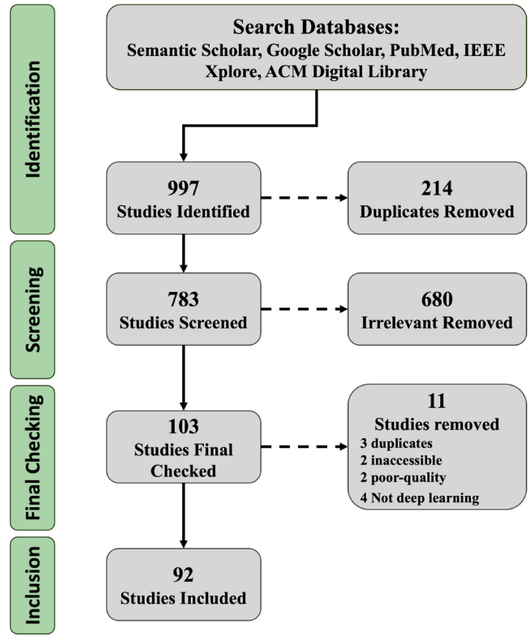


图 1. F 文献检索和选择流程的流程图 his 调查。

2. 文献检索与选择策略

为了确保覆盖的全面性和可重复性，本论文在五个主要学术数据库中进行了系统的文献检索和筛选程序：Semantic Scholar、Google Scholar、PubMed、IEEE Xplore 和 ACM Digital Library。检索旨在捕捉既有面向 Med-VQA 任务的研究，又有与之密切相关、明确涉及 Med-VQA 风格评估或问题设置的医学视觉-语言研究。检索查询围绕几个关键词组并通过同义词扩展构建：(i) “medical visual question answering”及其同义词“Med-VQA”、“MedVQA”、“VQA-Med”、“VQA Med”和“medical VQA”；(ii) “radiology visual question answering”；(iii) “clinical visual question answering”；以及 (iv) “medical vision-language model”结合

its synonyms “medical VLM”, “medical multimodal large language model”, “medical MLLM”, “medical pretraining”, and “medical foundation model”. In addition, “biomedical” and “biomedicine” are used as domain-related expansions of “medical” to improve retrieval completeness. The initial retrieval produced 997 records in total. Then, deduplication is performed, resulting in 783 unique papers (214 duplicates removed). Next, a relevance screening step is applied to remove papers outside the survey scope; this stage removed 680 records, primarily because they were non-multimodal, or focused on medical vision-language models without Med-VQA tasks, or were otherwise not aligned with Med-VQA or closely related clinical/radiology VQA settings, leaving 103 candidate papers. Finally, this search process included an accessibility and quality check to construct the reviewed paper set, excluding 3 remaining duplicates, 2 inaccessible papers, 2 poor-quality papers, and 4 papers that did not involve deep learning, yielding a final set of 92 papers included in this survey. The process is illustrated in Fig. 1.

3. Task-specific Med-VQA models

This section provides a systematic examination of task-specific models, which are explicitly designed for the Med-VQA task. As highlighted in the blue part of Fig. 2, these models prioritize domain-specific reasoning, interpretability, and computational efficiency, making them well-suited for constrained clinical environments. Unlike large VLMs that exhibit strong generalization across modalities and tasks, TSMs often sacrifice transferability in favour of tailored performance within narrow medical contexts. To contextualize their strengths and limitations relative to VLMs, this section categorizes task-specific approaches by core modelling strategies: (i) fusion algorithms for aligning visual and textual modalities, ranging from simple concatenation to transformer-based designs; (ii) external knowledge integration, leveraging medical captions, retrieval-augmented generation (RAG), and structured knowledge graphs; (iii) pre-training methods adopted to enhance generalization; (iv) augmentation-based strategies for data-scarce in medical domains; (v) multi-question learning to model intra-image question dependencies; and (vi) interpretable VQA methods that address the transparency and reliability needed for clinical adoption. Together, these approaches represent a vital category of Med-VQA systems that trade off broad adaptability for task specialization, low-latency inference, and explainability, enabling practical use in real-world healthcare scenarios.

3.1. Fusion algorithm

Fusion algorithms are essential to task-specific Med-VQA models, as they govern how visual and textual features are combined for clinical reasoning. These strategies range from simple concatenation to advanced attention-based and transformer architectures, each offering trade-offs between efficiency, semantic alignment, and model complexity. This section reviews key fusion approaches and their roles in enabling accurate, interpretable multimodal understanding in medical contexts.

Attention-Based Fusion: Attention mechanisms are commonly employed to explicitly align visual and textual features. For example, MedFuseNet (Sharma et al., 2021) and AMAM (Pan et al., 2022) utilize dual and co-attention modules to align question-driven and image-driven representations. These methods enable rich multimodal interactions but at the cost of increased computational complexity. Question-type-specific attention mechanisms (Liu et al., 2023; Wang et al., 2022) further improve performance by enforcing semantic alignment based on question categories, although their reliance on accurate question classification introduces additional failure points. Additionally, based on the question distribution, LGPFN (Du et al., 2025) uses a question-guided progressive multimodal fusion and language-gate mechanism to overcome the long-tail distribution of the Med-VQA dataset. Contrary to language guided attentions, to make the prediction better grounded on the image information, VG-CALF (Lameesa et al., 2025) leverages visual guided cross attention for multi-modal fusion. Recently, Transformer-based fusion models have significantly advanced the field. CMSA (Gong et al., 2021) leverages cross-modal self-attention within unified attention blocks, while SIA (Chen et al., 2023) integrates symmetric attention and cross-modal gating to model complex interactions. These models successfully excel in open-ended VQA. Some studies have adapted encoder-decoder transformer frameworks using pretrained vision transformers (ViTs) for vision and transformer-based encoders for text (Bazi et al., 2023), resembling general-purpose VLMs but tuned specifically for Med-VQA.

Multi-Scale Attention: Most attention-based methods rely on high-level global image features, often overlooking low-level local details. To address this, fine-grained and multi-scale fusion techniques have been developed. FGCVQA (Wu et al., 2023) applies cross-attention between global and lesion-specific visual features. MF^2-MVQA (Song et al., 2023) performs multi-stage fusion across several convolutional layers, enabling simultaneous preservation of semantic and regional features.

Multilayer Perceptrons: In pursuit of computational efficiency, Multilayer Perceptrons (MLP)-based fusion strategies have also been explored. A representative method, M-MixUp (Liu et al., 2022) demonstrates that concatenating multimodal features followed by MLPs, when augmented with multimodal MixUp, can achieve performance comparable to complex attention-based models. This approach is efficient and generalizable but may underperform in tasks requiring detailed semantic reasoning.

Fusion strategies in Med-VQA have progressed from basic concatenation to sophisticated attention-based and transformer-driven architectures. Advanced models offer stronger multimodal alignment and reasoning capabilities but come with higher computational costs. In contrast, lightweight architectures, when combined with effective augmentation, can achieve competitive performance, though they may fall short in handling complex clinical semantics.

3.2. External knowledge

In task-specific Med-VQA models, external knowledge serves as a crucial complement to limited in-domain training data, helping bridge the gap between visual cues and clinical reasoning. By embedding structured or unstructured medical information, these methods enhance answer accuracy and interpretability, especially in complex or ambiguous cases. This section reviews three major approaches: caption-aware models, which enrich semantic alignment using image-text descriptions; retrieval-augmented generation, which incorporates relevant external references during inference; and knowledge graph (KG)-based methods, which embed structured medical concepts for robust, traceable reasoning. These strategies reflect different trade-offs in complexity, data dependency, and deployment feasibility.

Caption-Aware VQA: Caption-aware methods exploit image captions to enrich textual semantics. One such method, Cong et al. (2022) uses attention maps from captioning tasks to guide focus toward semantically salient regions, and employs progressive bilinear interactions to improve modality alignment. MISS (Chen et al., 2024) introduces TransCap, a framework that synthetically generates image-text pairs from unimodal data using large language models, improving pretraining under limited supervision.

Retrieval-Augmented Generation: RAMM (Yuan et al., 2023) and FGRW (Liang et al., 2025) incorporates external knowledge by retrieving relevant image-text pairs from databases during fine-tuning. These are integrated into the VQA process via a retrieval-attention mechanism. This approach enriches model context and enhances accuracy, though it is sensitive to retrieval quality and requires a high-quality knowledge base.

Knowledge Graph-Based Methods: KG-based approaches embed structured medical knowledge directly into the model. TCC-MEM (Guo et al., 2022) combines targeted contrastive learning and multimodal entity matching to improve visual feature extraction and question

它的同义词包括“medical VLM”（医疗视觉语言模型）、“medical multimodal large language model”（医疗多模态大语言模型）、“medical MLLM”（医疗多模态大语言模型）、“medical pretraining”（医疗预训练）和“medical foundation model”（医疗基础模型）。此外，“biomedical”（生物医学）和“biomedicine”（生物医药）被用作“medical”的领域扩展，以提高检索的完整性。初始检索共产生997条记录。然后进行去重，得到783篇独立论文（去掉214条重复记录）。接下来，进行相关性筛选步骤，以删除不在调查范围内的论文；这一阶段删除了680条记录，主要是因为它们是非多模态的、或关注医疗视觉-语言模型但没有Med-VQA任务、或与Med-VQA或相关临床/影像学VQA设置不一致，从而留下103篇候选论文。最后，该检索过程包括可访问性和质量检查，以构建被审阅的论文集合，排除了3篇仍然重复的论文、2篇无法访问的论文、2篇质量较差的论文，以及4篇不符合要求的论文。

3. 任务特定的医疗视觉问答模型

本节系统地考察了针对特定任务的模型，这些模型是为Med-VQA任务专门设计的。如图2蓝色部分所示，这些模型优先考虑领域特定的推理能力、可解释性和计算效率，使其非常适合受限的临床环境。与在多种模态和任务中表现出强大泛化能力的大型视觉语言模型不同，TSM通常为了在狭窄的医疗环境中实现定制化性能而牺牲迁移能力。为了将其优势和局限性与VLM进行比较，本节按核心建模策略对特定任务的方法进行分类：(i) 用于对齐视觉和文本模态的融合算法，从简单的拼接到基于Transformer的设计；(ii) 外部知识整合，利用医疗标题、检索增强生成（RAG）和结构化知识图；(iii) 用于增强泛化能力的预训练方法。

3.1. 融合算法

融合算法对于特定任务的医学视觉问答（Med-VQA）模型至关重要，因为它们决定了视觉特征和文本特征如何结合以进行临床推理。这些策略从简单的拼接到先进的基于注意力和变换器的架构不等，每种方法在效率、语义对齐和模型复杂性之间提供不同的权衡。本节回顾了关键的融合方法及其在实现准确、可解释的医疗多模态理解中的作用。

基于注意力的融合：注意力机制通常用于显式对齐视觉和文本特征。例如，MedFuseNet（Sharma 等，2021）和 AMAM（Pan 等，2022）利用双重和共同注意力模块来对齐问题驱动和图像驱动的表示。这些方法能够实现丰富的多模态交互，但代价是计算复杂性增加。特定问题类型的注意力机制（Liu 等，2023；Wang 等，2022）通过基于问题类别强制语义对齐进一步提升性能，尽管它们对准确的问题分类的依赖引入了额外的失效点。此外，基于问题分布，LGPFN（Du 等，2025）使用问题引导的渐进多模态融合和语言门机制来克服 Med-VQA 数据集的长尾分布。与语言引导的注意力相反，为了使预测更好地基于图像信息，V

用于多模态融合的引导交叉注意力。近年来，基于Transformer的融合模型显著推动了该领域的发展。CMSA（Gong 等，2021）在统一注意力模块中利用跨模态自注意力，而SIA（Chen 等，2023）则整合对称注意力和跨模态门控以建模复杂的交互。这些模型在开放式视觉问答（VQA）中表现出色。一些研究已采用使用预训练视觉Transformer（ViTs）的编码器-解码器Transformer框架进行适配，用于视觉处理，并使用基于Transformer的编码器处理文本（Bazi 等，2023），类似于通用VLMs，但专门针对医疗领域视觉问答（Med-VQA）进行了调优。

多尺度注意力：大多数基于注意力的方法依赖于高层次的全局图像特征，往往忽略低层次的局部细节。为了解决这个问题，已经开发了细粒度和多尺度融合技术。FGCVQA（Wu 等，2023）在全局和病灶特定的视觉特征之间应用了交叉注意力。MF^2-MVQA（Song 等，2023）在多个卷积层上执行多阶段融合，使得语义特征和区域特征能够同时保留。

多层感知器：为了追求计算效率，也探索了基于多层感知器（MLP）的融合策略。一种具有代表性的方法 M-MixUp（Liu 等，2022）表明，将多模态特征进行拼接后输入 MLP，并结合多模态 MixUp，可以实现与复杂的基于注意力的模型相当的性能。这种方法高效且具有通用性，但在需要详细语义推理的任务中可能表现不佳。

Med-VQA中的融合策略已经从基本的拼接发展到复杂的基于注意力和变压器驱动的架构。高级模型提供了更强的多模态对齐和推理能力，但计算成本更高。相比之下，轻量级架构在结合有效增强方法时可以实现有竞争力的性能，尽管它们在处理复杂的临床语义方面可能有所不足。

3.2. 外部知识

在特定任务的医学视觉问答（Med-VQA）模型中，外部知识是对有限的领域内训练数据的关键补充，有助于弥合视觉线索与临床推理之间的差距。通过嵌入结构化或非结构化的医疗信息，这些方法提高了答案的准确性和可解释性，特别是在复杂或模糊的案例中。本节回顾了三种主要方法：图像字幕感知模型，通过使用图像-文本描述来丰富语义对齐；检索增强生成方法，在推理过程中整合相关的外部参考；以及基于知识图（KG）的方法，嵌入结构化的医疗概念以实现稳健且可追溯的推理。这些策略反映了复杂性、数据依赖性和部署可行性方面的不同权衡。

字幕感知的视觉问答（Caption-Aware VQA）：字幕感知方法利用图像字幕来丰富文本语义。其中一种方法，Cong 等（2022）使用字幕任务中的注意力图来引导关注语义上显著的区域，并采用渐进双线性交互来改善模态对齐。MISS（Chen 等，2024）引入了 TransCap，这是一个利用大语言模型从单模态数据合成生成图文对的框架，从而在有限监督下改善预训练。

检索增强生成：RAMM（Yuan 等，2023）和 FGRW（Liang 等，2025）通过在微调过程中从数据库中检索相关的图文对来融合外部知识。这些信息通过检索-注意力机制被整合到 VQA 过程。这种方法丰富了模型的上下文并提高了准确性，尽管它对检索质量敏感，并且需要高质量的知识库。

基于知识图谱的方法：基于KG的方法将结构化的医学知识直接嵌入模型中。TCC-MEM（Guo 等，2022）结合了目标对比学习和多模态实体匹配，以提高视觉特征提取和问题处理能力

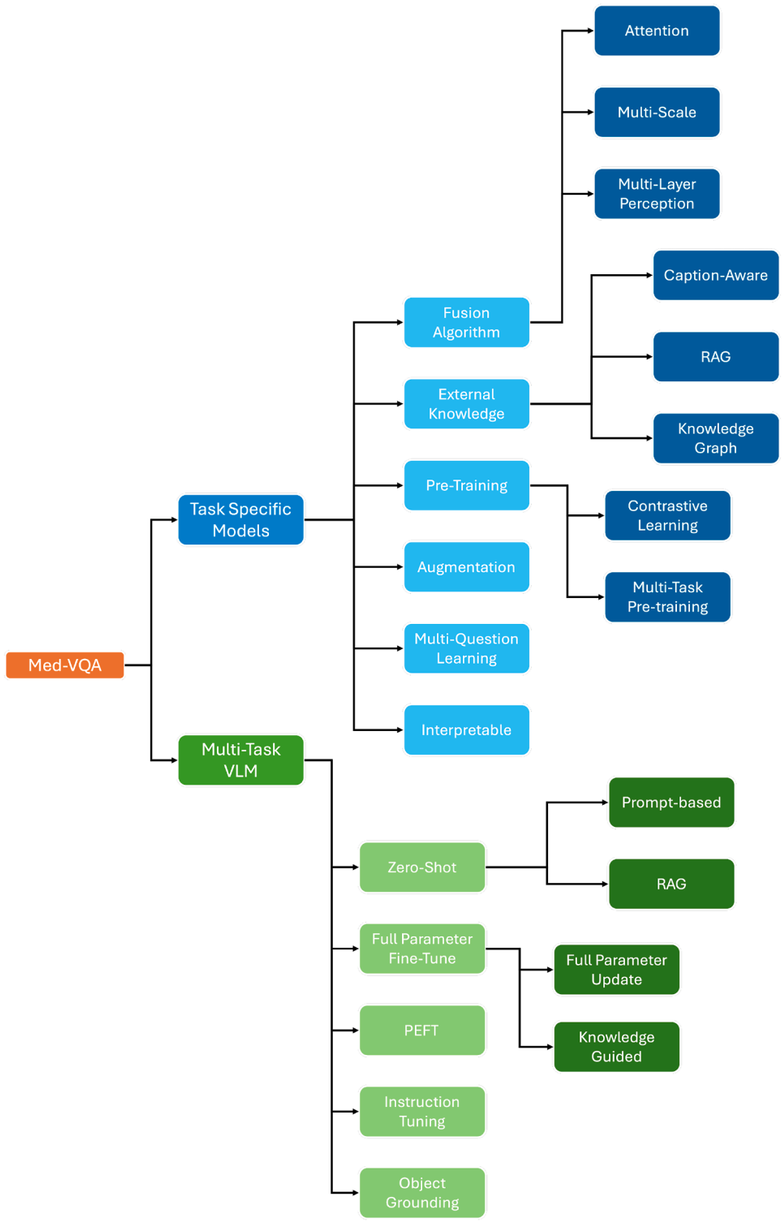


Fig. 2. Taxonomy of Med-VQA approaches. Task-specific models encompass modular innovations such as fusion algorithms, external knowledge integration, and pre-training strategies, whereas multi-task VLMs include zero-shot and fine-tuning paradigms, supported by techniques like prompt engineering, retrieval-augmented generation (RAG), and instruction tuning.

understanding. K-PathVQA (Naseem et al., 2023) enhances PathVQA by integrating a medical KG, enabling robust reasoning in cases where visual evidence is ambiguous. These models show improved generalization and performance, especially for open-ended questions, but require significant effort to construct and maintain medical knowledge bases. MedKLIP (Chen et al. , 2022) uses relation extraction tasks to extract triples from UMLS/Wikipedia reports to enhancing the medical knowledge during pre-training stage.

External knowledge techniques have become pivotal for task-specific Med-VQA. Caption-aware and RAG methods enrich image-question pairs with externally retrieved text or visuals, yielding sizeable gains in answer accuracy and clinical explainability, yet their success hinges on

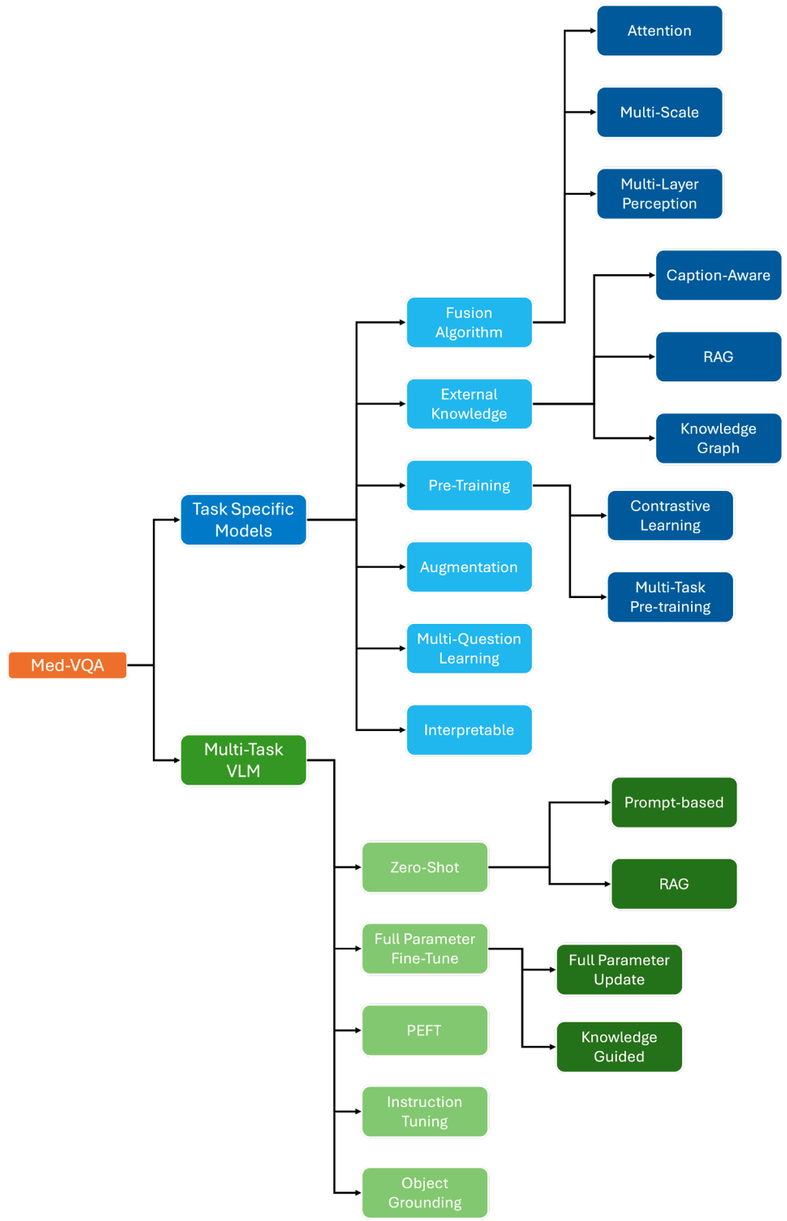


图2. Med-VQA 方法的分类。任务特定模型包括模块化创新，如融合算法、外部知识整合和预训练策略，而多任务 VLM 则包括零样本和微调范式，并采用例如提示工程、检索增强生成（RAG）和指令调优等技术支持。

理解。K-PathVQA（Naseem 等，2023）通过整合医学知识图谱增强了 PathVQA，使其在视觉证据模糊的情况下能够进行稳健推理。这些模型显示出改进的泛化能力和性能，尤其是在开放式问题上，但构建和维护医学知识库需要大量精力。MedKLIP（Chen 等，2022）使用关系抽取任务来提取

来自UMLS/维基百科的三元组报告，用于在预训练阶段增强医学知识。

外部知识技术已成为特定任务医疗视觉问答（Med-VQA）的关键。注意标题的和RAG方法通过使用外部检索的文本或视觉信息丰富图像-问题对，从而显著提高答案的准确性和临床可解释性，但它们的成功依赖于

Table 2
Comparison of task-specific models' pre-training methods by strategy and data.

Method	Pre-training Strategy	External Data
PubMedCLIP	ITC	ROCO
PMC-CLIP	MLM + ITC	PMC-OA
CPRD	CL + KD	–
CMCD	MLM + ITM + MIM + CL	ROCO + MedICaT
MM-VQA	MLM + CL	ROCO
MITER	CL + ITM	CheXpert + MIMIC-CXR + IU-Xray + ChestX-ray14 + COVID-19 + PadChest
CMSA	Image classification + Question-image matching	CheXpert
M2I2	MLM + ITM + ITC	ROCO + MedICaT + Image-CLEF2022
MUMC	MLM + ITM + ITC	Image-CLEF2022
MMBert	MLM	ROCO
MedKLIP	MLM + ITM + RE	ROCO + MedICaT + MIMIC-CXR
MISS	MLM + ITM + ITC	ROCO + MedICaT (augmented by ChatGPT)

Mask Language Modelling (MLM), ITM (Image-Text Matching), Image-Text Contrastive learning (ITC), Mask Image Modelling (MIM), Relation Extraction (RE), Contrastive learning (CL), Knowledge Distillation (KD).

reliable, domain-matched corpora. KG-based systems (e.g., TCC-MEM, multimodal relation graphs) embed curated medical ontologies directly into the inference pipeline, offering transparent and auditable reasoning but demanding heavy upfront graph construction and maintenance. In practice, selecting between RAG and KG strategies should weigh deployment constraints, such as data availability, compute budget, and the level of domain-specific precision and traceability required, which often pointing toward hybrid solutions that mix lightweight retrieval with targeted, high-value graph modules.

3.3. Pre-training methods

Given the limited size and high domain-specificity of Med-VQA datasets, pre-training has emerged as a foundational technique for enhancing task-specific model performance. By leveraging large-scale external medical corpora and adapting self-supervised learning objectives, these methods aim to learn robust, transferable multimodal representations–mitigating overfitting and improving generalization in low-resource settings. Unlike large VLMs that rely on broad general-domain pretraining, TSMs often adopt tailored strategies that align closely with clinical semantics and imaging features. Table 2 provides an overview of representative pre-training methods, categorized by learning objectives, external data sources, and modality alignment techniques.

Contrastive Learning: Contrastive learning has proven effective in aligning visual and textual modalities. CLIP (Radford et al., 2021) demonstrated the generalizability of such alignment by training on large-scale image-text pairs. To adapt CLIP to the medical domain, PubMedCLIP (Eslami et al., 2023) fine-tuned CLIP on PubMed image-text pairs, while PMC-CLIP (Lin et al., 2023) trained the model from scratch using 1.6 million images from PMC-OA, achieving state-of-the-art results on retrieval and VQA tasks. Other contrastive approaches exploit unlabelled radiological images. CPRD (Liu et al., 2021) trains modality-specific encoders on brain, chest, and abdominal scans using MoCo and distills them into a unified encoder. CMCD (Chen et al. , 2024) uses multiple teachers with cross-modal contrastive distillation for comprehensive representation learning. MM-VQA (Silva et al., 2023) combines supervised contrastive loss between image-text pairs (using sentence-Bert similarity) with masked language modelling (MLM) during pre-training on ROCO. MITER (Shu et al., 2024) proposes a two-stage contrastive training framework (uni-modal and cross-modal) to ensure strong intra- and inter-modal representation learning. AMiF (Zou & Yin, 2025a) and MVCM (Zou & Yin, 2025b) adopt more fine-graded unified alignment methods for heterogeneous modality alignments across multiple levels, modalities, views, and stages.

Multi-Task Pre-Training: Beyond contrastive learning, multi-task pre-training has emerged as a robust strategy. CMSA (Gong et al., 2021) pre-trains its ResNet-based encoder using classification, segmentation, and image-question matching tasks. M2I2 (Li et al., 2023) and MUMC (Li et al., 2023) combine masked image modelling, MLM, contrastive alignment, and image-text matching losses, resulting in superior downstream VQA performance. MMBert (Khare et al., 2021) adapts the Bert framework by converting multi-scale image features (from ResNet-152) into tokens and concatenating them with radiology captions. Pre-training is conducted using MLM by masking caption tokens. MedKLIP (Chen et al. , 2022) introduces knowledge-enhanced pre-training by extracting <entity, existence, location> triplets from reports, enriching them with descriptions from UMLS/Wikipedia, and aligning them with image regions via transformer decoders. MISS (Chen et al., 2024) treats Med-VQA as a generative task, where large language models generate captions for unimodal data, effectively bridging missing modalities in training.

Pre-training Datasets Pre-training strategies rely on external datasets to scale learning. CPRD is the only method that forgoes external data. In contrast, MITER uses six datasets, and PMC-CLIP pre-trains on 1.65M samples, the largest among all methods. Fig. 3 and Table 3 summarize dataset usage and size. These datasets are mostly X-ray images (especially chest radiographs) and other radiology modalities (e.g. CT, MRI, ultrasound). ROCO is most popular (9 methods), followed by MedICaT (6) and MIMIC-CXR (4), while most chest X-ray sets are used by only 1-2 methods. As for the dataset size, PMC-OA contains the largest number of images and MIMIC-CXR is the largest domain-specific dataset (chest X-ray).

This heavy reliance on chest X-ray datasets introduces biases in learned representations. Over half of the commonly used datasets are chest-focused, resulting in models overfitted to thoracic imaging patterns. This limits generalization to other modalities (e.g., CT, MRI, US) and anatomical regions (e.g., brain, abdomen). While diverse datasets such as PMC-OA and MedICaT span multiple modalities and body regions, they remain underutilized. This imbalance is a critical limitation. Models trained on chest-heavy datasets may excel in thoracic tasks but fail in broader applications. Thus, improved diversity in dataset design and adaptive pre-training strategies are essential. Future research should focus on curating diverse, balanced datasets and devising pre-training methods capable of mitigating modality- and anatomy-specific biases. Enabling transferability across domains and improving generalization is key to advancing Med-VQA toward clinically viable AI assistants.

Pre-training strategies are crucial for task-specific Med-VQA models to mitigate the challenges of limited annotated data and domain-specificity. Techniques such as contrastive learning, masked modelling, and multi-task pre-training are applied over large external datasets like ROCO, PMC-OA, and MIMIC-CXR, which have shown substantial improvements in feature alignment and downstream VQA performance. These methods allow TSMs to generalize better and achieve strong accuracy, particularly in complex clinical reasoning tasks. While they often

表2 按策略和数据对特定任务模型预训练方法的比较。

M方法	预训练 g 策略	外部 D ata
PubMedCLIP	ITC	ROCO
PMC-CLIP	MLM +IT C	PMC-OA
CPRD	CL+KD	–
CMCD 多模态视觉问答	MLM +ITM +MIM +CL	ROCO +MedICaT
MITER	MLM +CL CL+ITM	ROCO CheXpert +MIMIC-CXR +IU-Xr ay + ChestX-r ay14 +冠状病毒 ID-19 +P 广告箱
CMSA	I mage c lassificatio n +问题 n-im 年龄匹配	Ch专家
M2I2	MLM +ITM +ITC	ROCO +MedICa T +我 age-C LEF2022
MUMC	MLM +ITM +IT C	Image-CLEF2022
MMBert	MLM	ROCO
MedKLIP	MLM +ITM +RE	ROCO +MedICaT +MIMIC-CXR
MISS	MLM +ITM +IT C	ROCO +米edICaT (由 ChatGPT 增强)

面具语言 语言建模（MLM）、ITM（图文匹配）、图文对比学习（ITC）、掩码图像建模 (MIM), 关系 关系抽取（RE）、对比学习（CL）、知识蒸馏（KD）。

可靠的、与领域匹配的语料库。基于知识图谱（KG）的系统（例如 TCC-MEM、 多模态关系图）将精心策划的医疗本体直接嵌入推理流程中，提供透明和可审计的推理，但需要大量前期的图构建和维护。在实践中，在 RAG 与 KG 策略之间进行选择时，应权衡部署约束，例如数据可用性、计算预算以及所需的特定领域精度和可追溯性水平，这通常指向混合解决方案，即将轻量级检索与有针对性的高价值图模块相结合。

3.3. 预训练方法

鉴于Med-VQA数据集的规模有限且高度领域特定，预训练已经成为提升特定任务模型性能的基础性技术。通过利用大规模外部医学语料库并调整自监督学习目标，这些方法旨在学习稳健且可迁移的多模态表示——在低资源环境下减轻过拟合并提高泛化能力。与依赖广泛一般领域预训练的大型VLM不同，TSM通常采用与临床语义和影像特征紧密对齐的定制策略。表2概述了具有代表性的预训练方法，按学习目标、外部数据来源和模态对齐技术进行分类。

对比学习：对比学习已被证明在对齐视觉和文本模态方面有效。CLIP（Radford 等，2021）通过在大规模图像-文本对上训练，展示了这种对齐的通用性。为了将 CLIP 应用于医学领域，PubMedCLIP（Eslami 等，2023）在 PubMed 图像-文本对上对 CLIP 进行微调，而 PMC-CLIP（Lin 等，2023）使用 PMC-OA 的 160 万张图像从零开始训练模型，在检索和 VQA 任务上取得了最先进的结果。其他对比方法利用未标注的放射影像。CPRD（Liu 等，2021）使用 MoCo 在脑、胸部和腹部扫描上训练特定模态的编码器，并将其蒸馏成统一的编码器。CMCD（Chen 等，2024）使用多教师进行跨模态对比蒸馏，以实现全面的表示学习。MM-VQA（Silva 等，2023）结合了图像-文本对之间的监督对比损失（使用 sentence-Bert 相似度）

多任务预训练：除了对比学习，多任务预训练已经成为一种稳健的策略。CMSA（龚等，2021）

预训练其基于 ResNet 的编码器，使用分类、分割和图像-问题匹配任务进行训练。M2I2（Li 等, 2023）和 MUMC（Li 等, 2023）结合了掩码图像建模、MLM（掩码语言建模）、对比对齐和图文匹配损失，从而在下游 VQA 任务中表现优异。MMBert（Khare 等, 2021）通过将多尺度图像特征（来自 ResNet-152）转换为 token 并与放射学描述拼接，适配了 BERT 框架。预训练通过掩码语言建模（MLM）对描述 token 进行掩码训练。MedKLIP（Chen 等, 2022）引入了基于知识增强的预训练，通过从报告中提取<实体、存在、位置> 三元组，并使用 UMLS/Wikipedia 的描述进行丰富，再通过 Transformer 解码器将其与图像区域对齐。MISS（Chen 等, 2024）将 Med-VQA 视为生成任务，其中大型语言模型为单模态数据生成描述，有效弥合训练中缺失的模态。

预训练数据集 预训练策略依赖外部数据集来扩展学习。CPRD 是唯一不使用外部数据的方法。相比之下，MITER 使用六个数据集，而 PMC-CLIP 则在 165 万样本上进行预训练，是所有方法中规模最大的。图 3 和表 3 总结了数据集的使用情况和规模。这些数据集大多是 X 光图像（尤其是胸片）及其他放射学模式（例如 CT、MRI、超声波）。ROCO 最受欢迎（9 个方法使用），其次是 MedICaT（6 个）和 MIMIC-CXR（4 个），而大多数胸片数据集仅被 1-2 个方法使用。至于数据集规模，PMC-OA 包含的图像数量最多，而 MIMIC-CXR 是最大领域特定数据集（胸片）。

对胸部X光数据集的高度依赖会在学习的表征中引入偏差。常用数据集中超过一半集中于胸部，导致模型对胸腔影像模式过拟合。这限制了模型对其他模态（如CT、MRI、超声）和解剖区域（如大脑、腹部）的泛化能力。尽管像PMC-OA和MedICaT等多样化数据集涵盖多种模态和身体区域，但仍未得到充分利用。这种不平衡是一个关键限制。以胸部为主的数据集训练的模型在胸腔任务上可能表现出色，但在更广泛的应用中可能失败。因此，提高数据集设计的多样性和采用自适应预训练策略至关重要。未来的研究应重点关注策划多样化、平衡的数据集，并设计能够减轻模态和解剖特异性偏差的预训练方法。实现跨领域的可迁移性并提高泛化能力，是推动医学视觉问答向临床可行的AI助手发展的关键。

预训练策略对于特定任务的医学视觉问答（Med-VQA）模型至关重要，以缓解标注数据有限和领域特异性的挑战。对比学习、掩码建模和多任务预训练等技术被应用于大型外部数据集，如ROCO、PMC-OA和MIMIC-CXR，这些方法在特征对齐和下游VQA性能方面显示出显著改进。这些方法使任务特定模型（TSMs）能够更好地泛化并取得较高的准确性，尤其是在复杂的临床推理任务中。虽然它们常常

Table 3
Summary of external datasets used in pre-training methods.

Dataset	Imaging Modality	Body Region(s)
ROCO (Pelka et al., 2018)	Multi-modal radiology (X-ray, CT, MRI, etc.)	Various (no single focus)
PMC-OA (Lin et al., 2023)	Mixed biomedical figures	Various (no single focus)
MedICaT (Subramanian et al., 2020)	Mixed medical images	Various (no single focus)
CheXpert (Irvin et al., 2019)	X-ray	Chest
MIMIC-CXR (Johnson et al., 2019)	X-ray	Chest
IU X-ray (Demner-Fushman et al., 2016)	X-ray	Chest
ChestX-ray14 Wang et al. (2017)	X-ray	Chest
COVID-19 Radiography (Chowdhury et al., 2020)	X-ray	Chest
PadChest (Bustos et al., 2020)	X-ray	Chest
ImageCLEF2022 (Rückert et al., 2022)	Multi-modal radiology images	Various (no single focus)

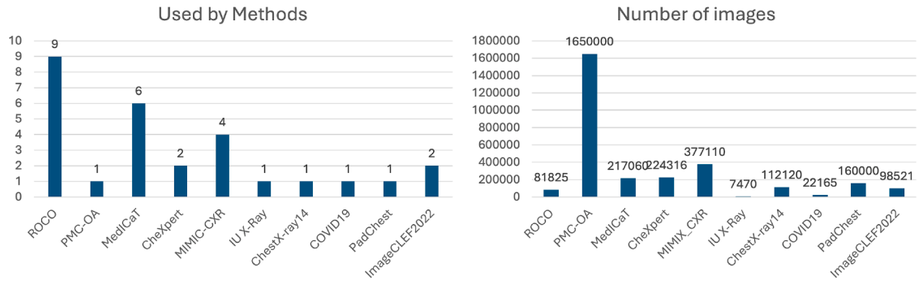


Fig. 3. Statistics of pre-training datasets.

require significant computational resources and access to large-scale, high-quality medical corpora, these pre-training approaches serve not only to enhance TSMs but also lay the groundwork for the development of large VLMs. When such resources are available, these strategies are expected to become the foundation of domain-adapted VLMs, enabling scalable, high-performance solutions across diverse medical VQA scenarios.

3.4. Augmentation-based methods

The scarcity of annotated data in the Med-VQA domain has motivated the adoption of various data augmentation strategies to expand the effective training distribution and improve model generalization. One prominent class of such strategies involves Mixup-based approaches. For instance, M-Mixup (Liu et al., 2022) adopts a standard MixUp technique to interpolate between pairs of training examples, thereby generating synthetic samples that regularize the decision boundary of the model. However, traditional MixUp methods may result in semantically implausible samples in the medical context. To address this limitation, VQAMix (Gong et al., 2022) proposes a conditional triplet MixUp strategy that enforces semantic consistency during sample generation. Specifically, it blends two image-question-answer (IQA) triplets only if they originate from the same anatomical region, thereby preventing the creation of medically implausible samples, such as a lung appearing in a brain CT scan. This constraint ensures that augmented examples maintain clinical plausibility and coherence. Beyond Mixup, other approaches have integrated augmentation with architectural innovations. Li et al. (2023b) introduces the Asymmetric Cross-Modal Attention (ACMA) network, which consists of two complementary attention pathways: an image-guided attention mechanism and a question-guided counterpart. Alongside this, they propose the Multimodal Augmented Mixup (MAM) strategy, which enhances standard MixUp by introducing additional multimodal perturbations, effectively enriching the training sample space.

In contrast to MixUp-based approaches, pseudo-labelling offers a label-efficient augmentation paradigm. PLMVQA (Yu et al., 2023) utilizes a self-training framework that generates pseudo-labels for unlabelled image-question pairs using a pretrained model. These pseudo-labelled samples are iteratively incorporated into the training set, enabling the exploitation of external unlabelled datasets without manual annotation. While this method significantly improves performance under limited supervision, its effectiveness is contingent upon the quality of the generated pseudo-labels. To mitigate the risk of propagating erroneous predictions, mechanisms such as confidence thresholds are employed to filter out low-confidence answers, thereby stabilizing the training dynamics.

3.5. Multi-question learning

Clinical diagnostic workflows often involve multiple interrelated questions about a single medical image, reflecting the structured and sequential reasoning employed by radiologists. To better emulate this process, recent works have explored multi-question learning (MQL) strategies that jointly model the dependencies among multiple QA pairs rather than treating each question-answer (QA) pair independently.

One such approach is hierarchical VQA, introduced by Pellegrini et al. (2023) in the Rad-ReStruct benchmark. This dataset provides structured radiology annotations where questions are organized hierarchically to mirror clinical reporting patterns. The proposed hi-VQA framework explicitly incorporates contextual dependencies by conditioning the answer to each question on the answers to previously addressed questions. For example, if an initial question confirms the presence of a lesion, this information is utilized in answering subsequent questions regarding its characteristics. This context-aware strategy enhances the consistency and clinical plausibility of the generated responses. Complementing the hierarchical approach, MMQL (Chen et al. , 2024) presents a model designed to jointly answer multiple questions in parallel. Rather than relying on a predefined question hierarchy, MMQL treats the entire set of questions associated with an image as joint input and produces all corresponding answers simultaneously. This formulation allows the model to leverage shared image features and capture inter-question correlations, such as the co-occurrence of related findings. To facilitate effective joint learning, MMQL introduces two technical innovations. First, it employs an iterative prompting mechanism in which

表 3 预训练方法中使用的外部数据集摘要。

D数据集	Imaging M模式	B身体部位
ROCO（佩 lk l., 2018）	多 - 模态放射学（ X-r 嗖, CT, M RI 等）	各种（无单数） le f ocus)
PMC-OA (Lin et al. 2023)	混合生物医学 fi数字	各种（无单数） le f ocus)
MedICaT（Subramanian 等）, 2020）	混合医学 im ages	各种（无单数） le f ocus)
Ch eXpert (Ir vin et al., 2019)	X-ray	Chest
MIMIC- CXR (Jo hnson 等 a l., 2019)	X-ray	Chest
IU X-r ay (Demner-F ush人和一个l., 2016)	X-ray	Chest
ChestX-ray14 王等 (2017)	X-ray	Chest
COVID-19 放射学（Chowdhury 等人） ., 2020)	X-ray	Chest
PadChest（Bustos 等人 , 2020)	X-ray	Chest
ImageCLEF2022（Rückert 等，2022）)	多重-m 口腔放射学 images	各种（无单数） le f ocus)

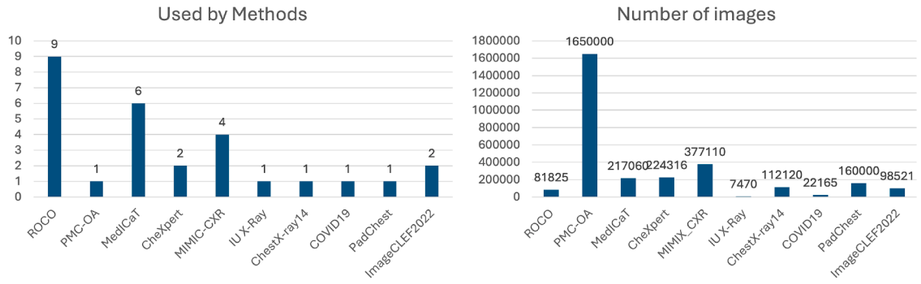


图 3. 预训练数据集统计。

需要大量的计算资源和对大规模、高质量医学语料库的访问，这些预训练方法不仅用于增强TSMs，同时也为大型VLMs的发展奠定了基础。当这些资源可用时，这些策略预计将成为领域适应型VLMs的基础，从而在各种医学VQA场景中实现可扩展的高性能解决方案。

3.4. 基于增强的方法

在Med-VQA领域，标注数据的稀缺促使研究人员采用各种数据增强策略，以扩大有效的训练分布并提升模型的泛化能力。其中一个突出的策略类别涉及基于Mixup的方法。例如，M-Mixup（Liu et al., 2022）采用标准的MixUp技术在训练样本对之间进行插值，从而生成用于正则化模型决策边界的合成样本。然而，传统的MixUp方法可能在医学语境中产生语义上不合理的样本。为了解决这一局限性，VQAMix（Gong et al., 2022）提出了一种条件三元组MixUp策略，在样本生成过程中保持语义一致性。具体来说，它仅在两个图像-问题-答案（IQA）三元组来自同一解剖区域时进行混合，从而防止生成医学上不合理的样本，例如肺部出现在脑部CT扫描中。

与基于 MixUp 的方法相比，伪标签提供了一种标签高效的增强范式。PLMVQA（Yu 等，2023）利用自训练框架为未标记的数据生成伪标签使用预训练模型对图像-问题对进行伪标签标注。这些伪标签样本会被迭代地纳入训练集，从而在无需人工标注的情况下利用外部未标注的数据集。虽然该方法在有限监督下显著提升了性能，但其有效性依赖于生成的伪标签的质量。为了减轻传播错误预测的风险，会采用置信度阈值等机制来过滤低置信度的答案，从而稳定训练过程。

3.5. 多问题学习

临床诊断工作流程通常涉及关于单一医学影像的多个相互关联的问题，这反映了放射科医生所采用的结构化和顺序推理。为了更好地模拟这一过程，近期的研究探索了多问题学习（MQL）策略，这种策略联合建模多个问答（QA）对之间的依赖关系，而不是将每个问答（QA）对独立对待。

其中一种方法是分层VQA，由Pellegrini等人（2023）在Rad-ReStruct基准中提出。该数据集提供了结构化的放射学标注，其中问题按照层级组织，以反映临床报告的模式。所提出的hi-VQA框架通过将每个问题的答案依赖于之前问题的答案来显式地整合上下文依赖。例如，如果一个初始问题确认了病变的存在，这一信息将在回答后续关于其特征的问题时被利用。这种上下文感知策略增强了生成响应的一致性和临床合理性。为了补充分层方法，MMQL（Chen等人，2024）提出了一个旨在并行回答多个问题的模型。与依赖预定义问题层级的方法不同，MMQL将与图像相关的整个问题集视为联合输入，并生成所有

already-answered questions can serve as input prompts for answering the remaining questions. To prevent cascading errors, an entropy-based pruning strategy is applied to discard unreliable predictions from being reused as prompts. Second, MMQL applies a question shuffle augmentation during training, which enforces order-invariance and encourages the model to remain agnostic to the question sequence. This design ensures robustness and flexibility across varying question orders. Evaluation under a "patient-level" metric–requiring all questions for a given image to be correctly answered–demonstrates that MMQL outperforms single-question models in both accuracy and consistency. Importantly, MMQL does not assume any fixed structure among questions, making it adaptable to diverse clinical scenarios and question sets.

3.6. Interpretable and trustworthy VQA methods

In the medical domain, interpretability and trustworthiness are paramount, as clinical decisions demand not only accurate outputs but also transparent justifications. To this end, a growing body of research in Med-VQA has focused on enhancing model transparency, logical consistency, and explanation quality.

One line of research improves interpretability by explicitly modulating the model's attention to prioritize question semantics. For instance, Vu et al. (2020) propose a question-centric VQA architecture, termed QC-MLB, which emphasizes the influence of the question in the image-text fusion process. Specifically, they modify the bilinear pooling operation by squaring and re-weighting the question feature vectors prior to fusion. This adjustment ensures that the question's semantic content exerts a stronger influence on the joint embedding, thereby aligning the model's reasoning process more closely with the clinical inquiry. This design is motivated by empirical observations in Med-VQA that question-only models often outperform image-only models, indicating that many medical questions are themselves highly informative and contain strong priors about the likely answer. A complementary direction focuses on ensuring logical consistency between answers to semantically related questions. Tascon-Morales et al. (2019) argue that trustworthy VQA systems should avoid contradictory responses and reflect medically valid reasoning. They introduce a consistency-preserving training framework that incorporates domain-specific logical rules into the loss function. For example, in a diabetic retinopathy dataset, a negative response to "Are there microaneurysms present?" should logically preclude subsequent affirmations of moderate or severe disease, as such grades presuppose the existence of microaneurysms. Their method enforces such rules by applying constraint-based penalty terms during training. While this approach promotes internal consistency, it relies on predefined medical knowledge to specify the interdependencies between questions. As such, it may not generalize to scenarios where the logical relationships are unknown or not explicitly encoded. Similarly, Cycle-VQA (Fan et al., 2025) use keeps the consistency by adding a backward reasoning path that let the image and prediction are "pointing" to the corresponding question.

Beyond improving reasoning structure, other works have sought to enhance spatial interpretability by allowing models to handle localized queries. Most conventional VQA systems treat the entire image as the visual input, limiting their capability to answer region-specific questions. To address this limitation, Tascon-Morales et al. (2022) propose a model capable of resolving spatial references within questions. For instance, in answering "What is the abnormality in the upper lobe of the right lung?", their model links the spatial language in the question to specific image regions. This is achieved by integrating spatial attention mechanisms that direct the model's focus toward the referenced sub-region, while still leveraging global image context. This yields two key interpretability benefits: (1) it enables fine-grained, region-specific interrogation of the image, aligning with how clinicians examine localized findings, and (2) it produces inherently more transparent attention distributions, clarifying which part of the image influenced the model's decision. However, the applicability of this method is contingent upon the availability of datasets annotated with region-specific questions, thereby increasing data collection complexity. In addition to architectural and training-based strategies, some efforts improve interpretability through the generation of explanatory outputs. Bai et al. (2023) introduce a transformer-based Med-VQA framework that not only predicts answers but also produces visual explanations using attention visualization. Their model adopts a dual-encoder structure–comprising separate encoders for image and question–and an autoregressive decoder for answer generation. During inference, the model's attention distributions over both image regions and question tokens are retained and visualized as heatmaps. These attention maps serve as post hoc explanations, allowing users to inspect which parts of the input influenced the generated answer. This form of transparency is particularly valuable in clinical applications, where visual explanations can assist in validating model outputs or identifying potential reasoning failures.

Finally, Hu et al. (2024a) create a large chest X-ray VQA dataset with more complex questions involving disease names, locations, and severities, and they propose a graph-based reasoning approach to tackle it. Their model constructs three types of relationship graphs, spatial relationships between detected image regions, semantic relationships between key terms in the question, and an implicit knowledge graph linking medical concepts, and then learns to traverse these graphs to infer the answer. For each question type, the model can follow a different "reasoning path" through the nodes. This method offers interpretable reasoning chains: the sequence of graph nodes and edges used can be viewed as a rationale for the answer.

4. Advances in large vision-language models for Med-VQA

Vision-Language Models have brought a paradigm shift to the Med-VQA domain by enabling powerful multimodal reasoning through large-scale pre-training on diverse image-text corpora. Unlike TSMs, VLMs are designed for general-purpose applications and offer strong zero-shot and transfer learning capabilities, often without requiring extensive task-specific annotations. This section systematically reviews how VLMs are applied to Med-VQA tasks, with a focus on five key strategies. As highlighted in the green segment of Fig. 2, it first examines zero-shot methods, which utilize pretrained general-domain knowledge to perform medical question answering without fine-tuning, highlighting their advantages and limitations in specialized clinical scenarios. Then, it explores full parameter update fine-tuned approaches, which adapt VLMs to medical contexts through supervised updates, improving domain alignment and diagnostic accuracy. Followed by low resource fine-tune methods (parameter-efficient fine-tune). Moreover, instruction tuning methods which enhancing the reasoning capability and the downstream tasks flexibility are discussed. Finally, explainable methods the prediction is grounded on actual anatomy on medical image are introduced.

4.1. Zero-shot methods

Zero-shot Med-VQA methods avoid task-specific fine-tuning by leveraging large pre-trained models and auxiliary information sources. These methods fall into two primary categories: (1) **Prompt-Based Generative Models**, and (2) **Retrieval-Augmented Models**.

Prompt-Based Generative Models: Med-Flamingo (Moor et al., 2023) exemplifies this approach by adapting the Flamingo architecture, a 9B-parameter autoregressive transformer with cross-attention for images, to the medical domain. This model is pre-trained on a corpus of paired medical images and text (from textbooks and publications). In deployment, Med-Flamingo is used in a few-shot prompting setup (rather than task-specific fine-tuning). Med-Flamingo excels in generating nuanced free-text answers and rationales and achieves up to 20% improvement in clinician-rated quality over baselines. However, it requires carefully crafted prompts and remains prone to hallucinations due to open-ended generation. Similarly, RadFM (Wu et al., 2025a) is

已回答的问题可以作为输入提示，用于回答剩余的问题。为了防止级联错误，采用基于熵的剪枝策略，以丢弃不可靠的预测，避免其被重复用作提示。其次，MMQL 在训练过程中应用了问题打乱增强，这强制实现顺序不变性，并鼓励模型对问题顺序保持中立。这种设计确保了在不同问题顺序下的稳健性和灵活性。基于"患者级"指标的评估——要求对给定图像的所有问题都正确回答——表明 MMQL 在准确性和一致性方面均优于单问题模型。重要的是，MMQL 不假设问题之间存在任何固定结构，使其能够适应多样的临床场景和问题集合。

3.6. 可解释且可靠的 VQA 方法

在医疗领域，可解释性和可信性是至关重要的，因为临床决策不仅需要准确的结果，还需要透明的理由。为此，越来越多的 Med-VQA 研究集中于提高模型的透明性、逻辑一致性和解释质量。

一项研究通过显式调节模型的注意力以优先考虑问题语义来提高可解释性。例如，Vu 等人（2020）提出了一种以问题为中心的 VQA 架构，称为 QC-MLB，该架构强调问题在图像-文本融合过程中的影响。具体来说，他们通过在融合之前对问题特征向量进行平方和重新加权来修改双线性池化操作。这一调整确保了问题的语义内容对联合嵌入产生更强的影响，从而使模型的推理过程更贴近临床问题。这一设计的动机来自 Med-VQA 中的经验观察，即仅使用问题的模型通常优于仅使用图像的模型，这表明许多医学问题本身信息量很大，并且包含有关可能答案的强先验。一个补充方向则侧重于确保语义相关问题的答案之间逻辑一致性。

除了改善推理结构之外，其他研究还通过允许模型处理局部查询来增强空间可解释性。大多数传统的视觉问答（VQA）系统将整个图像作为视觉输入，这限制了它们回答特定区域问题的能力。为了解决这一限制，Tascon-Morales 等人（2022）提出了一种能够解析问题中空间引用的模型。例如，在回答"右肺上叶的异常是什么？"时，他们的模型将问题中的空间语言与特定图像区域联系起来。这是通过整合空间注意机制实现的，该机制引导模型将注意力集中在引用的子区域，同时仍利用全局图像上下文。这带来了两个关键的可解释性优势：（1）它能够对图像进行细粒度、区域特定的查询，与临床医生检查局部发现的方式一致；（2）它产生本质上更透明的注意力分布。

带有特定区域问题标注的数据集的可用性，从而增加了数据收集的复杂性。除了基于架构和训练的策略之外，一些研究通过生成解释性输出提高了可解释性。Bai 等人（2023）提出了一种基于 Transformer 的 Med-VQA 框架，不仅可以预测答案，还可以通过注意力可视化生成视觉解释。他们的模型采用双编码器结构——包含用于图像和问题的独立编码器——以及用于答案生成的自回归解码器。在推理过程中，模型对图像区域和问题词元的注意力分布都会被保留并可视化为热力图。这些注意力图作为事后解释，使用户能够检查哪些输入部分影响了生成的答案。这种形式的透明性在临床应用中尤为有价值，因为视觉解释可以帮助验证模型。

最后，Hu 等人（2024a）创建了一个大型胸部 X 光问答（VQA）数据集，其中包含涉及疾病名称、位置和严重程度的更复杂问题，并提出了一种基于图的推理方法来解决这些问题。他们的模型构建了三种类型的关系图：检测到的图像区域之间的空间关系、问题中关键术语之间的语义关系，以及连接医学概念的隐性知识图，然后学习遍历这些图来推断答案。对于每种问题类型，模型可以通过节点遵循不同的"推理路径"。这种方法提供了可解释的推理链：使用的图节点和边的序列可以被视为答案的理由。

4. 医学视觉问答大型视觉-语言模型的进展

视觉-语言模型通过在多样化的图文语料库上进行大规模预训练，实现了强大的多模态推理能力，从而在医学视觉问答（Med-VQA）领域带来了范式转变。与TSM不同，VLM被设计用于通用应用，并提供强大的零样本和迁移学习能力，通常无需大量特定任务的标注。本节系统回顾了VLM在Med-VQA任务中的应用，重点关注五种关键策略。如图2绿色部分所示，首先考察零样本方法，这类方法利用预训练的通用领域知识在无需微调的情况下执行医学问答，突出了它们在特定临床场景中的优势和局限性。随后，探讨了全参数更新微调方法，通过监督更新将VLM适应于医学环境，从而提升领域对齐性和诊断准确性。接着是低资源微调方法（参数有效）。

4.1. 零样本方法

零样本医学视觉问答（Med-VQA）方法通过利用大型预训练模型和辅助信息来源来避免针对特定任务的微调。这些方法主要分为两类：(1) 基于提示的生成模型，(2) 检索增强模型。

基于提示的生成模型：Med-Flamingo（Moor 等，2023）通过将 Flamingo 架构（一个具有图像跨注意力的 9B 参数自回归变换器）适配到医疗领域，体现了这一方法。该模型在配对的医学图像和文本语料库（来自教材和出版物）上进行预训练。在部署时，Med-Flamingo 以少样本提示设置（而非特定任务微调）使用。Med-Flamingo 擅长生成细致的自由文本答案和推理，并在临床医生评估的质量上相比基线模型提升了最多 20%。然而，它需要精心设计的提示，并且由于开放式生成仍容易出现幻觉。同样，RadFM（Wu 等，2025a）是

a generative encoder-decoder model designed specifically for radiology tasks. It integrates a 3D Vision Transformer, a Perceiver-based fusion module, and a MedLLaMA-13B decoder. Trained on a large-scale corpus (MedMD, approximately 16 million medical image-text pairs and RadMD, a refined subset containing about 3 million expert-reviewed radiology cases), RadFM handles both 2D and 3D inputs and demonstrates strong zero-shot VQA performance on datasets like PMC-VQA and SLAKE.

Retrieval-Augmented Models: In contrast, MKGF (Wu et al., 2025b) enhances frozen large language models (e.g. LLaVA-med-7B or HuatuoGPT-Vision-7B) with a structured Multi-Modal Knowledge Graph (MMKG) of medical images, diseases, and organs. A learned retriever maps input queries to relevant nodes in the MMKG, and retrieved evidence is injected into the model's input. This grounding reduces hallucinations and improves factuality in zero-shot settings. MKGF shows performance gains on VQA-RAD and SLAKE, with the added benefit of evidence transparency. However, it is limited by graph coverage and retrieval accuracy, and its multi-stage pipeline introduces inference latency.

Zero-shot Med-VQA methods bypass task-specific fine-tuning by leveraging the general-domain knowledge embedded in large pretrained models. These approaches are appealing for rapid deployment and low-resource settings, but they often face challenges in clinical accuracy, domain-specificity, and interpretability. Prompt-based generative models enable flexible response generation but are sensitive to prompt design and prone to hallucinations. Retrieval-augmented models improve factual grounding by injecting external evidence but depend heavily on retrieval quality and structured knowledge sources. Overall, zero-shot methods offer promising generalization, yet lack the domain adaptation necessary for reliable clinical use–underscoring the need for fine-tuning or hybrid strategies in high-stakes medical applications.

4.2. Full parameter fine-tuned methods

Fine-tuning methods are the cornerstone of high performance Med-VQA systems, enabling pretrained models to adapt to domain-specific tasks. These approaches differ in architecture, data requirements. This paper group the methods into two categories: (1) Full parameter update models, (2) Knowledge-Guide Models. The former group adapts the general VLM or pre-trained Medical VLM to downstream tasks with full parameter update, while the later group also adopts this paradigm but using auxiliary data for domain knowledge injection.

Full Parameter Update: These methods construct foundational V + L representations and adapt them across multiple downstream tasks. However, compare to zero-shot methods, these methods often requires full gradient update for downstream tasks. MedViLL (Moon et al., 2022) uses a ResNet-50 backbone for image encoding and a 12-layer Bert transformer for joint multimodal embedding. It introduces a bidirectional auto-regressive attention mask during pre-training, combining causal and bidirectional dependencies to balance generative and discriminative tasks. MedViLL is pre-trained on large-scale CXR-report datasets like MIMIC-CXR and Open-I and fine-tuned on classification, retrieval, report generation, and VQA. The flexibility of this architecture and its multi-task capabilities demonstrate the utility of unified modelling in clinical natural language processing and vision tasks. Similarly, BiomedCLIP (Zhang et al., 2024b) scales the contrastive vision-language paradigm by training on PMC-15M, a 15-million biomedical image-caption dataset. Its architecture is a CLIP-style dual encoder (ViT-B and PubMedBert) with domain-specific augmentations (high-resolution input, patch dropout). It serves as an effective foundation for zero-shot scenarios like image-text retrieval and image classification, though its generative capabilities require an external fusion/decoder module. However, for Med-VQA tasks it still fine-tuned the model using the MITER (Shu et al., 2024) framework. Further, M^3AE (Chen et al., 2024b) extend the diversity for pre-training tasks by pre-training with masked modelling on both modalities. The architecture consists of a vision encoder + pixel decoder and a Bert-based language encoder + word-piece decoder. Intermediate-level features are reconstructed from heavily masked inputs. The pre-training objective aligns high-level visual and textual abstractions, which enhances performance on classification, retrieval, and VQA, especially in low-data regimes. Though large and training-intensive, M^3AE excels at producing semantically aligned multimodal embeddings. PTUnifier (Chen et al. , 2023) expands on this paradigm by incorporating soft prompts for input flexibility. Its dual-encoder backbone (CLIP-ViT-B for images and RoBerta-base for text) feeds into a 6-layer fusion transformer. Through dynamically selected visual/textual prompt tokens, it unifies input formats (image-only, text-only, image-text pairs) and downstream task types (classification, retrieval, captioning, VQA). The key innovation is in the learnable prompt pool (1024 prompts) enabling effective transfer across tasks while minimizing architectural changes.

Knowledge-Guided Models: These methods inject domain-specific knowledge explicitly into the vision-language pipeline. But similar to previous group of methods, these methods need full parameter updates. MMCAP (Yan et al. , 2024) constructs a multimodal knowledge graph using UMLS concepts and image-caption pairs. A GAT fuses this graph with learned image and text embeddings. A transformer decoder generates answers conditioned on both fused features and a Type-Conditional Prompt that encodes question intent. MMCAP reduces hallucination and boosts clinical accuracy, especially in low-data scenarios. However, it is resource-intensive and constrained by graph coverage. BioMedGPT (Zhang et al., 2024a) is a compact encoder-decoder model (182M parameters) trained on 25 diverse biomedical tasks. It demonstrates strong performance on classification and VQA benchmarks despite its size. Its design allows low-resource deployment, and its generative fine-tuning makes it more grounded than zero-shot models, though its capacity may limit performance on highly complex or reasoning-heavy queries. Med-CoE (Qiu et al., 2025) uses GPT3.5 to generate chain of evidence (CoE) reasoning process, and further fine-tune a Flan-Alpaca by < Image, Quetion, Answer > triples and CoE data for explainable Med-VQA system.

4.3. Parameter-efficient fine-tuning models

Parameter-Efficient Fine-Tuning (PEFT) methods reduce training and memory costs by updating only small parts of large models. MILE (Chen et al., 2024) explores four PEFT strategies: LoRA, Prefix Tuning, IA3, and P-Tuning v2 on a VLM composed of ViT-base, a joint image-text encoder, and transformer decoder. LoRA-Tuning and Prefix-Tuning exhibit the best tuning effects, achieving comparable performance to global fine-tuning models while reducing training costs by 40%. LayerNorm-tuning (Chen et al. , 2024) updates only normalization parameters in large multimodal transformers. Experiments show that only tuning LayerNorm layers of Joint Text Multimodal (JTM) encoder can achieves near-parity (78.6% vs. 82.0%) with full fine-tuning on SLAKE, with about 60% parameter updates. This approach provides extreme efficiency and scalability, particularly valuable for low-resource institutions. VQA-Adapter (Liu et al., 2023) inserts lightweight adapters into frozen CLIP encoders. A multi-stage label-smoothing regularization prevents overfitting on small Med-VQA datasets. This method requires only 2.38% updated parameters and outperforms full tuning on both closed- and open-ended tasks. It provides a simple yet effective way to update large pretrained models without catastrophic forgetting. Prefix Tuning for VQA (Van Sonsbeek et al. , 2023) prepends learned visual "prefix" tokens to the input of a language model (e.g., GPT-2). Visual features are transformed by a small adapter network into prefix embeddings, which condition the language model for open-ended Med-VQA generation. It achieves superior BLEU, F1, and BertScore across PathVQA, SLAKE, and OVQA benchmarks, demonstrating efficiency and fluency in free-form answer generation.

一个专门为放射学任务设计的生成型编码器-解码器模型。它集成了3D视觉变换器、基于Perceiver的融合模块和MedLLaMA-13B解码器。在大规模语料库上训练（MedMD，约1600万医学图像-文本对，以及RadMD，一个经过精炼的子集，包含约300万经过专家审核的放射学病例），RadFM可处理2D和3D输入，并在PMC-VQA和SLAKE等数据集上展示出强大的零样本视觉问答性能。

增强检索模型：相比之下，MKGF（Wu 等，2025b）通过结构化的多模态知识图（MMKG，包含医学影像、疾病和器官）增强了冻结的大型语言模型（例如 LLaVA-med-7B 或 HuatuoGPT-Vision-7B）。一个学习到的检索器将输入查询映射到 MMKG 中的相关节点，检索到的证据被注入模型的输入。这种基础支持减少了幻觉现象并提高了零样本情况下的事实准确性。MKGF 在 VQA-RAD 和 SLAKE 数据集上显示了性能提升，同时增加了证据透明性。然而，它受限于知识图的覆盖范围和检索准确性，其多阶段管道还会带来推理延迟。

零样本医用视觉问答（Med-VQA）方法通过利用大型预训练模型中嵌入的通用领域知识来规避特定任务的微调。这些方法对于快速部署和低资源环境具有吸引力，但它们在临床准确性、领域特定性和可解释性方面常常面临挑战。基于提示的生成模型能够灵活生成响应，但对提示设计敏感且容易产生幻觉。增强检索的模型通过引入外部证据来提高事实依据性，但高度依赖检索质量和结构化知识来源。总体而言，零样本方法提供了有前景的泛化能力，但缺乏可靠临床使用所需的领域适应性——这凸显了在高风险医疗应用中进行微调或采用混合策略的必要性。

4.2. 全参数微调方法

微调方法是高性能医学视觉问答（Med-VQA）系统的基石，使预训练模型能够适应特定领域的任务。这些方法在架构和数据需求上有所不同。本文将这些方法分为两类：（1）全参数更新模型，（2）知识引导模型。前一类通过全参数更新将通用视觉语言模型（VLM）或预训练的医学 VLM适应下游任务，而后一类也采用这一范式，但使用辅助数据注入领域知识。

全参数更新：这些方法构建了基础的 V +L 表示，并将其适应于多个下游任务。然而，与零-shot 方法相比，这些方法通常需要对下游任务进行全梯度更新。MedViLL（Moon 等, 2022）使用 ResNet-50 作为图像编码骨干，并使用 12 层 BERT Transformer 进行联合多模态嵌入。它在预训练期间引入了双向自回归注意力掩码，结合因果和双向依赖，以平衡生成和判别任务。MedViLL 在大规模 CXR-报告数据集如 MIMIC-CXR 和 Open-I 上进行预训练，并在分类、检索、报告生成和视觉问答（VQA）任务上进行微调。该架构的灵活性及其多任务能力展示了统一建模在临床自然语言处理和视觉任务中的实用性。类似地，BiomedCLIP（Zhang 等, 2024b）通过训练扩大了对比视觉-语言范式。一个视觉编码器 + 像素解码器和基于Bert的语言编码器 + 词片段解码器。从高度遮蔽的输入中重建中级特征。预训练目标是对齐高级视觉和文本抽象，这提高了分类、检索和视觉问答（VQA）的性能，特别是在数据量低的情况下。尽管规模大且训练密集，M^3AE 在生成语义对齐的多模态嵌入方面表现出色。PTUnifier（Chen 等人，2023）在此范式基础上进行扩展，通过引入软提示以增加输入的灵活性。其双编码器骨干（用于图像的 CLIP-ViT-B 和用于文本的 RoBerta-base）输入到一个6层融合变换器。通过动态选择的视觉/文本提示标记，它统一了输入格式（仅图像、仅文本、图像-文本对）和下游任务类型（分类、检索、生成描述、VQA）。关键创新在于可学习的提示池（1024个提示），使跨任务的有效迁移成为可能，同时最小化

知识引导模型：这些方法将特定领域的知识显式地注入视觉-语言流程中。但与前一组方法类似，这些方法需要完全的参数更新。MMCAP（Yan 等，2024）使用 UMLS 概念和图像-字幕对构建多模态知识图。GAT 将该图与学习到的图像和文本嵌入融合。一个 Transformer 解码器在融合特征和编码问题意图的类型条件提示的条件下生成答案。MMCAP 减少了幻觉现象并提升了临床准确性，特别是在数据量较少的情况下。然而，它资源消耗大，并受限于图覆盖范围。BioMedGPT（Zhang 等，2024a）是一个紧凑的编码器-解码器模型（1.82亿参数），在25个多样化的生物医学任务上训练。尽管其体积较小，但在分类和 VQA 基准上表现出强劲性能。其设计允许低资源部署，其生成式微调使其更加可靠。

4.3. 参数高效微调模型

参数高效微调（PEFT）方法通过仅更新大型模型的小部分参数来降低训练和内存成本。MILE（Chen 等，2024）在由 ViT-base、联合图文编码器和 Transformer 解码器组成的 VLM 上探索了四种 PEFT 策略：LoRA、前缀调优（Prefix Tuning）、IA3 和 P-Tuning v2。LoRA 调优和前缀调优表现出最佳的调优效果，在训练成本降低 40% 的同时，实现了与全量微调模型相当的性能。LayerNorm 调优（Chen 等，2024）仅更新大型多模态 Transformer 中的归一化参数。实验表明，在联合文本多模态（JTM）编码器中仅调优 LayerNorm 层，就能在 SLAKE 上实现接近全量微调的性能（78.6% 对比 82.0%），并更新约 60% 的参数。这种方法提供了极高的效率和可扩展性，对于资源有限的机构尤为有价值。VQA-Adapter（Liu 等，2023）在冻结的 CLIP 编码器中插入轻量级适配器。一个多阶段

4.4. Instruction-tuning

Instruction tuning in medical VLM and Med-VQA settings refers to a training-time paradigm that updates model parameters using instruction-response formatted data, with the goal of improving instruction following and task transfer across diverse medical tasks (e.g., diagnosis support, report generation, and question answering). This differs from task-specific fine-tuning, which further adapts a model to a particular dataset or objective and is primarily used to maximize performance on a targeted benchmark. By improving a model's general instruction-following capability, instruction tuning can also benefit zero-shot/few-shot deployment at inference time (i.e., prompt-only use without parameter updates), often yielding more robust reasoning behaviour and better downstream adaptability. In practice, many medical VLM pipelines apply instruction tuning after pre-training but before task-specific fine-tuning, using either full-parameter updates or PEFT to balance performance and resource cost. Med-Gemini (Saab et al., 2024) provides a representative example of task adaptation: the authors introduce several variants, including *Med-Gemini-L 1.0* (adapting Gemini Ultra via self-training with integrated web search), *Med-Gemini-M 1.5* (instruction tuning Gemini Pro on a mixture of Med-VQA and medical classification tasks), and *Med-Gemini-S 1.0* (augmenting the compact Gemini Nano with task-specific encoders, e.g., a Flamingo-style cross-attention encoder for raw ECG waveform input). These variants follow the standard Gemini transformer design with separate vision and text components connected through cross-modal attention; importantly, their reported gains come primarily from training-time adaptation (e.g., tuning on task-specific instructions and/or self-training) and inference-time augmentation (e.g., retrieval via web search in Med-Gemini-L). Similarly, LLaVA-Med (Li et al., 2023a) adapts a 13B LLaVA model in two stages–biomedical figure-caption alignment (15M pairs) followed by instruction-tuned QA (60K GPT-4-generated pairs)–and MLeVLM (Xu et al., 2024) adopts a comparable pipeline while introducing an attention-based feature-alignment module and training on GPT-4-generated instruction datasets.

4.5. Object grounding and explainable AI in medical VQA with vision-language models

Explainability and visual grounding are essential capabilities in Medical VLMs for building clinical trust and ensuring diagnostic transparency. Recent studies have explored various strategies–such as region-of-interest (ROI) annotations, pixel-level supervision, expert model integration, and multimodal explanation datasets–to enable interpretable reasoning in Med-VQA tasks.

ROI-guided Learning: R-LLaVA (Chen et al., 2025a) introduces doctor-annotated bounding boxes into the visual input space using CLIP embeddings, allowing the model to explicitly attend to diagnostic regions during training. By reconstructing datasets with ROI-based questions (e.g., region localization and description), the model learns spatial alignment between questions and anatomical areas. This approach enhances answer precision without modifying the model architecture, proving effective across multiple Med-VQA benchmarks.

Expert-Guided Reasoning: VILA-M3 (Nath et al., 2025) integrates task-specific expert models (e.g., segmentation or classification networks) into a general-purpose VLM pipeline. Through expert-guided instruction tuning, the VLM invokes these pretrained models during inference via modality-specific prompts, allowing precise spatial grounding and clinically validated outputs. This hybrid design supports both 2D and 3D inputs, yielding strong performance on segmentation, VQA, and report generation.

Pixel-Level Grounding: To support fine-grained explainability, MedPLIB (Huang et al., 2025) proposes a mixture-of-experts framework trained with both image-text and pixel-grounding supervision. It enables direct interaction through pixel-level prompts (e.g., bounding boxes, masks), supporting complex image understanding tasks. Similarly, MIMO (Chen et al., 2025b) is a unified medical vision-language model that takes visual-referring multimodal inputs (text + visual cues) and produces pixel-grounded multimodal outputs, linking generated medical terms to image regions, supported by the 895K-sample MIMOSeg dataset. These models demonstrate the importance of spatially resolved understanding for trustworthy medical AI.

Explainable Benchmarks. The GEMeX dataset (Liu et al., 2025a) provides a large-scale, multimodal benchmark with visual and textual explanations for each QA pair. Each item includes bounding boxes and rationales, covering four question types (open-ended, closed-ended, single-choice, multi-choice). Models evaluated on GEMeX must jointly generate accurate answers, textual justifications, and visual region annotations. The dataset also introduces dedicated metrics for answer correctness, explanation quality, and localization, encouraging development of interpretable VQA models.

5. Datasets and evaluation metrics

5.1. Datasets

In this subsection, the commonly used datasets are discussed:

VQA-MED: The VQA-MED dataset series comprises four variants, initially introduced as **VQA-Med-2018** during the ImageCLEF 2018 challenge (Hasan et al., 2018). As the first publicly available Med-VQA dataset, it employs a semi-automatic pipeline to generate question-answer (QA) pairs from image descriptions. Initially, a rule-based system produces candidate QA pairs through sentence simplification, answer phrase identification, and question generation, followed by two rounds of manual validation by annotators, including clinical experts, ensuring semantic correctness and clinical relevance. **VQA-MED-2019** (Ben Abacha et al., 2019), inspired by VQA-RAD (Lau et al., 2018), encompasses four primary question categories–modality, plane, organ system, and abnormalities–with pattern-based questions validated from naturally posed clinical inquiries. Classification tasks cover the first three categories, while abnormality-related queries necessitate generative approaches. **VQA-MED-2020** (Ionescu et al. , 2020), presented in ImageCLEF 2020, focuses exclusively on abnormality-related questions, including 330 frequent abnormality-related queries. Additionally, this dataset introduced the visual question generation task in medical imaging, comprising 1,001 images and 2,400 manually refined questions. **VQA-MED-2021** (Ben Abacha et al., 2021) follows the structure of VQA-MED-2020, maintaining the same training dataset but featuring newly curated validation and test sets, manually reviewed by clinicians.

VQA-RAD: Proposed in 2018, VQA-RAD (Lau et al., 2018) is specifically designed for radiology applications, featuring images balanced across head, chest, and abdomen categories sourced from MedPix. Clinical practitioners created unguided, free-form, and templated questions, with manual validation ensuring clinical relevance. Answers are categorized into closed or open-ended types. Despite its moderate size, VQA-RAD provides essential insights into realistic question formulation and expectations for Med-VQA systems.

RadVisDial: RadVisDial (Kovaleva et al., 2020), the first publicly available radiology visual dialogue dataset, introduces a more interactive and complex scenario than traditional VQA datasets. Derived from MIMIC-CXR (Johnson et al., 2019), the dataset includes structured reports annotated with 14 labels representing various abnormalities. It comprises silver-standard dialogues synthetically generated from text reports and gold-standard dialogues from interactive exchanges between radiologists, focusing on the practicalities of context and patient history in dialogue-based clinical scenarios.

PathVQA: PathVQA (He et al., 2020) explores the VQA domain within pathology, utilizing images and associated captions sourced from electronic textbooks and digital libraries. QA pairs generated semi-automatically from image descriptions undergo manual validation. Questions span seven categories (what, where, when, who, how, how many/much, yes/no), with open-ended queries constituting

4.4. 指令调优

在医疗视觉语言模型（VLM）和医疗视觉问答（Med-VQA）场景中，指令调优指的是一种训练时范式，它使用指令-响应格式的数据更新模型参数，目标是改善模型的指令执行能力和跨各种医疗任务（如诊断支持、报告生成和问答）的任务迁移能力。这不同于特定任务的微调，后者进一步将模型适应特定的数据集或目标，并主要用于最大化在目标基准上的性能。通过提升模型的通用指令执行能力，指令调优还可以在推理时实现零样本/少样本部署（即仅使用提示而无需更新参数），通常能够产生更稳健的推理行为和更好的下游适应能力。在实际操作中，许多医疗VLM流程在预训练后但在特定任务微调前应用指令调优，可以采用全参数更新或参数高效微调（PEFT）来平衡性能和

同样，MIMO（Chen 等, 2025b）是一种统一的医学视觉-语言模型，它接受视觉指向的多模态输入（文本 +视觉线索）并生成像素级多模态输出，将生成的医学术语与图像区域关联，该模型得到了895K样本的MIMOSeg数据集的支持。这些模型展示了空间解析理解对于可信医学人工智能的重要性。

可解释的基准。GEMeX 数据集（Liu 等人, 2025a）提供了一个大规模的多模态基准，每个问答对都包含视觉和文本解释。每个条目包括边界框和理由，涵盖四种问题类型（开放式、封闭式、单选、多选）。在GEMeX 上评估的模型必须联合生成准确的答案、文本理由和视觉区域标注。该数据集还引入了专门的评估指标，用于答案正确性、解释质量和定位，鼓励开发可解释的 VQA 模型。

5. 数据集和评估指标

5.1. 数据集

在本小节中，将讨论常用的数据集：

VQA-MED：VQA-MED 数据集系列包含四个变体，最初在 2018 年 ImageCLEF 挑战赛期间作为 VQA-Med-2018 推出（ Hasan 等，2018 ）。作为首个公开可用的医学视觉问答（Med-VQA）数据集，它采用半自动化流程从图像描述中生成问答（QA）对。最初，一个基于规则的系统通过句子简化、答案短语识别和问题生成来生成候选问答对，随后经过两轮人工验证，包括临床专家在内的标注者确保其语义正确性和临床相关性。VQA-MED-2019（Ben Abacha 等，2019）受 VQA-RAD（Lau 等，2018）的启发，涵盖四个主要问题类别——成像方式、扫描平面、器官系统和异常——其模式化问题来源于自然提出的临床疑问并经过验证。分类任务涵盖前三个类别，而与异常相关的问题则需要生成式方法。
VQA-MED-2020（I

4.5. 医学视觉问答中基于视觉-语言模型的对象定位和可解释人工智能

可解释性和视觉定位是医疗视觉语言模型（Medical VLMs）中至关重要的能力，用于建立临床信任并确保诊断透明性。近期研究探索了多种策略——如感兴趣区域（ROI）标注、像素级监督、专家模型整合以及多模态解释数据集——以实现医疗视觉问答（Med-VQA）任务中的可解释推理。

基于ROI的引导学习：R-LLaVA（Chen 等, 2025a）通过在视觉输入空间中引入医生标注的边界框并使用CLIP嵌入，使模型在训练过程中能够明确关注诊断区域。通过使用基于ROI的问题（如区域定位和描述）重建数据集，模型学习问题与解剖区域之间的空间对齐。这种方法在不修改模型架构的情况下提高了答案的精确性，并在多个Med-VQA基准测试中证明了其有效性。

专家引导推理：VILA-M3（Nath 等, 2025）将任务特定的专家模型（例如分割或分类网络）集成到通用 VLM 流水线中。通过专家引导的指令调优，VLM 在推理过程中通过特定模态的提示调用这些预训练模型，从而实现精确的空间定位和临床验证的输出。这种混合设计支持二维和三维输入，在分割、视觉问答（VQA）和报告生成方面表现优异。

像素级定位：为了支持细粒度的可解释性，MedPLIB（Huang 等, 2025）提出了一种混合专家框架，该框架通过图像-文本和像素定位监督进行训练。它通过像素级提示（例如，边界框、掩码）实现直接交互，从而支持复杂的图像理解任务。

VQA-RAD：VQA-RAD（Lau 等，2018）于2018年提出，专门为放射学应用设计，特点是在头部、胸部和腹部类别的图像之间保持平衡，这些图像来源于 MedPix。临床医生创建了无指导的自由形式和模板化问题，并通过人工验证确保临床相关性。答案被分类为封闭式或开放式类型。尽管规模适中，VQA-RAD 为 Med-VQA 系统的实际问题设计和预期提供了重要的见解。

RadVisDial：RadVisDial（Kovaleva 等，2020）是首个公开可用的放射学视觉对话数据集，其引入了比传统 VQA 数据集更具互动性和复杂性的场景。该数据集源自 MIMIC-CXR（Johnson 等，2019），包含带有 14 个标签的结构化报告，这些标签代表各种异常情况。它包括从文本报告合成生成的银标准对话，以及源自放射科医生之间互动交流的金标准对话，重点关注基于对话的临床场景中上下文和病历的实际应用。

PathVQA：PathVQA（He 等，2020）探索了病理学领域的 VQA，使用来自电子教材和数字图书馆的图像及相关说明。由图像描述半自动生成的问答对经过人工验证。问题涵盖七类（什么、哪里、何时、谁、如何、多少、是/否），其中开放式问题占比

approximately half of the dataset. Closed-ended yes/no questions are balanced to prevent bias, following guidelines consistent with American Board of Pathology certification exam standards.

SLAKE: SLAKE (Liu et al., 2021) integrates semantic annotations with structured medical knowledge bases, employing images sourced from publicly available datasets (Kavur et al., 2021; Simpson et al., 2019; Wang et al., 2017). Expert clinicians annotated these images with segmentation masks and bounding boxes. Additionally, SLAKE includes a knowledge graph containing approximately 2,600 English and Chinese triples, enabling formulation of questions requiring external knowledge. Question categories are systematically balanced to minimize dataset bias.

Kvasir-VQA: Kvasir-VQA (Gautam et al., 2024) extends endoscopy imaging datasets, HyperKvasir and Kvasir-Instrument, by including comprehensive QA annotations across the gastrointestinal tract, covering both normal and pathological conditions (e.g., polyps, esophagitis, ulcerative colitis) and medical instrumentation. The dataset contains 6,500 annotated images with multiple QA pairs, covering six question categories: yes/no, single-choice, multiple-choice, colour, location, and numerical counts. Collaborative expert annotation ensures clinical relevance and accuracy, and a subset of this dataset was utilized in the ImageCLEF 2023 challenge (Ionescu et al., 2023).

P-VQA: The patient-centric dataset P-VQA (Huang et al., 2023) provides a holistic coverage of medical imaging scenarios throughout patient care–from consultation and diagnostic imaging to therapeutic advice. Spanning 20 prevalent diseases, the dataset includes 2,169 images and 24,800 QA pairs, constructed in layperson language to address practical patient concerns such as symptoms, procedures, and test outcomes. Additionally, a medical knowledge graph consisting of 419 entities supports patient-oriented reasoning tasks, distinguishing it from clinician-focused datasets.

OVQA: The Orthopedics Visual Question Answering (OVQA) dataset (Huang et al. , 2022) addresses orthopedic-specific imaging, comprising 2,001 images (X-rays and CT slices) and 19,020 QA pairs. Semi-automatically generated using physician-approved templates, questions reflect clinically frequent inquiries, subsequently validated by experienced orthopedic surgeons. OVQA thus provides a highly accurate, clinically relevant dataset tailored explicitly for orthopedic diagnostic tasks.

OmniMedVQA: OmniMedVQA (Hu et al., 2024b) is a large-scale benchmark designed to evaluate medical VLMs across a broad range of real clinical imaging scenarios. It aggregates data from 73 existing medical datasets spanning 12 imaging modalities and 20 anatomical regions, aiming to test whether a model's VQA capability generalizes beyond a single organ, disease group, or modality. The dataset is positioned as a comprehensive evaluation resource rather than a narrow task benchmark, and experimental results reported by the creators suggest that many current large VLMs still struggle with robust medical visual reasoning on this benchmark. In practice, OmniMedVQA is most useful for probing coverage and transfer (e.g., how performance shifts across modality/body region) and for highlighting failure modes that might be hidden when evaluation is limited to a single-domain Med-VQA dataset.

DermaVQA: DermaVQA (Yim et al., 2024) targets dermatology-oriented visual question answering by pairing user-generated dermatology images with clinical-style textual queries and corresponding answers, motivated by real-world tele-dermatology / asynchronous messaging workflows. Unlike many Med-VQA benchmarks dominated by radiology, DermaVQA focuses on skin conditions and supports multilingual settings (reported languages include Chinese, English, and Spanish) to reflect how patients actually communicate symptoms and context. The dataset is intended to enable evaluation of multimodal models on dermatology question answering and response generation, and baseline experiments emphasize the challenges of grounding answers in subtle visual patterns while also handling noisy, user-authored descriptions and language variation. This makes DermaVQA especially relevant for studying robustness to uncontrolled imaging conditions and for assessing whether multimodal assistants can provide coherent, context-aware responses in real patient-facing dermatology scenarios.

Medical-CXR-VQA: Medical-CXR-VQA (Hu et al., 2025) is a large-scale chest X-ray VQA dataset curated to support question answering over CXR images, with an emphasis on scaling supervision using LLM-assisted construction. It is distributed via PhysioNet and is explicitly framed as an LLM-enhanced resource for chest X-ray visual question answering, helping models learn to answer clinically relevant questions grounded in radiographic evidence. Compared with smaller, manually authored radiology VQA sets, Medical-CXR-VQA is designed to increase both the volume and variety of QA pairs for chest radiographs, making it useful for training and benchmarking modern multimodal models that require large datasets. Because it is centered on chest imaging, it is particularly suitable for studying disease presence (or location) questions on chest X-rays and for evaluating how well a model's textual responses align with radiographic findings under a high-data regime.

MIMIC-CXR-VQA: MIMIC-CXR-VQA (Aas-Alas et al. , 2025) is a large-scale VQA dataset built on top of the MIMIC-CXR ecosystem, designed to be more complex and diverse than template-light chest X-ray QA benchmarks. The dataset creators highlight that questions can involve set and logical operations, and that the dataset uses a broad collection of templates to diversify question forms and reasoning types. It is reported to be derived mainly from MIMIC-CXR-JPG and Chest ImaGenome, both available through PhysioNet, which helps tie questions to structured radiographic concepts and annotations.

5.2. Discussion on datasets

The eleven benchmark datasets reviewed in this survey encompass a wide spectrum of imaging modalities, dataset scales, and clinical applications. A detailed overview is provided in Tables 4 and 5. Dataset sizes range from the relatively modest ImageCLEF VQA-MED-2018 collection, which comprises 2,866 radiology images and 6,413 QA pairs, to the extensive MIMIC-CXR-VQA dataset, which includes 254,934 chest X-ray images paired with 3,247,512 QAs. Radiology, especially chest and head imaging via X-ray, CT, and MRI, dominates the landscape. Notable exceptions include PathVQA, which focuses on pathology (5,000 microscopic images and over 32,800 QA pairs), Kvasir-VQA in gastrointestinal endoscopy (6,500 annotated images), DermaVQA in Dermatology (3,434 images and 1,488 QA pairs), and OVQA targeting orthopedic domains using X-rays and CT images (2,001 images and 19,020 QA pairs). However, imaging domains such as ophthalmology, and pediatrics remain under-represented. Several datasets incorporate multimodal imaging. For example, VQA-MED-2019/2020/2021 and P-VQA span X-ray, CT, MRI, and ultrasound, while datasets like SLAKE and VQA-RAD concentrate solely on radiological modalities. This diversity in modalities enhances opportunities for training and evaluating VQA models across varying clinical scenarios.

The provenance and construction of Med-VQA question-answer pairs vary markedly across datasets. Early benchmarks such as VQA-RAD and SLAKE primarily rely on clinician- or expert-authored questions that closely reflect natural clinical inquiry. By contrast, the ImageCLEF VQA-MED series (2018-2021) commonly adopts semi-automatic pipelines, generating questions from rule-based templates derived from image captions and then applying expert validation to ensure correctness. RadVisDial follows a hybrid design: its silver-standard split is produced via automated question generation from radiology reports, whereas its gold-standard split consists of manually curated, expert-driven dialogues. Patient-centric resources such as P-VQA and OVQA similarly construct QA pairs from hospital records in a semi-automatic manner, using physician-curated templates to preserve clinical plausibility. More recently, large-scale datasets (e.g., OmniMedVQA, Medical-CXR-VQA, and MIMIC-CXR-VQA) increasingly leverage large language models to generate candidate QA pairs, followed by clinician review or expert annotation for verification. Collectively, these choices reflect a persistent trade-off between scalability and clinical validity: automated or

大约一半的数据集。封闭式的是/否问题经过平衡，以防止偏见，遵循与美国病理学委员会认证考试标准一致的指南。

SLAKE：SLAKE（Liu 等, 2021）将语义注释与结构化医疗知识库结合，使用来自公开可用数据集的图像（Kavur 等, 2021；Simpson 等, 2019；Wang 等, 2017）。专家临床医生对这些图像进行了分割掩码和边界框的注释。此外，SLAKE 包含一个知识图谱，包含约 2,600 个中英文三元组，使得可以构建需要外部知识的问题。问题类别经过系统平衡，以尽量减少数据集偏差。

Kvasir-VQA：Kvasir-VQA（Gautam 等，2024）通过在整个胃肠道中包含全面的问答注释，扩展了内镜影像数据集 HyperKvasir 和 Kvasir-Instrument，涵盖正常和病理状况（如息肉、食管炎、溃疡性结肠炎）以及医疗器械。该数据集包含 6,500 张带有多个问答对的注释图像，涵盖六类问题：是/否、单选、多选、颜色、位置和数字计数。专家协作注释确保了临床相关性和准确性，该数据集的一部分被用于 ImageCLEF 2023 挑战赛（Ionescu 等，2023）。

P-VQA：以患者为中心的数据集 P-VQA（Huang 等，2023）提供了涵盖整个患者护理过程中医疗影像场景的全面内容——从会诊和诊断影像到治疗建议。该数据集涵盖 20 种常见疾病，包括 2,169 张图像和 24,800 个问答对，这些问答以通俗语言构建，以回应患者的实际关切，如症状、程序和检测结果。此外，由 419 个实体组成的医学知识图谱支持面向患者的推理任务，使其区别于以临床医生为中心的数据集。

OVQA：骨科视觉问答（OVQA）数据集（Huang 等人，2022）针对骨科特定影像，包括 2,001 张图像（X 光片和 CT 切片）和 19,020 对问答对。该数据集使用经过医生批准的模板半自动生成，问题反映了临床常见的咨询，随后由有经验的骨科医生进行验证。因此，OVQA 提供了一个高度准确、临床相关的数据集，专门用于骨科诊断任务。

OmniMedVQA：OmniMedVQA（Hu 等人，2024b）是一个大型基准，旨在评估医学视觉语言模型（VLM）在各种真实临床影像场景中的表现。它汇集了来自 73 个现有医学数据集的数据，涵盖 12 种影像模态和 20 个解剖区域，旨在测试模型的 VQA 能力是否能够超越单一器官、疾病组或模态。该数据集被定位为一个综合评估资源，而不是一个狭义的任务基准，创作者报告的实验结果表明，许多现有大型 VLM 在该基准上进行稳健医学视觉推理仍存在困难。在实践中，OmniMedVQA 最有用的方面是用于探测覆盖范围和知识迁移（例如，性能在不同模态/身体区域之间的变化），以及突出在仅限单域医学 VQA 数据集评估时可能隐藏的失败模式。

DermaVQA：DermaVQA（Yim 等，2024）针对皮肤科方向的视觉问答，其方法是将用户生成的皮肤科图像与临床风格的文本查询及相应答案配对，灵感来源于真实世界的远程皮肤科/异步消息工作流程。与许多以放射学为主的医学问答基准不同，DermaVQA 专注于皮肤病状况，并支持多语言环境（已报告语言包括中文、英文和西班牙文），以反映患者实际交流症状和背景的方式。该数据集旨在支持皮肤科问答和响应生成的多模态模型评估，基准实验强调在细微视觉模式中给出答案的难度，同时处理噪声的用户撰写描述和语言差异。这使得 DermaVQA 尤其适合研究对不受控成像条件的鲁棒性，以及评估

研究多模态助手是否能够在实际面对患者的皮肤科场景中提供连贯、具备上下文意识的回应。

Medical-CXR-VQA：Medical-CXR-VQA（Hu 等，2025）是一个大规模的胸部 X 光视觉问答（VQA）数据集，旨在支持对胸部 X 光图像的问答，特别强调使用大语言模型（LLM）辅助构建来扩展监督。该数据集通过 PhysioNet 发布，并明确定位为一个 LLM 增强的胸部 X 光视觉问答资源，帮助模型学习基于影像证据回答临床相关问题。与较小、手工编写的放射学 VQA 数据集相比，Medical-CXR-VQA 旨在增加胸片问答对的数量和多样性，使其适用于培训和评估需要大规模数据集的现代多模态模型。由于它以胸部影像为中心，因此特别适合研究胸片上的疾病存在（或位置）问题，并评估模型在高数据环境下文本回答与影像发现的一致性。

MIMIC-CXR-VQA：MIMIC-CXR-VQA（Aas-Alas 等，2025）是一个大型的视觉问答（VQA）数据集，建立在 MIMIC-CXR 生态系统之上，设计上比轻模板胸片问答基准更加复杂和多样化。数据集的创建者强调，问题可能涉及集合运算和逻辑运算，并且该数据集使用了广泛的模板集合，以丰富问题形式和推理类型。据报道，该数据集主要源自 MIMIC-CXR-JPG 和 Chest ImaGenome，这些数据都可以通过 PhysioNet 获得，这有助于将问题与结构化的放射学概念和注释关联起来。

5.2. 数据集讨论

本次综述中评估的十一个基准数据集涵盖了广泛的成像模式、数据集规模和临床应用。表 4 和表 5 提供了详细概述。数据集规模从相对较小的 ImageCLEF VQA-MED-2018 集合（包含 2,866 张放射学图像和 6,413 个问答对）到庞大的 MIMIC-CXR-VQA 数据集（包括 254,934 张胸部 X 光图像配对 3,247,512 个问答对）不等。放射学，尤其是通过 X 光、CT 和 MRI 进行的胸部和头部成像，占据了主导地位。值得注意的例外包括 PathVQA，专注于病理学（5,000 张显微镜图像和超过 32,800 个问答对）、用于胃肠道内镜的 Kvasir-VQA（6,500 张带注释图像）、用于皮肤科的 DermaVQA（3,434 张图像和 1,488 个问答对）、以及针对骨科领域利用 X 光和 CT 图像的 OVQA（2,001 张图像和 19,020 个问答对）。然而，影像学领域如眼科和儿科仍然代表性不足。若干数据

Med-VQA 问答对的来源和构建在各个数据集中差异显著。早期的基准数据集如 VQA-RAD 和 SLAKE 主要依赖临床医生或专家撰写的问题，这些问题紧密反映了自然的临床询问。相比之下，ImageCLEF VQA-MED 系列（2018-2021）通常采用半自动化流程，从图像说明中基于规则的模板生成问题，然后应用专家验证以确保正确性。Rad-VisDial 采用混合设计：其银标准部分是通过从放射学报告自动生成问题得到的，而其金标准部分则由手工策划、专家驱动的对话组成。以患者为中心的资源如 P-VQA 和 OVQA 也以半自动化方式从医院记录构建问答对，使用医生策划的模板以保持临床合理性。更近一些，大规模数据集（例如 OmniMedVQA、Medical-CXR-VQA 和 MIMIC-CXR-VQA）正在日益

Table 4
The statistic of different Med-VQA datasets.

Dataset	Images (Train/Val/Test)	QA pairs (Train/Val/Test)	Modality	Location
VQA-MED-2018	2866(2278/324/264)	6413(5413/500/500)	XR/CT/MRI/US/MA/GI/AG/PT	Brain/Chest/Abdomen
VQA-MED-2019	4200(3200/500/500)	15292(12792/2000/500)	XR/CT/MRI/US/MA/GI/AG/PT	Brain/Chest/Abdomen
VQA-MED-2020	5000(4000/500/500)	5000(4000/500/500)	XR/CT/MRI/US/MA/GI/AG/PT	Brain/Chest/Abdomen
VQA-MED-2021	5000(4000/500/500)	5000(4000/500/500)	XR/CT/MRI/US/MA/GI/AG/PT	Brain/Chest/Abdomen
VQA-RAD	315(315/315/315)	3515(3062/0/451)	XR/CT/MRI	Brain/Chest/Abdomen
RadVisDial	133348(123287/2064/8000)	1261480(1232870/20640/8000)	XR	Chest
PathVQA	4998(2499/1499/1000)	32799(17325/9462/6012)	Pathology	–
SLAKE	642(450/96/96)	14028(11907/1060/1061)	XR/CT/MRI	Brain/Chest/Abdomen
Kvasir-VQA	6500(-)	84500(-)	GI	GI
P-VQA	2169(1526/318/435)	24800(17336/2575/4889)	XR/CT/MRI/US	Brain/Chest/Abdomen
OVQA	2001(2000/1235/1234)	19020(15216/1902/1902)	XR/CT	Head/Chest/Hand/Leg
OmniMedVQA	118010(-)	127995(-)	XR/CT.....(12 modalities)	Head/Eye.....(20 locations)
DermaVQA	3434(-)	1488(-)	Dermatology	Skin
Medical-CXR-VQA	215547(172651/21534/21363)	780014(623662/78228/78124)	XR	Chest
MIMIC-CXR-VQA	254934(249006/2098/3830)	3247512(3157795/26627/63090)	XR	Chest

Abbreviation: X-ray (XR), Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Ultrasound (US), Mammography (MA), Gastrointestinal (GI), Angiography (AG), Positron Emission Tomography (PT), Knowledge Graph (KG), Validation(Val).

Table 5
The attributes of different Med-VQA datasets.

Dataset	Question Type	Annotation	Availability	Additional Annotation
VQA-MED-2018	Loc/Finding/Yes-No/Other	Rule-based(Manually Checked)	Open-source on Train and Val	–
VQA-MED-2019	Modal/Plane/Organ Sys/Abnl	Rule-based(Manually Checked)	Open-source	–
VQA-MED-2020	Modal/Plane/Organ Sys/Abnl	Rule-based(Manually Checked)	Open-source on Train and Val	–
VQA-MED-2021	Modal/Plane/Organ Sys/Abnl	Rule-based(Manually Checked)	Open-source	–
VQA-RAD	Modal/Plane/Organ Sys/Abnl/Loc/Find-ing/Color/Size/Num/Other	Manually(Rule-based Paraphrase)	Open-source	–
RadVisDial	Dialog	Manually(Rule-based Fliter)	Open-source	–
PathVQA	Find-ing/Loc/Num/When/Whose/How	Model-based(Manually Checked)	Open-source	–
SLAKE	Modal/Plane/Organ Sys/Abnl/Loc/Find-ing/Color/Size/Shape/Qual-ity/KG	Rule-based(Manually Checked)	Open-source	KG/Segmentation
Kvasir-VQA	Abnl/Loc/Find-ing/Color/Size/Num/Other	Manually	Open-source	Segmentation
P-VQA	Diagnosis/Treatment/Advice	Rule-based	Open-source	KG
OVQA	Modal/Plane/Organ Sys/Abnl/Finding/Other	Model-based(Manually Checked)	Open-source	–
OmniMedVQA	Abnl/Loc/Find-ing/Color/Size/Num/Other	Manually	Open-source	Segmentation
DermaVQA	Modal/Abnl/Finding/Diagno-sis/Other	Model-based(Manually Checked, Partially)	Open-source	–
Medical-CXR-VQA	Modal/Plane/Abnl/Loc/Finding	Model-based(Manually Checked, Partially)	Open-source	–
MIMIC-CXR-VQA	Abnl/Finding/Yes-No	Model-based(Manually Checked, Partially)	Provide Generation Code	–

Abbreviation: Validation(Val), Location(Loc), Modal(Modality), Abnormality(Abnl), Number(Num), Knowledge Graph (KG).

LLM-assisted generation enables rapid expansion and broader coverage, while clinician-authored questions offer higher domain relevance and linguistic naturalness. In this context, semi-automated pipelines coupled with rigorous expert validation appear to provide a practical and increasingly popular compromise.

In addition to QA pairs, several datasets provide rich auxiliary annotations. SLAKE is notable for its comprehensive metadata, including segmentation masks, bounding boxes, and a structured bilingual knowledge graph containing over 2,600 triples. PathVQA's questions are grouped for pathology reasoning, though it does not provide pixel labels. P-VQA also incorporates a disease-focused knowledge graph for reasoning tasks. Kvasir-VQA offers annotations for disease subtype classification, anatomical landmarks, and instrument detection. These enhancements enable more complex VQA tasks such as spatial reasoning, organ identification, and cross-modal retrieval. SLAKE's bilingual support and segmentation annotations are especially valuable for addressing linguistic diversity and fine-grained localization.

Given the nascent state of Med-VQA research, existing datasets primarily address radiology and pathology. However, there is significant potential in other medical fields such as ophthalmology and dermatology, where established databases could support future VQA tasks. Under-represented areas call for new datasets: for instance, dermatology, ophthalmology, or pediatric imaging remain largely uncovered. The use of semi-automatic generation and language models (e.g. LLaMA or GPT-4) has begun to scale annotation, but balancing quality and quantity is crucial. Tools like MVQAS (Bai et al., 2021) and LabelBox have been developed to facilitate data collection and semi-automated annotation, offering valuable resources for expanding this research area. Also, the combination of crowdsourcing platforms and clinician review can produce high-quality QA data. Moreover, the success of knowledge-enhanced datasets (SLAKE, P-VQA) points to integrating medical ontologies into VQA. Future work should aim for more diverse modalities and domains, multi-language support, and richer annotations (e.g. temporal data or cross-modality EHR links) to fully capture clinical complexity. In

表 4 不同医学视觉问答数据集的统计。

D数据集	Im年龄（训练/验证/测试）	QA对（训练/验证/测试）	M模式	Location
VQA-MED-2018	2866(2278/324/264)	6413(5413/500/500)	X光/CT/MRI/超声/胆靶/胃肠/血管造影/正电子发射断层	Br胸/腹
VQA-MED-2019	4200(3200/500/500)	15292(12792/2000/500)	XR/CT/MRI/US/MA/GI/AG/PT	Br头/胸/腹
VQA-MED-2020	5000(4000/500/500)	5000(4000/500/500)	X光/CT/MRI/超声/胆靶/胃肠/血管造影/正电子发射断层	Br头/胸/腹
VQA-MED-2021	5000(4000/500/500)	5000(4000/500/500)	X光/CT/MRI/超声/胆靶/胃肠/血管造影/正电子发射断层	Br头痛/胸部/腹部
VQA-RAD	315(315/315/315)	3515(3062/0/451)	X射线/CT/MRI	Br疼痛/胸部/腹部
放射学视觉对话	133348(123287/2064/8000)	1261480(1232870/20640/8000)	XR	Chest
PathVQA	4998(2499/1499/1000)	32799(17325/9462/6012)	Pathology	–
缓解	642(450/96/96)	14028(11907/1060/1061)	X射线/CT/MRI	Br疼痛/胸部/腹部
克瓦西尔VQA	6500(-)	84500(-)	GI	GI
P-VQA	2169(1526/318/435)	24800(17336/2575/4889)	X光/CT/MRI/超声	Br疼痛/胸部/腹部
OVQA	2001(2000/1235/1234)	19020(15216/1902/1902)	XR/CT	头部/胸部/Hand/Leg
OmniMedVQA	118010(-)	127995(-)	XR/CT.....(12 种模态)	头/眼睛.....(20 个地点)
DermaVQA	3434(-)	1488(-)	Derm学论	Skin
M医学胸片-VQA	215547(172651/21534/21363)	780014(623662/78228/78124)	XR	Chest
MIMIC-CXR-VQA	254934(249006/2098/3830)	3247512(3157795/26627/63090)	XR	Chest

缩写：X射线（XR）、C 计算机断层扫描（CT）、磁共振成像（MRI）、超声（US）、乳腺摄影（MA）、胃肠(胃肠道), 血管造影, 静脉注射 正电子发射断层扫描（PET）、知识图谱（KG）、验证（Val）。

表 5 不同 Med-VQA 数据集的属性。

D数据集	任务 io n T ype	Annotation	可用性 ty	A附加公告 otation
VQA-MED-2018	Loc/F寻找/是-否/其他	Rule-基于（人工检查）	开源 o n 训练和验证 l	–
VQA-MED-2019	Modal/Plane/Organ Sys/Abnl	Rule-基于（人工检查）	Open-源	–
VQA-MED-2020	M模式/平面/器官系统/ Abnl	Rule-基于（人工检查）	Open-源 o n 训练和验证 l	–
VQA-MED-2021	M卵巢/平面/器官系统/ Abnl	Rule-基于（人工检查）	Open-源	–
VQA-RAD	M调式/平面/器官系统/异常/位置/查找-种类/颜色/尺寸/数量/其他	手动（基于规则的改写）	开源	–
RadVisDial	Dialog	M手动（基于规则的过滤器）	开源	–
PathVQA	查找-ing/地点/数字/什么时候/谁的/如何	M基于模型（人工检查）	开源	–
缓解	调式/平面/器官系统/异常/位置/查找-种类/颜色/尺寸/形状/质量重量/公斤	Rule-基于（人工检查）	开放式源 rce	K G/分割
克瓦西尔VQA	Abnl/位置/查找-种类/颜色/尺寸/数量/其他	M每年	开放式 rce	分段 ion
P-VQA	诊断/治疗/建议	Rule-基于	开放式源 rce	KG
OVQA	调式/平面/器官系统/异常/发现/其他	模型-b ased（人工检查）	开源	–
OmniM edVQA	异常/位置/查找-种类/颜色/尺寸/数量/其他	Manually	开放式 rce	分段 ion
DermaVQA	方式/异常/发现/诊断-姐妹/其他	M基于模型(Manuall y C hecked, Partially)	开放式源 rce	–
医学胸片视觉问答 A	Modal/Plane/Abnl/位置/查找	M odel-b 基础（人工检查，部分）	开放式源 rce	–
MIMIC-CXR-VQA	异常/发现 g/是-否	M odel-b 基础（人工检查，部分）	提供生成代码	–

Abbr缩写：验证(Val)、位置(Loc)、模式(Modality)、异常(Abnl)、数量(Num)、知识图谱(KG) .

大型语言模型辅助生成能够实现快速扩展和更广的覆盖范围，而临床医生撰写的问题则具有更高的领域相关性和语言自然性。在这种情况下，半自动化流程结合严格的专家验证似乎提供了一种实用且日益流行的折衷方案。

除了问答对之外，一些数据集还提供丰富的辅助标注。SLAKE 以其全面的元数据而著称，包括分割掩码、边界框以及包含超过 2,600 个三元组的结构化双语知识图谱。PathVQA 的问题按病理推理进行分组，但它并不提供像素标签。P-VQA 也结合了以疾病为中心的知识图谱用于推理任务。Kvasir-VQA 提供了疾病亚型分类、解剖标志和器械检测的标注。这些增强功能使得更复杂的 VQA 任务成为可能，如空间推理、器官识别和跨模态检索。SLAKE 的双语支持和分割标注对于应对语言多样性和精细定位尤其有价值。

鉴于医学视觉问答（Med-VQA）研究仍处于初期阶段，现有的数据集主要涉及放射学和病理学。然而，在其他医学领域如眼科学和皮肤科学仍存在巨大潜力，这些领域的成熟数据库可以支持未来的VQA任务。代表性不足的领域需要新的数据集：例如，皮肤学、眼科或儿科影像仍然在很大程度上未被覆盖。半自动生成和语言模型（如LLaMA或GPT-4）的使用已经开始规模化标注，但平衡质量与数量至关重要。像MVQAS（Bai等, 2021）和LabelBox这样的工具已被开发用于数据收集和半自动标注，为扩展这一研究领域提供了宝贵资源。此外，众包平台与临床医生审查相结合也可以产生高质量的问答数据。此外，知识增强型数据集（SLAKE，P-VQA）的成功表明了将医学本体整合到VQA中的潜力。未来工作

sum, current datasets provide a varied foundation, but expanding underrepresented domains and leveraging advanced annotation pipelines will be key to advancing Med-VQA.

5.3. Evaluation metric

Evaluation in Med-VQA spans across multiple paradigms depending on whether the task is framed as classification, text-generation, or open-ended answering evaluated via vision-language models. Below section describes each group, the commonly used metrics, their mathematical definitions, and their practical interpretations.

5.3.1. Classification-based metrics

These metrics are primarily used when the answers come from a predefined set of categories (e.g., yes/no, multiple choice).

- **Accuracy (Acc)**:

  $$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

  Accuracy represents the overall correctness of predictions by computing the ratio of correctly classified instances (both positive and negative) to the total number of instances.
- **Recall**:

  $$\text{Recall} = \frac{TP}{TP + FN}$$

  Recall (also known as sensitivity) measures the capability of the model to identify all relevant (positive) instances. High recall indicates few false negatives.
- **Specificity**:

  $$\text{Specificity} = \frac{TN}{TN + FP}$$

  Specificity quantifies how well the model identifies negative instances. It is especially important in domains where false positives are costly.
- **Precision**:

  $$\text{Precision} = \frac{TP}{TP + FP}$$

  Precision assesses the accuracy of the positive predictions. A high precision score implies that most predicted positives are actually correct.
- **F1 Score**:

  $$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

  The F1 score is the harmonic mean of precision and recall, offering a balance between the two when they are in tension. It is particularly useful for imbalanced datasets.

5.3.2. Generation-based metrics

These metrics are used when the model is allowed to generate free-form natural language answers. They evaluate the linguistic or semantic similarity between the predicted and reference answers.

- **BLEU (Bilingual Evaluation Understudy)**:

  $$\text{BLEU-N} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

  BLEU (Papineni et al., 2002) calculates the precision of n-gram matches between the generated and reference text, with a brevity penalty (BP) to discourage overly short outputs. Commonly used in machine translation, it is limited in capturing semantic equivalence.
- **WUPS (Wu-Palmer Similarity)**:

  $$\text{WUPS} = \frac{1}{N}\sum_{i=1}^{N} \max_j \text{WuP}(a_i, g_j)$$

  WUPS measures semantic similarity between predicted and ground-truth answer words using WordNet hierarchies. It accounts for partial correctness by considering the taxonomic distance between concepts.
- **WBSS (Word-based Semantic Similarity)**:

  WBSS (Malinowski & Fritz, 2014) computes cosine similarity between word embeddings of the predicted and ground-truth answers. It evaluates semantic closeness even when the lexical forms differ.
- **CBSS (Character-based Semantic Similarity)**:

  Similar to WBSS but operates at the character level, CBSS is useful for capturing fine-grained similarities in domain-specific terminology, particularly in radiological or clinical contexts.
- **MPT (Medical Phrase Token Matching)**:

  MPT is a domain-specific metric that performs token-level matching using curated medical phrase sets. It is designed to better reflect correctness in clinical semantics, recognizing that multiple surface forms may convey the same diagnosis.
- **METEOR (Metric for Evaluation of Translation with Explicit ORdering)** (Banerjee & Lavie, 2005):

  $$\text{METEOR} = F_{mean} \cdot (1 - Penalty)$$

  METEOR aligns words using stemming, synonymy, and paraphrase matching, and considers both precision and recall. It often correlates better with human judgment than BLEU in open-ended medical tasks.

5.3.3. Metrics for VLM-based methods

With the emergence of powerful vision-language models (VLMs), new evaluation metrics have been developed that directly use the token-level outputs produced by these models.

- **Token Accuracy**:

  $$\text{Token Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\{t_i^{pred} = t_i^{gt}\}$$

  Token accuracy measures the fraction of predicted tokens that exactly match the corresponding ground-truth tokens. It is useful when VQA is formulated as a token-by-token classification problem.
- **Token F1 Score**:

  $$\text{Token F1} = \frac{2 \cdot \text{Token Precision} \cdot \text{Token Recall}}{\text{Token Precision} + \text{Token Recall}}$$

  This metric accounts for partial matches at the token level and is particularly relevant for answers that involve multi-token medical terms or phrases.

5.3.4. Discussion on evaluation metric

Evaluation metrics in Med-VQA are essential for assessing model performance across a variety of task formats, including classification, free-form generation, and multimodal reasoning. Each metric provides distinct strengths and limitations depending on the nature of the question-answering task. Classification metrics such as Accuracy, Precision, Recall, and F1-score offer clear, interpretable evaluations for closed-ended questions (e.g., yes/no, multiple-choice). These metrics are particularly effective in scenarios where answers belong to a predefined discrete set, making them suitable for categorically constrained tasks. Their numeric outputs provide straightforward comparisons among models and serve as baseline indicators of task success. However, these metrics fall short when applied to free-form or semantically rich answers, as they cannot accommodate lexical variations or partial correctness. To address these shortcomings, generation-based metrics have been adopted, particularly for open-ended tasks. Metrics such as BLEU and METEOR evaluate textual similarity by comparing n-gram overlap and alignments between generated and reference answers. While they provide a more flexible evaluation mechanism, they tend to penalize valid paraphrases and may not accurately reflect semantic correctness, especially in the specialized medical domain. Furthermore, these metrics are sensitive to

总的来说，目前的数据集提供了多样化的基础，但扩展欠代表的领域并利用先进的标注流程将是推动医疗视觉问答发展的关键。

5.3. 评估指标

在医疗视觉问答（Med-VQA）中，评估涵盖了多种范式，这取决于任务是被界定为分类、文本生成，还是通过视觉语言模型评估的开放式回答。以下部分描述了每一类群体、常用指标、它们的数学定义及其实际意义。

5.3.1. 基于分类的度量

这些指标主要在答案来自预定义类别时使用（例如，是/否，多项选择）。

- **准确率 (Acc)：**

  $$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

  准确性通过计算正确分类的实例（包括正例和负例）占总实例数的比率，来表示预测的整体正确性。
- **回忆：**

  $$\text{Recall} = \frac{TP}{TP + FN}$$

  召回率（也称为敏感性）衡量模型识别所有相关（正）实例的能力。高召回率表明假阴性很少。
- **特异性：**

  $$\text{Specificity} = \frac{TN}{TN + FP}$$

  特异性量化模型识别负实例的能力。在假阳性代价高的领域，这一点尤其重要。
- **精度：**

  $$\text{精度} = \frac{TP}{TP + FP}$$

  精确率评估正向预测的准确性。高精确率得分意味着大多数预测为正的结果实际上是正确的。
- **F1 分数：**

  $$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

  F1 分数是精确率和召回率的调和平均数，在两者存在矛盾时提供平衡。它对于不平衡的数据集特别有用。

5.3.2. 基于生成的指标

当模型被允许生成自由形式的自然语言回答时，会使用这些指标。它们评估预测答案与参考答案之间的语言或语义相似性。

- **BLEU（双语评估替代法）：**

  $$\text{BLEU-N} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

  BLEU（Papineni 等，2002）计算生成文本与参考文本之间 n-gram 匹配的精确度，并使用简短惩罚（BP）以防止输出过短。它常用于机器翻译，但在捕捉语义等价性方面有限。
- **WUPS（Wu-Palmer 相似度）：**

  $$\text{WUPS} = \frac{1}{N}\sum_{i=1}^{N} \max_j \text{WuP}(a_i, g_j)$$

  WUPS 使用 WordNet 层级结构来衡量预测答案词与真实答案词之间的语义相似性。它通过考虑概念之间的分类距离来考虑部分正确性。
- **WBSS（基于词汇的语义相似度）：**

  WBSS（Malinowski & Fritz, 2014）计算预测答案和真实答案的词嵌入之间的余弦相似度。即使词汇形式不同，它也能评估语义上的接近程度。
- **CBSS（基于字符的语义相似度）：**

  与 WBSS 类似，但在字符层面上运行，CBSS 对于捕捉特定领域术语的精细相似性非常有用，特别是在放射学或临床环境中。
- **MPT（医疗短语匹配）：**

  MPT 是一个特定领域的指标，使用精心整理的医学短语集进行词元级匹配。它旨在更好地反映临床语义的正确性，认识到多种表面形式可能传达相同的诊断信息。
- METEOR（带显式排序的翻译评估度量）（Banerjee & Lavie, 2005）：

  $$\text{METEOR} = F_{mean} \cdot (1 - Penalty)$$

  METEOR 使用词干、同义词和意译匹配来对齐单词，并同时考虑精确率和召回率。在开放式医疗任务中，它通常比 BLEU 与人工判断的相关性更高。

5.3.3. 基于*VLM*的方法的指标

随着强大的视觉-语言模型（VLMs）的出现，已经开发出新的评估指标，直接使用这些模型生成的令牌级输出。

- **令牌准确率：**

  $$\text{Token Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\{t_i^{pred} = t_i^{gt}\}$$

  令牌准确率衡量预测的令牌与对应的真实令牌完全匹配的比例。当VQA被构建为逐令牌分类问题时，它是有用的。
- **词元 F1 分数：**

  $$\text{Token F1} = \frac{2 \cdot \text{Token Precision} \cdot \text{Token Recall}}{\text{Token Precision} + \text{Token Recall}}$$

  该指标考虑了在词元层面的部分匹配，对于涉及多词元医学术语或短语的答案特别相关。

5.3.4. 评估指标讨论

Med-VQA 中的评估指标对于评估模型在各种任务格式下的性能至关重要，包括分类、自由生成和多模态推理。每种指标根据问答任务的性质提供不同的优点和局限性。分类指标如准确率（Accuracy）、精确率（Precision）、召回率（Recall）和 F1 分数为封闭式问题（例如是/否、多项选择题）提供了清晰、可解释的评估。这些指标在答案属于预定义离散集合的场景中特别有效，使其适用于分类受限的任务。它们的数值输出能够直接比较不同模型，并作为任务成功的基准指标。然而，当应用于自由形式或语义丰富的答案时，这些指标就显得不足，因为它们无法容纳词汇变化或部分正确性。为了应对这些缺点，已经采用了基于生成的指标，p

syntactic variations and do not explicitly account for domain-specific language nuances. Domain-aware metrics like WUPS, WBSS, CBSS, and MPT attempt to incorporate semantic knowledge into evaluation. These approaches often rely on external ontologies or medical knowledge bases to gauge conceptual similarity between predicted and reference answers. For instance, WUPS uses WordNet hierarchies, while CBSS and WBSS leverage domain-specific knowledge graphs. Although these methods offer improved alignment with expert judgment, they are constrained by the scope and accuracy of the external resources employed, which are not always tailored to the medical domain. Recently, VLM-based metrics have emerged as a promising alternative. These leverage pretrained large-scale models to evaluate answers based on learned semantic embeddings or token-level alignment. Such approaches often demonstrate improved agreement with human evaluation by capturing deeper contextual and semantic relationships. However, their opacity and potential biases–particularly when the same VLM is used for both generation and evaluation–pose concerns for interpretability and fairness.

Cross-applying metrics across task types offers additional insights but must be approached with caution. For instance, applying classification metrics to generated responses can yield easily interpretable scores but may ignore legitimate semantic variants, unfairly penalizing the model. Conversely, using generation-based or VLM-based metrics in classification settings allows recognition of semantically correct variations but introduces ambiguity and reduces the clarity of evaluation. In summary, no single metric suffices for evaluating all facets of Med-VQA tasks. Instead, employing a combination of metrics tailored to specific task formats is recommended. Classification metrics provide clarity for closed-ended tasks, generation metrics capture linguistic variation for open-ended responses, and domain-aware or VLM-based methods add semantic depth. Together, these evaluation strategies offer a more comprehensive and reliable assessment framework for advancing Med-VQA research.

6. Comparative analysis: Task-specific models vs. vision-language models

6.1. Quantitative evaluation of task-specific models

Based on Table 6, across the Fusion Algorithm category, recent transformer-based fusion methods substantially outperform earlier CNN-LSTM models. For example, Q2ATransformer (Swin-Transformer visual backbone, Bert language backbone, and transformer fusion) achieves state-of-the-art accuracy on the VQA-RAD and PathVQA benchmarks. Specifically, Q2ATransformer reaches 80.5% on VQA-RAD (overall accuracy) and 74.6% on PathVQA (overall accuracy), well above older models that relied on simple CNN features and BAN or CMSA fusion (which typically score below 75%). Vision-language transformers like VLMVQA (ViT backbone) also score highly (75.4% on VQA-RAD overall), indicating that modern architectures yield the best performance in this group. In contrast, earlier BAN or CMSA methods (e.g. Caption-Aware VQA, CMSA models) achieve around 75% on VQA-RAD. In the External Knowledge group, models that incorporate medical domain knowledge or retrieval show strong gains on clinical datasets. The retrieval-augmented model RAMM (ViT visual backbone with retrieval fusion) reports state-of-the-art performance on Med-VQA 2019 (82.1% accuracy) and also leads SLAKE (86.1%). By contrast, Caption-Aware VQA (ResNet152 + Bert with BAN fusion) and TCC-MEM (ResNet50 + LSTM) achieve lower results (75%/82% on VQA-RAD/SLAKE). K-PathVQA (ResNet50 + BioELMO with cross-attention fusion) excels on PathVQA close-ended questions (94.6% closed) but only 42.1% on open questions, highlighting how specialized knowledge can drastically boost closed-form accuracy. Overall, external-knowledge models top this category on medical datasets, outperforming fusion-only models when domain context is needed. The Augmentation methods show mixed results. M-Mixup (Pooling Tokens visual features with MLP fusion) achieves 82.9% on PathVQA overall - the highest across all groups - suggesting its visual augmentation helps pathology questions. ACMA-MAM (ResNet34 + LSTM with complex attention) scores 76.1% on VQA-RAD and 83.1% on SLAKE, competitive with the top fusion-based models. In contrast, VQA-Mix (MEVF backbone) performs poorly on open-ended PathVQA (only 13.4%), indicating some augmentation schemes may hurt open-answer tasks. Finally, the Multi-Question Learning models (Rad-ReStruct and MMQL) are only evaluated on VQA-RAD and SLAKE; MMQL (Swin Backbone) achieves 77.1% on VQA-RAD overall and 86.7% on SLAKE, but achieve state-of-the art performances on the SLAKE dataset. In summary, the table shows that transformer-based vision backbones and sophisticated fusion (especially cross-attention or transformer fusion) consistently yield the highest accuracies across datasets, while simpler CNN/Bert setups lag behind.

Vision Backbone: The data and statistics from Fig. 4 and Table 6 reveal clear patterns relating architecture choices to performance and popularity trends. Transformer-based encoders (Swin-Transformer, ViT) are increasingly favored, reflecting their superior capacity to capture global image context. In particular, ViT accounts for the second-largest share (35%) among backbones and saw a noticeable rise in adoption from 2023 to 2025. Swin-Transformer, although less prevalent overall (5%), was introduced in 2023 and sustained into 2024. Both architectures outperform traditional CNNs on benchmarks like VQA-RAD (e.g., 81%-85% on closed accuracy), supporting the observation that transformers offer enhanced visual-semantic reasoning in Med-VQA. Despite this trend, ResNet remains the most dominant backbone by usage (38%), with consistent adoption from 2020 to 2023. This suggests its robustness and compatibility with advanced multimodal fusion methods. Notably, ResNet152 (e.g., in FGCVQA) and ResNet34 (e.g., in ACMA-MAM) achieve competitive accuracy when integrated with sophisticated reasoning modules, highlighting that CNNs still hold value under the right configurations. Meanwhile, newer CNN variants such as EfficientNet and MEVF make up smaller proportions (8% and 5%, respectively), with scattered temporal usage. Overall, the trend clearly favours transformer-based visual backbones, but high-performing CNNs remain viable alternatives when paired with strong architectural enhancements.

Language Backbone: Base on the data and statistics from Fig. 5 and Table 6, the evolution of language encoders in Med-VQA reveals a clear transition from traditional recurrent architectures to transformer-based models. Bert dominates both in frequency (66%) and in temporal adoption, peaking sharply in 2023 and maintaining a leading role across all datasets, VQA-RAD, PathVQA, and SLAKE, where it consistently supports top-tier performance (e.g., 81.2%-86.9% overall accuracy on SLAKE). Its contextualized embeddings and bidirectional encoding seem particularly beneficial for understanding nuanced clinical questions. RNN-based encoders (including LSTM and GRU) hold a significant early presence (31%) and were widely used between 2020 and 2022. However, their prevalence has declined over time, reflecting their limitations in capturing long-range dependencies compared to transformers. BioELMO, though tailored to biomedical domains, accounts for only 3% of use and appears in just a few 2022-2023 models with moderate performance. Linear encoders are similarly rare (6%) and exhibit no clear trend in adoption or performance advantage.

Fusion Method: Recent advances in Med-VQA strongly favour fusion strategies that facilitate fine-grained cross-modal interaction. Base on the data and statistics from Fig. 6 and Table 6, Transformer-based fusion methods dominate (27%), with a steep rise in usage from 2022 to 2023, and consistently appear in top-performing models across benchmarks. Models like Q2ATransformer and MCCAN, which leverage transformer decoders or attention-driven query learning, demonstrate superior performance, particularly on open-ended questions. Cross-attention mechanisms also play a central role, accounting for 32% of usage and showing sustained adoption from 2022 onward. These methods explicitly model interactions between image regions and question tokens, enabling enhanced semantic alignment. ACMA and MISS exemplify this trend, achieving state-of-the-art results on VQA-RAD and SLAKE us-

句法变体并未明确考虑特定领域语言的细微差别。像 WUPS、WBSS、CBSS 和 MPT 这样的领域感知指标试图将语义知识纳入评估。这些方法通常依赖外部本体或医学知识库来衡量预测答案与参考答案之间的概念相似度。例如，WUPS 使用 WordNet 层次结构，而 CBSS 和 WBSS 则利用特定领域的知识图谱。尽管这些方法在与专家判断的一致性方面有所提升，但它们受到所使用外部资源范围和准确性的限制，而这些资源并不总是针对医学领域定制的。最近，基于 VLM 的指标作为一种有前景的替代方案出现。这些方法利用预训练的大规模模型根据学习到的语义嵌入或词元级对齐来评估答案。这类方法通过捕捉更深层次的上下文和语义关系，通常表现出与人工评估更高的一致性。

在不同任务类型之间交叉应用指标可以提供额外的见解，但必须谨慎。例如，将分类指标应用于生成的回答可以得到易于解释的分数，但可能会忽略合法的语义变体，从而不公平地惩罚模型。相反，在分类设置中使用基于生成或基于VLM的指标可以识别语义上正确的变体，但会引入模糊性并降低评估的清晰度。总之，没有单一指标能够评估Med-VQA任务的所有方面。相反，建议使用针对特定任务形式定制的指标组合。分类指标为封闭式任务提供清晰度，生成指标捕捉开放式回答的语言变异，而领域感知或基于VLM的方法则增加语义深度。结合这些评估策略，可以为推动Med-VQA研究提供更全面、更可靠的评估框架。

MLP 融合在 PathVQA 上总体达到 82.9% —— 在所有组中最高 —— 表明其视觉增强有助于病理学问题。ACMA-MAM（带复杂注意力的 ResNet34 + LSTM）在 VQA-RAD 上得分 76.1%，在 SLAKE 上得分 83.1%，与顶级融合模型竞争。相比之下，VQA-Mix（MEVF 骨干）在开放式 PathVQA 上表现不佳（仅为 13.4%），说明某些增强方案可能会对开放式回答任务产生不利影响。最后，多问题学习模型（Rad-ReStruct 和 MMQL）仅在 VQA-RAD 和 SLAKE 上进行评估；MMQL（Swin 骨干）在 VQA-RAD 上总体达到 77.1%，在 SLAKE 上达到 86.7%，但在 SLAKE 数据集上实现了最新的最佳性能。总之，表格显示基于 Transformer 的视觉骨干和复杂融合（尤其是交叉注意力或 Transformer 融合）在各数据集上始终能产生最高的准确率，而更简单的 CNN/Bert 设置则落后。

视觉骨干：图4和表6的数据和统计显示了架构选择与性能及流行趋势之间的明显模式。基于Transformer的编码器（Swin-Transformer，ViT）越来越受到青睐，反映出它们在捕捉全局图像上下文方面的卓越能力。特别是，ViT在骨干网络中占第二大份额（35%），并且从2023年到2025年采用率明显上升。Swin-Transformer虽然总体上较少（5%），但在2023年引入并持续使用到2024年。这两种架构在基准测试（如VQA-RAD）中的表现都优于传统CNN（例如封闭准确率为81%-85%），支持了Transformer在Med-VQA中提供增强视觉语义推理能力的观点。尽管有此趋势，ResNet仍然是使用最广的骨干（38%），从2020年至2023年保持稳定采用。这表明其稳健性以及与先进多模态融合方法的兼容性。值得注意的是，ResNet152（例如在FGCVQ

6. 比较分析：任务特定模型与视觉-语言模型

6.1. 任务特定模型的量化评估

根据表6，在融合算法类别中，基于最近的变压器的融合方法在性能上显著优于早期的CNN-LSTM模型。例如，Q2ATransformer（Swin-Transformer视觉骨干、Bert语言骨干和变压器融合）在VQA-RAD和PathVQA基准上实现了最先进的准确率。具体来说，Q2ATransformer在VQA-RAD上的总体准确率达到80.5%，在PathVQA上的总体准确率达到74.6%，远高于依赖简单CNN特征和BAN或CMSA融合的旧模型（其得分通常低于75%）。类似VLMVQA（ViT骨干）的视觉-语言变压器也表现出高分（VQA-RAD总体准确率为75.4%），表明现代架构在这一组中提供了最佳性能。相比之下，早期的BAN或CMSA方法（例如Caption-Aware VQA、CMSA模型）在VQA-RAD上只能达到约75%。在外部知识组中，结合医学领域知识或检索的模型在临床数据集上表现出显著提升。

语言骨干：基于图5和表6的数据和统计，Med-VQA中语言编码器的发展显示出从传统循环架构向基于Transformer模型的明显转变。BERT在使用频率（66%）和时间采纳上占据主导地位，在2023年达到峰值，并在所有数据集（VQA-RAD、PathVQA和SLAKE）中保持领先地位，其在SLAKE上的整体准确率始终保持顶尖水平（例如81.2%-86.9%）。其上下文化的嵌入和双向编码对于理解细微的临床问题尤其有益。基于RNN的编码器（包括LSTM和GRU）在早期占有显著份额（31%），并在2020年至2022年间被广泛使用。然而，随着时间推移，它们的普遍性有所下降，反映出相比Transformer在捕捉长程依赖方面的局限性。BioELMO虽然针对生物医学领域，但使用比例仅为3%，且仅出现在少数案例中。

融合方法：最近在医学视觉问答（Med-VQA）方面的进展强烈倾向于促进细粒度跨模态交互的融合策略。根据图6和表6的数据和统计，基于Transformer的融合方法占主导地位（27%），从2022年到2023年的使用率急剧上升，并在各类基准测试中持续出现在表现最佳的模型中。像Q2ATransformer和MCCAN这样的模型，利用Transformer解码器或基于注意力的查询学习，展示了卓越的性能，尤其在开放式问题上表现突出。跨注意力机制也起着核心作用，占使用量的32%，并从2022年起持续被采用。这些方法明确建模图像区域与问题词之间的交互，实现了增强的语义对齐。ACMA和MISS就是这一趋势的典型代表，在VQA-RAD和SLAKE上取得了最先进的结果。

Table 6
The performance comparison for task-specific models on four common datasets.

Method	Name	Year	Visual Backbone	Language backbone	Fusion	VQA-Med-2019	VQA-RAD			SLAKE			PathVQA		
						Overall	Open	Close	Overall	Open	Close	Overall	Open	Close	Overall
Fusion Algorithm	MedFuseNet (Sharma et al., 2021)	2021	VGG16/SenseNet121/ResNet152	Bert/XLNet & LSTM	attention & MFB	–	–	–	–	–	–	–	–	63.6	–
	CMSA (Gong et al., 2021)	2021	ResNet34	LSTM	CMSA	–	61.5	80.9	73.2	73.8	82	77	–	–	–
	AMAM (Pan et al., 2022)	2022	MAML &CDAE	GRU	BAN	–	63.8	80.3	73.3	–	–	–	18.2	84.4	50.2
	QTR (Wang et al., 2022)	2022	ResNet34	LSTM	CMSA	–	60.9	82.7	74.1	–	–	–	–	–	–
	SIA (Chen et al., 2023)	2023	MMQ	LSTM	cross-attention	–	64.3	81	74.7	–	–	–	13.9	83.4	48.7
	FGCVQA (Wu et al., 2023)	2023	ResNet152	Bert	cross-attention	–	66.5	84.6	77.4	–	–	–	–	–	–
	MF2-MVQA (Song et al., 2023)	2023	EfficientNetv2	Bert	attention	69.7	63.7	80.1	73.6	–	–	–	–	–	–
	Q2ATransformer (Liu et al., 2023)	2023	Swin-Transformer	Bert	Transformer	–	79.2	81.2	80.5	–	–	–	54.9	88.9	74.6
	VLMVQA (Bazi et al., 2023)	2023	ViT	Bert	Transformer	–	71.5	82.5	75.4	–	–	–	58.3	84.6	67.1
	LGPFN (Du et al., 2025)	2025	ResNet-50	Bert	attention	–	78.2	81.6	80.4	81.1	88.7	84.1	53.4	91.9	75.7
	VG-CALF (Lameesa et al., 2025)	2025	ViT	Bert	cross-attention	–	67.0	85.5	76.1	81.4	83.8	83.3	–	–	–
External Knowledge	Caption Aware VQA (Cong et al., 2022)	2022	ResNet152	Bert	BAN	–	69.8	79.8	75.8	80.2	86.1	82.5	–	–	–
	TCC-MEM (Pellegrini et al., 2023)	2022	ResNet50	LSTM	BAN	–	–	–	–	79.2	86.8	82.8	–	–	–
	RAMM (Chen et al., 2024)	2023	ViT	Bert	cross-attention	82.1	63.7	79.8	73.4	82.5	91.6	96.1	–	–	–
	K-PathVQA (Naseem et al., 2023)	2023	ResNet50	BioELMO	cross-attention	–	–	–	–	–	–	–	42.1	94.6	69
	MISS (Chen et al., 2024)	2024	ViT	Bert	cross-attention	–	71.8	80.4	76.1	81.5	82.9	82	–	–	–
	FGRW (Liang et al., 2025)	2025	ViT	Bert	cross-attention	–	69.8	82.0	77.2	87.3	89.7	85.4	–	–	–
Per-training	MMBert (Khare et al., 2021)	2021	ResNet152	Bert	Transformer	67.2	63.1	77.9	72.0	–	–	–	–	–	–
	CPRD (Liu et al., 2021)	2021	MEVF	LSTM	BAN	–	61.1	80.4	72.7	81.2	83.4	82.1	–	–	–
	ARL (Chen et al. , 2022)	2022	ViT	Bert	cross-attention	80.3	67.6	86.8	79.2	81.9	91.4	85.6	–	–	–
	MUMC (Li et al., 2023)	2023	ViT	Bert	Transformer	–	71.5	84.2	79.2	–	–	84.9	39.0	90.4	65.1
	M2I2 (Li et al., 2023)	2023	ViT	Bert	Transformer	–	66.5	83.5	76.8	74.7	91.1	81.2	36.3	88.0	62.2
	PubMedCLIP (Eslami et al., 2023)	2023	ResNet50	RNN	BAN	–	60.1	80.0	71.8	77.7	81.3	79.1	–	–	–
	PMC-CLIP (Lin et al., 2023)	2023	ResNet50	Bert	Transformer	–	67.0	84,0	77.6	81.9	88.0	84.3	–	–	–
	MM-VQA (Silva et al., 2023)	2023	EfficientNetv2	Bert	RealFormer	62.8	–	–	–	–	–	–	–	–	–
	CMCD (Chen et al. , 2024)	2024	ViT	Bert	cross-attention	–	67.7	80.6	78.7	82.4	87.6	84.3	–	–	–
	MITER (Shu et al., 2024)	2024	ViT	Bert	cross-attention	–	59.4	80.5	72.1	79.2	84.4	81.2	–	–	–
	AMiF (Zou & Yin, 2025a)	2025	ViT	Bert	cross-attention	83.2	72.0	87.3	80.5	84.8	90.2	86.7	–	–	–
	MVCM (Zou & Yin, 2025b)	2025	ViT	Bert	cross-attention	–	68.5	90.8	80.9	83.7	91.8	86.9	39.8	90.5	65.3
Augmentation	M-Mixup (Liu et al., 2022)	2022	Pooling Tokens	Linear	MLPs	–	53.1	81.3	70.2	–	–	–	79.9	87.5	82.9
	VQA-Mix (Gong et al., 2022)	2022	MEVF	LSTM	BAN	–	56.6	79.6	70.4	–	–	–	13.4	83.5	46.6
	ACMA-MAM (Li et al., 2023b)	2023	ResNet34	LSTM	ACMA	–	63.6	84.4	76.1	80.8	86.7	83.1	22	85.5	53.8
	PLMVQA (Yu et al., 2023)	2023	ViT	Bert	Transformer	–	62.6	80.5	73.4	80	87.7	83	–	–	–
Multi-Question Learning	Rad-ReStruct (Pellegrini et al., 2023)	2023	EfficientNet-b5	RadBert	Transformer	–	–	–	76.3	–	–	–	–	–	–
	MMQL (Chen et al. , 2024)	2024	Swin-Transformer	Bert	Transformer	–	64.3	85.7	77.1	84.2	90.6	86.7	–	–	–
Interpretable	QC-MLB (Vu et al., 2020)	2020	ResNet152	Word2Vec & Linear	Attention & MLB	–	60.33	–	–	–	–	–	–	–	–
Interpretable	Cycle-VQA (Fan et al., 2025)	2025	ResNet50	Bert	BAN	–	–	–	73.1	–	–	82.0	–	–	–

Table 6
The performance comparison for task-specific models on four common datasets.

Method	Name	Year	Visual Backbone	Language backbone	Fusion	VQA-Med-2019	VQA-RAD			SLAKE			PathVQA		
						Overall	Open	Close	Overall	Open	Close	Overall	Open	Close	Overall
Fusion Algorithm	MedFuseNet (Sharma et al., 2021)	2021	VGG16/SenseNet121/ResNet152	Bert/XLNet & LSTM	attention & MFB	–	–	–	–	–	–	–	–	63.6	–
	CMSA (Gong et al., 2021)	2021	ResNet34	LSTM	CMSA	–	61.5	80.9	73.2	73.8	82	77	–	–	–
	AMAM (Pan et al., 2022)	2022	MAML &CDAE	GRU	BAN	–	63.8	80.3	73.3	–	–	–	18.2	84.4	50.2
	QTR (Wang et al., 2022)	2022	ResNet34	LSTM	CMSA	–	60.9	82.7	74.1	–	–	–	–	–	–
	SIA (Chen et al., 2023)	2023	MMQ	LSTM	cross-attention	–	64.3	81	74.7	–	–	–	13.9	83.4	48.7
	FGCVQA (Wu et al., 2023)	2023	ResNet152	Bert	cross-attention	–	66.5	84.6	77.4	–	–	–	–	–	–
	MF2-MVQA (Song et al., 2023)	2023	EfficientNetv2	Bert	attention	69.7	63.7	80.1	73.6	–	–	–	–	–	–
	Q2ATransformer (Liu et al., 2023)	2023	Swin-Transformer	Bert	Transformer	–	79.2	81.2	80.5	–	–	–	54.9	88.9	74.6
	VLMVQA (Bazi et al., 2023)	2023	ViT	Bert	Transformer	–	71.5	82.5	75.4	–	–	–	58.3	84.6	67.1
	LGPFN (Du et al., 2025)	2025	ResNet-50	Bert	attention	–	78.2	81.6	80.4	81.1	88.7	84.1	53.4	91.9	75.7
	VG-CALF (Lameesa et al., 2025)	2025	ViT	Bert	cross-attention	–	67.0	85.5	76.1	81.4	83.8	83.3	–	–	–
External Knowledge	Caption Aware VQA (Cong et al., 2022)	2022	ResNet152	Bert	BAN	–	69.8	79.8	75.8	80.2	86.1	82.5	–	–	–
	TCC-MEM (Pellegrini et al., 2023)	2022	ResNet50	LSTM	BAN	–	–	–	–	79.2	86.8	82.8	–	–	–
	RAMM (Chen et al., 2024)	2023	ViT	Bert	cross-attention	82.1	63.7	79.8	73.4	82.5	91.6	96.1	–	–	–
	K-PathVQA (Naseem et al., 2023)	2023	ResNet50	BioELMO	cross-attention	–	–	–	–	–	–	–	42.1	94.6	69
	MISS (Chen et al., 2024)	2024	ViT	Bert	cross-attention	–	71.8	80.4	76.1	81.5	82.9	82	–	–	–
	FGRW (Liang et al., 2025)	2025	ViT	Bert	cross-attention	–	69.8	82.0	77.2	87.3	89.7	85.4	–	–	–
Per-training	MMBert (Khare et al., 2021)	2021	ResNet152	Bert	Transformer	67.2	63.1	77.9	72.0	–	–	–	–	–	–
	CPRD (Liu et al., 2021)	2021	MEVF	LSTM	BAN	–	61.1	80.4	72.7	81.2	83.4	82.1	–	–	–
	ARL (Chen et al. , 2022)	2022	ViT	Bert	cross-attention	80.3	67.6	86.8	79.2	81.9	91.4	85.6	–	–	–
	MUMC (Li et al., 2023)	2023	ViT	Bert	Transformer	–	71.5	84.2	79.2	–	–	84.9	39.0	90.4	65.1
	M2I2 (Li et al., 2023)	2023	ViT	Bert	Transformer	–	66.5	83.5	76.8	74.7	91.1	81.2	36.3	88.0	62.2
	PubMedCLIP (Eslami et al., 2023)	2023	ResNet50	RNN	BAN	–	60.1	80.0	71.8	77.7	81.3	79.1	–	–	–
	PMC-CLIP (Lin et al., 2023)	2023	ResNet50	Bert	Transformer	–	67.0	84,0	77.6	81.9	88.0	84.3	–	–	–
	MM-VQA (Silva et al., 2023)	2023	EfficientNetv2	Bert	RealFormer	62.8	–	–	–	–	–	–	–	–	–
	CMCD (Chen et al. , 2024)	2024	ViT	Bert	cross-attention	–	67.7	80.6	78.7	82.4	87.6	84.3	–	–	–
	MITER (Shu et al., 2024)	2024	ViT	Bert	cross-attention	–	59.4	80.5	72.1	79.2	84.4	81.2	–	–	–
	AMiF (Zou & Yin, 2025a)	2025	ViT	Bert	cross-attention	83.2	72.0	87.3	80.5	84.8	90.2	86.7	–	–	–
	MVCM (Zou & Yin, 2025b)	2025	ViT	Bert	cross-attention	–	68.5	90.8	80.9	83.7	91.8	86.9	39.8	90.5	65.3
Augmentation	M-Mixup (Liu et al., 2022)	2022	Pooling Tokens	Linear	MLPs	–	53.1	81.3	70.2	–	–	–	79.9	87.5	82.9
	VQA-Mix (Gong et al., 2022)	2022	MEVF	LSTM	BAN	–	56.6	79.6	70.4	–	–	–	13.4	83.5	46.6
	ACMA-MAM (Li et al., 2023b)	2023	ResNet34	LSTM	ACMA	–	63.6	84.4	76.1	80.8	86.7	83.1	22	85.5	53.8
	PLMVQA (Yu et al., 2023)	2023	ViT	Bert	Transformer	–	62.6	80.5	73.4	80	87.7	83	–	–	–
Multi-Question Learning	Rad-ReStruct (Pellegrini et al., 2023)	2023	EfficientNet-b5	RadBert	Transformer	–	–	–	76.3	–	–	–	–	–	–
	MMQL (Chen et al. , 2024)	2024	Swin-Transformer	Bert	Transformer	–	64.3	85.7	77.1	84.2	90.6	86.7	–	–	–
Interpretable	QC-MLB (Vu et al., 2020)	2020	ResNet152	Word2Vec & Linear	Attention & MLB	–	60.33	–	–	–	–	–	–	–	–
Interpretable	Cycle-VQA (Fan et al., 2025)	2025	ResNet50	Bert	BAN	–	–	–	73.1	–	–	82.0	–	–	–

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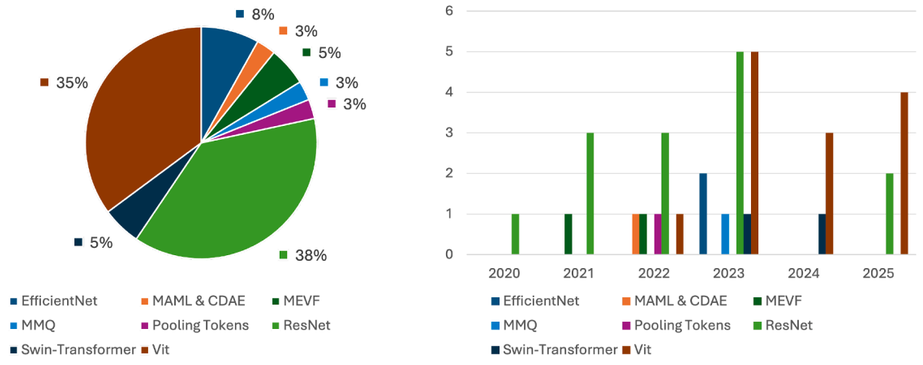


Fig. 4. Statistics of adopted visual backbones.

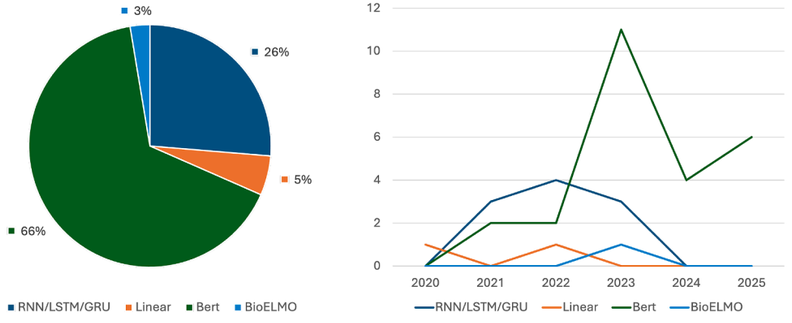


Fig. 5. Statistics of adopted language backbones.

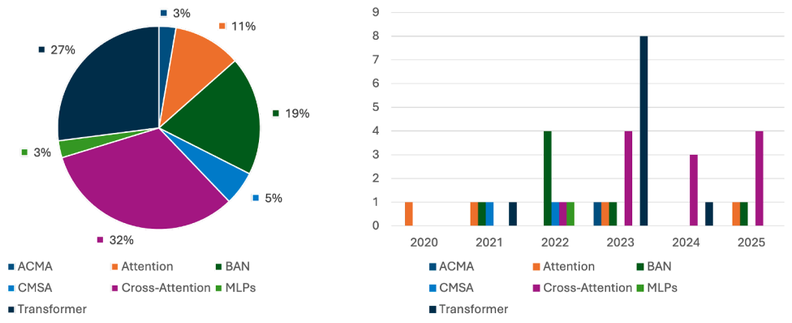


Fig. 6. Statistics of adopted fusion methods.

ing structured cross-modal fusion. Earlier methods like BAN (19%) and CMSA (5%) were more prevalent prior to 2022 but are increasingly supplanted by transformer and cross-attention approaches. Attention-based MLPs and simple attention fusion have limited adoption (≤ 11%) and exhibit no notable gains in recent works. Overall, the data confirm a clear shift: high-performing models consistently adopt architectures that enable deep, structured multimodal reasoning, with transformer-based and cross-attention-based fusion emerging as the most effective paradigms.

Question Type: Nearly every model shows much higher accuracy on closed-ended (Yes/No or short-answer) questions than on open-ended questions. Noticeably, PathVQA closed accuracies often exceed 85%, whereas open-end accuracy remains below 60% for most models. This gap reflects the greater difficulty of free-form answers. Models that incorporate more knowledge or candidate-answer reasoning (like Q2ATransformer) partially close this gap on open questions.

Anomalies include methods tailored to specific datasets: e.g., K-PathVQA's high closed accuracy on pathology images (94.6%) comes at the cost of very low open accuracy (42.1%), illustrating a trade-off. Likewise, VQA-Mix shows an unusually low 13.4% open accuracy on PathVQA, suggesting its augmentation strategy is not well suited to pathology questions. In general, performance on the newer SLAKE dataset is improved by models with domain adaptation (e.g. RAMM, ACMA) indicating that medical-specialized embeddings help gastrointestinal imaging.

Insufficiency: Several promising interpretable or knowledge-driven models are absent from the above comparison because they report results on non-standard or private datasets. For example, most of the

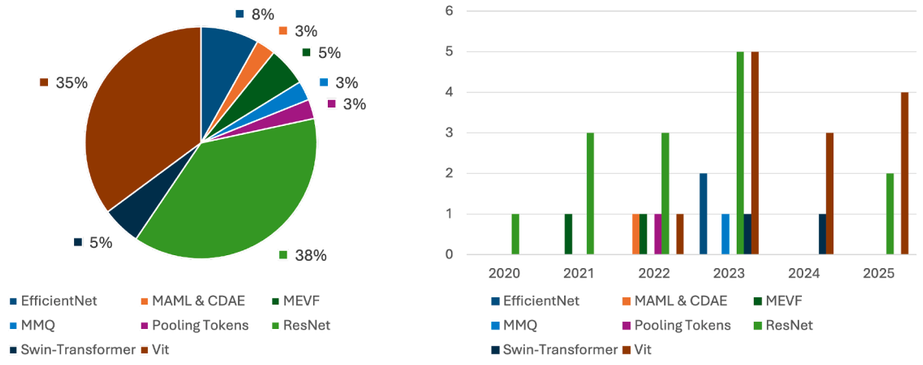


图 4. 所采用的视觉主干网络统计。

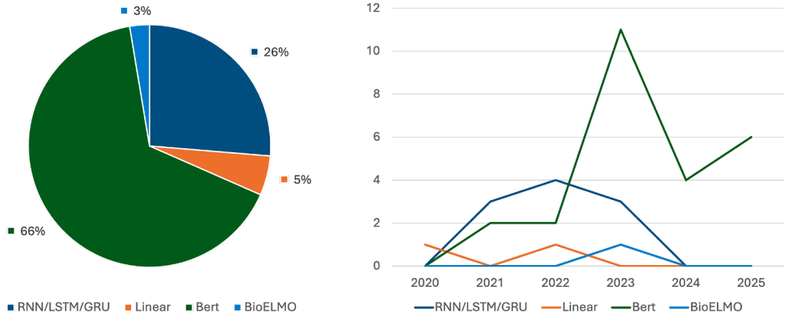


图5. 所采用语言骨干的统计。

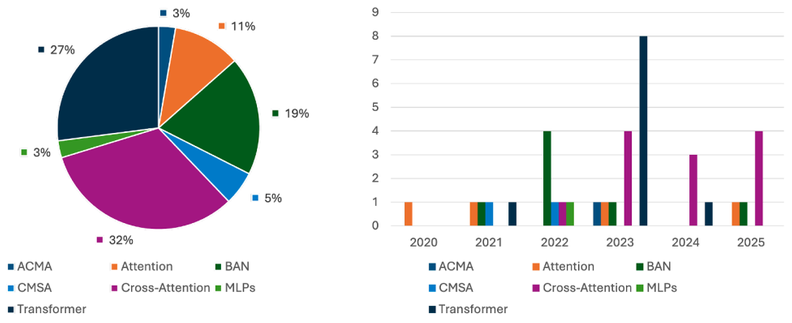


图6. 所采用的融合方法统计。

进行结构化的跨模态融合。早期的方法如 BAN（19%）和 CMSA（5%）在 2022 年之前更为普遍，但正逐渐被 Transformer 和跨注意力方法取代。基于注意力的 MLP 和简单注意力融合的应用有限（≤ 11%），在近期的研究中没有显著的提升。总体而言，数据表明了一个明确的趋势：高性能模型始终采用能够实现深度、结构化多模态推理的架构，而基于 Transformer 和跨注意力的融合成为最有效的范式。

问题类型：几乎每个模型在封闭式（是/否或简答）问题上的准确率都明显高于开放式问题。值得注意的是，PathVQA 的封闭式准确率通常超过 85%，而大多数模型的开放式准确率仍低于 60%。这一差距反映了自由回答的更高难度。模型

那些整合了更多知识或候选答案推理（如 Q2ATransformer）的模型，在开放性问题上部分地缩小了这一差距。

异常情况包括针对特定数据集量身定制的方法：例如，K-PathVQA在病理图像上的高封闭准确率（94.6%）是以开放准确率非常低（42.1%）为代价的，这说明了权衡。同样，VQA-Mix在PathVQA上的开放准确率异常低，仅为13.4%，表明其增强策略不太适合病理问题。总体而言，对较新的SLAKE数据集的性能，通过具有领域适应能力的模型（如RAMM、ACMA）得到了提升，这表明医学专用的嵌入有助于胃肠道影像分析。

不足：由于在非标准或私人数据集上报告结果，上述比较中缺少若干有前途的可解释或知识驱动模型。例如，大多数的

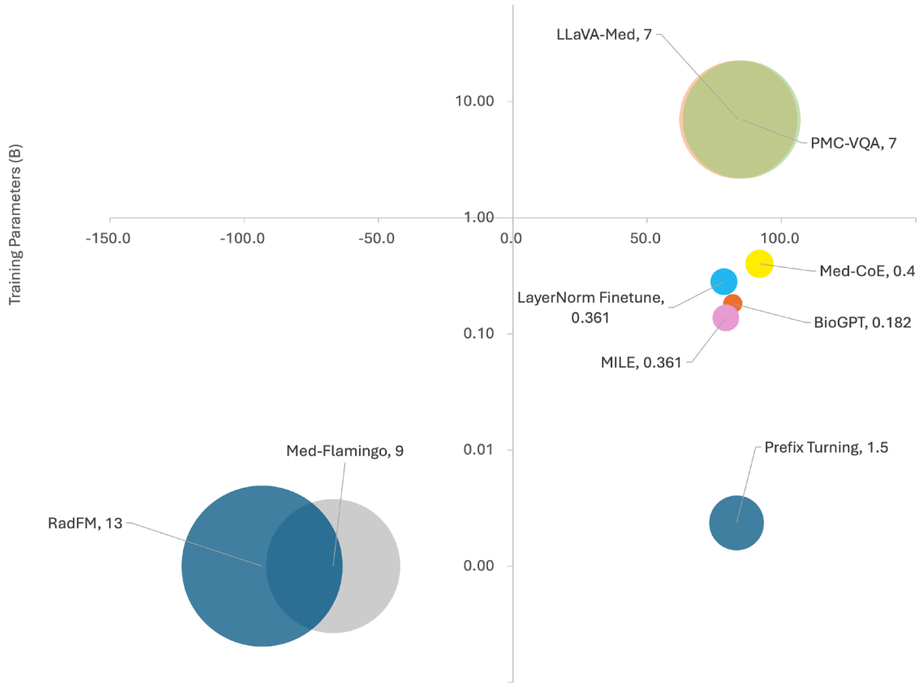


Fig. 7. The parameter size, trainable parameters and the SLAKE performances of various medical vision-language models. The X-axis denotes the performances on SLAKE, positive values represents the accuracy on SLAKE, while negative values represents Bert Similarity, as RadFM and Med-Flamingo do not reports accuracy on SLAKE. Y-axis represents the trainable parameters. Noticeably, RadFM and Med-Flamingo is zero-shot methods with zero trainable parameters. The size of the bubble represents the total parameter size of corresponding models.

interpretable methods. These gaps imply that many explainable or pretrained-VQA approaches lack unified evaluation. The absence of such methods suggests two future needs: first, standardizing benchmarks so that novel interpretable models (knowledge graphs, ROI-driven models) can be fairly compared. This could involve releasing public versions of custom datasets or including new question types in established benchmarks. Second, bridging methodologies by adapting interpretable techniques to existing datasets. For instance, applying graph-based reasoning or external knowledge modules to VQA-RAD/SLAKE would illuminate their real benefit. Addressing these gaps will enable a more comprehensive evaluation of explainability and generalization in Med-VQA, moving beyond mere accuracy on narrow tests.

6.2. Quantitative evaluation of vision-language models in Med-VQA task

Table 7 illustrates how these factors influence model performance across benchmarks such as SLAKE, PathVQA, and VQA-RAD.

Model Size: As shown in Fig. 7, the examined methods span a broad range of parameter scales, from compact models such as BioMedGPT (182M) and instruction-tuned variants like MILE and LayerNorm Fine-tuning (361M), to large-scale architectures including Med-Flamingo (9B), LLaVA-Med (7B), and RadFM (13B). However there is no strong evidence that large models perform better than Task-Specific models. For example, smaller models such as BioMedGPT remain competitive when sufficiently fine-tuned (71.6% on VQA-RAD outperforms 7B PMC-VQA model), witch offering efficient alternatives suitable for deployment in constrained environments.

Learning Paradigm, Zero-Shot vs. Fine-Tuning: In previous subsection, the VLM model are broadly categorized into zero-shot/few-shot and fine-tuning paradigms. Zero-shot methods, such as GPT-4V, Med-Flamingo, and MGKF, do not update model parameters during downstream deployment. These approaches rely on in-context learning or retrieval-based reasoning. Despite the advantage of avoiding task-specific training, they generally underperform on structured VQA tasks. For example, Med-Flamingo yields only 20% accuracy on VQA-RAD, and GPT-4V achieves limited token recall (41) and modest semantic similarity (Bert-Sim 66.9) on SLAKE. Conversely, fine-tuned models demonstrate substantial gains in both accuracy and text-generation quality. Methods such as PTUnifier, BiomedCLIP, and M^3AE consistently outperform zero-shot counterparts. In particular, BiomedCLIP and PTUnifier achieve accuracy scores of 85.4 and 82.5, respectively, on SLAKE, highlighting the effectiveness of task-specific adaptation.

Tuning Strategy, Full vs. PEFT: In addition to the learning paradigm, the granularity of parameter tuning plays a critical role in performance and efficiency. Full fine-tuning methods, such as PTUnifier, M^3AE, and MMCAP, update all model parameters, resulting in higher computational demands but also leading to superior task adaptation. BiomedCLIP, for example, achieves 85.4 accuracy on SLAKE, and MMCAP reaches 84.8, placing them among the top-performing systems. Alternatively, PEFT methods update only a small fraction of model weights. VQA-Adapter modifies just 2.38% of parameters using adapter modules and yet achieves a VQA-RAD accuracy of 74.8%. LayerNorm Fine-tuning updates normalization layers exclusively and maintains a close margin to full fine-tuning (78.6% vs. 82.0% accuracy on SLAKE). Similarly, MILE leverages low-rank adaptation (LoRA) and prefix-tuning modules to achieve 79.3% SLAKE accuracy with minimal overhead.

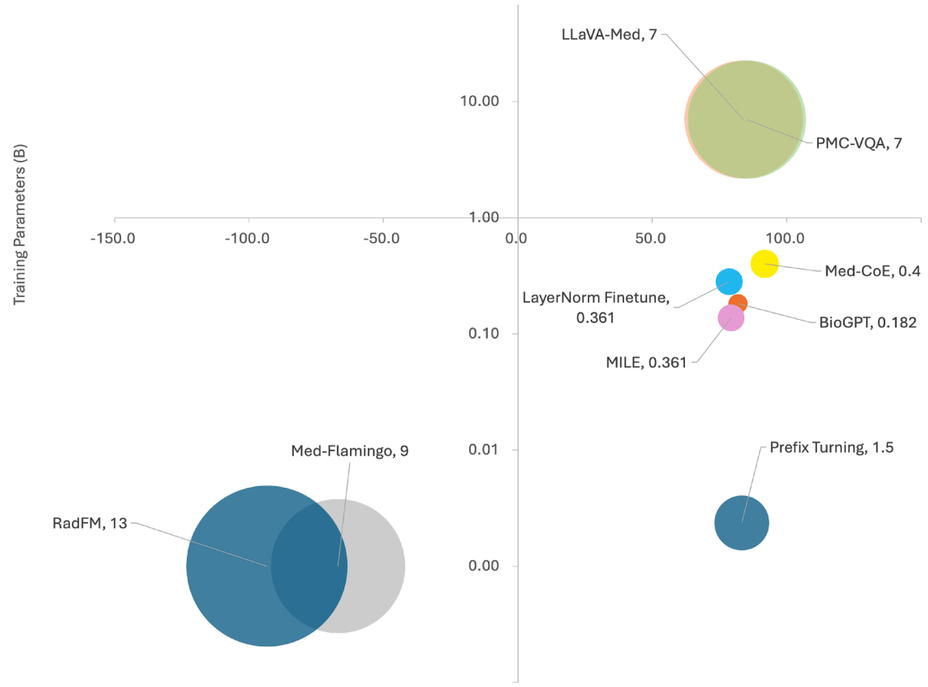


图7. 各种医疗视觉-语言模型的参数规模、可训练参数和SLAKE性能。X轴表示在SLAKE上的性能，正值表示在SLAKE上的准确率，而负值表示Bert相似度，因为RadFM和Med-Flamingo没有报告在SLAKE上的准确率。Y轴表示可训练参数数量。值得注意的是，RadFM和Med-Flamingo是零样本方法，具有零可训练参数。气泡的大小表示相应模型的总参数规模。

可解释的方法。这些差距表明，许多可解释或预训练的VQA方法缺乏统一的评估。缺乏此类方法暗示了未来有两个需求：首先，标准化基准，以便可以公平地比较新型可解释模型（知识图、基于ROI的模型）。这可能涉及发布自定义数据集的公开版本或在现有基准中加入新问题类型。其次，通过将可解释技术应用于现有数据集来弥合方法学差距。例如，将基于图的推理或外部知识模块应用于VQA-RAD/SLAKE，将揭示其实际效益。解决这些差距将有助于更全面地评估Med-VQA中的可解释性和泛化能力，而不仅仅局限于狭窄测试中的准确率。

6.2. Med-VQA任务中视觉-语言模型的定量评估

表 7 展示了这些因素如何影响模型在 SLAKE、PathVQA 和 VQA-RAD 等基准上的性能。

模型规模：如图7所示，所研究的方法涵盖了广泛的参数规模，从紧凑型模型如BioMedGPT（182M）和经过指令微调的变体如MILE和LayerNorm微调（361M），到大规模架构包括Med-Flamingo（9B）、LLaVA-Med（7B）和RadFM（13B）。然而，没有强有力的证据表明大型模型的表现优于特定任务模型。例如，经过充分微调的小型模型如BioMedGPT仍然具有竞争力（在VQA-RAD上达到71.6%，超过7B PMC-VQA模型），同时提供了适合在受限环境下部署的高效替代方案。

学习范式：零样本与微调：在前一个小节中，VLM 模型大致分为零样本/少样本和微调范式。零样本方法，如 GPT-4V、Med-Flamingo 和 MGKF，在下游部署期间不更新模型参数。这些方法依赖上下文学习或基于检索的推理。尽管避免了特定任务训练的优势，但它们在结构化 VQA 任务上的表现通常不如人意。例如，Med-Flamingo 在 VQA-RAD 上的准确率仅为 20%，GPT-4V 在 SLAKE 上的标记回忆率有限（41），语义相似度适中（Bert-Sim 66.9）。相反，微调模型在准确率和文本生成质量上都显示出显著提升。PTUnifier、BiomedCLIP 和 M^3AE 等方法始终优于零样本对应模型。特别是，BiomedCLIP 和 PTUnifier 在 SLAKE 上分别取得了 85.4 和 82.5 的准确率，凸显了任务特定微调的有效性。

调优策略，全量与PEFT：除了学习范式外，参数调优的粒度在性能和效率方面也起着关键作用。全量微调方法，如PTU-nifier、M^3AE和MMCAP，会更新所有模型参数，导致计算需求较高，但也带来更优的任务适应性。例如，BiomedCLIP在SLAKE上达到85.4的准确率，而MMCAP达到84.8，使它们位列性能最佳的系统之一。另一方面，PEFT方法仅更新小部分模型权重。VQA-Adapter使用适配器模块仅修改2.38%的参数，却在VQA-RAD上取得了74.8%的准确率。LayerNorm微调专门更新归一化层，并与全量微调保持接近的差距（SLAKE上的准确率为78.6% vs. 82.0%）。类似地，MILE利用低秩适配（LoRA）和前缀调优模块，在几乎不增加负担的情况下实现了79.3%的SLAKE准确率。

Table 7
The performance comparison for vision-language models on three common datasets.

Method	Name	Year	Parameters	Tuned Parameters	SLAKE					VQA-RAD				
					Token Recall	Token-F1	Accuracy	Bert-Sim	BLEU	Accuracy	Token Recall	Bert-Sim	BLEU	ROUGE
Zero Shot/Few Shot	GPT4-V	2023	–	None	–	41	–	–	–	–	–	–	–	–
	Med-Flamingo (Moor et al., 2023)	2023	9B	None	–	–	–	–	–	20	–	73.4	36	38.6
	RadFM (Wu et al., 2025a)	2023	13B	None	–	–	–	67.8	–	–	–	81.5	52.2	52.7
	MGKF (Wu et al., 2025b)	2025	7B	None	74.3	–	–	–	–	–	71.7	–	–	–
Fine-tune	MedVILL (Moon et al., 2022)	2022	–	Full	–	–	–	–	–	70.7	–	–	–	–
	PTUnifier (Chen et al., 2023)	2023	–	Full	–	–	–	–	–	78.3	–	–	–	–
	BioMedCLIP (Zhang et al., 2024b)	2023	–	Full	–	–	–	–	–	75.2	–	–	–	–
	BioGPT (Zhang et al., 2024a)	2023	182M	Full	–	–	57.9	–	–	71.6	–	–	–	–
	LLaVA-Med (Li et al., 2023a)	2023	7B	Full	–	–	–	–	–	–	–	–	–	–
	Prefi Tuning (Van Sonsbeek et al., 2023)	2023	1.5B	Part	–	–	63.6	78.5	70.3	–	–	–	–	–
	VQA-Adapter (Liu et al., 2023)	2023	–	Part	–	–	–	–	–	75.8	–	–	–	–
	PMC-VQA (Zhang et al., 2023)	2023	7B	Full	–	–	–	–	–	51.6	–	–	–	–
	M3AE (Eslami et al., 2023)	2024	–	Full	–	–	–	–	–	77	–	–	–	–
	MMCAP (Yan et al., 2024)	2024	–	Full	–	–	–	–	–	77.8	–	–	–	–
	LayerNorm Finetune (Chen et al., 2024)	2024	361M	Part	–	–	–	–	–	–	–	–	–	–
	MILE (Chen et al., 2024)	2024	361M	Part	–	–	–	–	–	–	–	–	–	–
	Med-Gemini (Saab et al., 2024)	2024	–	–	–	87.5	–	–	–	–	–	–	–	–
	Med-CoE (Qiu et al., 2025)	2025	400M	Full	–	–	91.8	–	–	–	–	–	–	–

These results suggest that carefully designed PEFT strategies can provide high performance with significantly reduced training costs, making them appealing for real-world deployment.

Evaluation Datasets and Metrics: Evaluation of VLM-based Med-VQA methods is conducted across multiple benchmark datasets using a diverse set of metrics. This heterogeneity in both datasets and evaluation protocols presents significant challenges in drawing direct performance comparisons between methods. The most commonly used datasets include SLAKE, PathVQA, and VQA-RAD, and several types of metrics are used to assess performance on Med-VQA, including Accuracy, Token-Level Recall and Token-F1, BertScore and BLEU, ROUGE, etc. In the compiled results (Table 7), not all methods are evaluated on all datasets or with all metrics. For example, Med-Gemini reports high token-level F1 on SLAKE (87.5) and PathVQA (64.7), but does not report VQA-RAD accuracy. Conversely, MGKF is evaluated on VQA-RAD using token recall (71.7) rather than accuracy. Similarly, some models like RadFM and GPT-4V are evaluated using BLEU and Bert-Sim, emphasizing semantic fidelity over strict answer matching. The absence of uniform evaluation protocols across models and datasets complicates direct performance comparison. Some methods are assessed only on a subset of benchmarks, while others use distinct versions of the same datasets or different scoring strategies. For example, SLAKE accuracy is reported for MedViLL, PTUnifier, and M^{3}AE, but token-F1 is only available for Med-Gemini. VQA-RAD performance is presented using accuracy for some models and BLEU for others, making it difficult to draw consistent conclusions. Additionally, some models prioritize different output types (e.g., retrieval vs. generation), which affects the suitability of certain metrics. For retrieval-based models like MGKF, accuracy is computed based on retrieved factual answers, whereas generative models such as Med-Gemini or MLeVLM are judged on free-text coherence and correctness.

Overall, the analysis underscores that fine-tuned VLMs, particularly those leveraging full parameter updates or well-designed PEFT strategies, achieve markedly better performance than zero-shot approaches. While large models such as Med-Gemini lead in absolute metrics, smaller models such as BioMedGPT and PEFT-based variants like MILE demonstrate competitive effectiveness under resource constraints. The trade-off between generalization (via zero-shot capabilities) and precision (via fine-tuning) remains a critical consideration in system design. As Med-VQA applications move toward real-world clinical deployment, the choice of tuning granularity and model scale will continue to shape the feasibility and reliability of such systems. Further, to ensure fair and comprehensive assessment of Med-VQA systems, future work should promote standardized evaluation settings that include unified benchmark subsets and a consistent set of metrics for both closed and open-ended VQA. Incorporating both surface-level (e.g., accuracy, BLEU) and semantic (e.g., BertScore, token-F1) metrics would provide a more holistic view of model behavior. Furthermore, establishing a shared leaderboard with consistent prompts, formats, and data partitions could significantly enhance the reliability of reported comparisons.

6.3. Evaluating the trade-offs: Task-specific models vs. large vision-language models

Modern transformer-fusion architectures consistently top the task-specific models' leader boards. For instance, a recent Swin + Bert + Transformer model (Q2ATransformer) achieves ≈80.5% accuracy on VQA-RAD, far above earlier CNN + BAN/CMSA systems (70-75%). On other datasets, specialized models likewise lead: e.g. transformer-based MMQL reached 86.7% on SLAKE (Table 6) out performs all the VLM methods in Table 7. TSMs with external knowledge (retrieval or knowledge-augmented methods) further boost accuracy on clinical data. In one case, a RAG model (RAMM) exceeded 91.6% on SLAKE. Augmentation-based models have shown mixed results: some (like M-Mixup) yield very high pathology-question performance (≈82.9% on PathVQA), while others can hurt open-ended accuracy

Table 7
The performance comparison for vision-language models on three common datasets.

Method	Name	Year	Parameters	Tuned Parameters	SLAKE					VQA-RAD				
					Token Recall	Token-F1	Accuracy	Bert-Sim	BLEU	Accuracy	Token Recall	Bert-Sim	BLEU	ROUGE
Zero Shot/Few Shot	GPT4-V	2023	–	None	–	41	–	–	–	–	–	–	–	–
	Med-Flamingo (Moor et al., 2023)	2023	9B	None	–	–	–	–	–	20	–	73.4	36	38.6
	RadFM (Wu et al., 2025a)	2023	13B	None	–	–	–	67.8	–	–	–	81.5	52.2	52.7
	MGKF (Wu et al., 2025b)	2025	7B	None	74.3	–	–	–	–	–	71.7	–	–	–
Fine-tune	MedVILL (Moon et al., 2022)	2022	–	Full	–	–	–	–	–	70.7	–	–	–	–
	PTUnifier (Chen et al., 2023)	2023	–	Full	–	–	–	–	–	78.3	–	–	–	–
	BioMedCLIP (Zhang et al., 2024b)	2023	–	Full	–	–	–	–	–	75.2	–	–	–	–
	BioGPT (Zhang et al., 2024a)	2023	182M	Full	–	–	57.9	–	–	71.6	–	–	–	–
	LLaVA-Med (Li et al., 2023a)	2023	7B	Full	–	–	–	–	–	–	–	–	–	–
	Prefi Tuning (Van Sonsbeek et al., 2023)	2023	1.5B	Part	–	–	63.6	78.5	70.3	–	–	–	–	–
	VQA-Adapter (Liu et al., 2023)	2023	–	Part	–	–	–	–	–	75.8	–	–	–	–
	PMC-VQA (Zhang et al., 2023)	2023	7B	Full	–	–	–	–	–	51.6	–	–	–	–
	M3AE (Eslami et al., 2023)	2024	–	Full	–	–	–	–	–	77	–	–	–	–
	MMCAP (Yan et al., 2024)	2024	–	Full	–	–	–	–	–	77.8	–	–	–	–
	LayerNorm Finetune (Chen et al., 2024)	2024	361M	Part	–	–	–	–	–	–	–	–	–	–
	MILE (Chen et al., 2024)	2024	361M	Part	–	–	–	–	–	–	–	–	–	–
	Med-Gemini (Saab et al., 2024)	2024	–	–	–	87.5	–	–	–	–	–	–	–	–
	Med-CoE (Qiu et al., 2025)	2025	400M	Full	–	–	91.8	–	–	–	–	–	–	–

这些结果表明，经过精心设计的 PEFT 策略可以在显著降低训练成本的同时提供高性能，使其在实际部署中具有吸引力。

评估数据集和指标：基于VLM的Med-VQA方法的评估是在多个基准数据集上使用多种指标进行的。数据集和评估协议的多样性为直接比较各方法的性能带来了重大挑战。最常用的数据集包括SLAKE、PathVQA和VQA-RAD，并使用多种类型的指标来评估Med-VQA的性能，包括准确率、词元级召回率和词元F1、BertScore和BLEU、ROUGE等。在汇总结果（表7）中，并非所有方法都在所有数据集或使用所有指标上进行评估。例如，Med-Gemini在SLAKE（87.5）和PathVQA（64.7）上报告了较高的词元级F1，但没有报告VQA-RAD的准确率。相反，MGKF在VQA-RAD上使用词元召回（71.7）进行评估，而非准确率。同样，一些模型如RadFM和GPT-4V使用BLEU和Bert-Sim进行评估，更强调语义一致性而非严格的答案匹配。

总体而言，分析强调了经过微调的视觉-语言模型（VLMs），尤其是那些利用完整参数更新或精心设计的PEFT策略的模型，性能明显优于零样本方法。虽然诸如Med-Gemini等大型模型在绝对指标上领先，但较小的模型如BioMedGPT以及基于PEFT的变体如MILE在资源受限的情况下也显示出竞争力。通过零样本能力实现的泛化与通过微调实现的精确性之间的权衡仍然是系统设计中的关键考虑因素。随着Med-VQA应用向真实临床部署发展，调优粒度和模型规模的选择将继续影响此类系统的可行性和可靠性。此外，为确保对Med-VQA系统进行公平且全面的评估，未来的工作应推动标准化的评估设置，其中包括统一的基准子集以及对封闭式和开放式VQA任务使用一致的指标。

6.3. 评估权衡：任务特定模型与大型视觉语言模型

现代的变压器融合架构在各类特定任务模型排行榜上持续领先。例如，最近的 Swin +Bert +Transformer 模型（Q2ATransformer）在 VQA-RAD 上实现了 ≈80.5% 的准确率，远高于之前的 CNN + BAN/CMSA 系统（70-75%）。在其他数据集上，专门化模型同样领先：例如，基于变压器的 MMQL 在 SLAKE 上达到了 86.7%（表 6），超过了表 7 中的所有 VLM 方法。具备外部知识（检索或知识增强方法）的 TSM 进一步提升了临床数据的准确率。在一个案例中，RAG 模型（RAMM）在 SLAKE 上超过了 91.6%。基于增强的数据模型结果不一：一些（如 M-Mixup）在病理学问题上的表现非常高（在 PathVQA 上为 ≈82.9%），而另一些可能会降低开放式问题的准确率。

(e.g. VQA-Mix scored only 13% on open questions). In summary, the TSMs approaches achieve roughly 75-86% accuracy across Med-VQA benchmarks (varying by dataset and method). The only exception is Med-CoE (a large VLM) which achieve 91.8 accuracy on SLAKE which is the highest among all methods (regardless of TSMs or VLMs), and largely overpass those VLM methods.

Fine-tuned medical VLMs show performance close to that of TSMs. Table 7 reports that most fine-tuned VLM systems reach around 85% on SLAKE and 75% on VQA-RAD, which is very close to the best TSMs. For example, the 7B-parameter PMC-VQA model achieves 85.2% on SLAKE. However, zero-shot/few-shot Med-VLMs: Prompting or retrieval with frozen large models typically underperforms fine-tuned variants. For example, a zero-shot Med-VLM (GPT-4V) only attains 20% accuracy on VQA-RAD (vs. 80.5% for Q2ATransformer). Other zero-shot VLMs similarly lag behind supervised models. Current Med-VLMs often generate free-form text answers, while TSMs use classification with fixed answer sets. TSMs are more likely to produce exact matches with ground truth, leading to higher evaluation scores under strict metrics like accuracy. Also, the comparison is not fair, as Med-VLMs may generate semantically correct but lexically different answers (e.g., “no abnormality” vs. “normal”), which get penalized by metrics like BLEU or token accuracy. TSMs, by contrast, are optimized for these metrics. However, the exact match (accuracy) is the lower bound of token-level metrics. For example, ’lung cancer’ vs. ’lung edema’ may yield 0.5 for token accuracy but 0 for exact match.

The performance data reveal clear trade-offs between TSMs and large VLMs. **Generalization vs. specialization:** Large VLM-based systems tend to generalize better to new or open-ended questions. By contrast, smaller fusion-based TSMs are often finely tuned to a given dataset or question type and may not transfer as well. However, well-designed TSMs (e.g. BiomedCLIP, which injects domain contrastive learning) can match large models on many tasks. **Interpretability and reliability:** Large generative models are essentially black boxes, which hinders interpretability. Increasing model capacity comes at the expense of both interpretability and computational efficiency. In medical settings this is critical, as generative models can hallucinate plausible-sounding but incorrect medical facts. By contrast, TSMs typically produce answers from a fixed vocabulary or selected image regions, making them inherently more predictable. To mitigate hallucination, recent Med-VQA methods integrate knowledge grounding or human oversight: e.g. retrieval-augmented generation (RAG) systems and confidence scoring are used to constrain answers. In short, TSMs are generally easier to explain (their reasoning modules and attention can be visualized) and less prone to fantasy, whereas large Med-VLMs require additional mechanisms (chains-of-thought, expert validation) to ensure factuality. **Efficiency and deployment:** Another key trade-off is compute cost. Large medical VLMs demand far more resources; top models such as LLaVA-Med (13B) require substantial GPU/TPU memory and long training times. In contrast, specialized models often use 100-400M parameters and can be trained or run on modest hardware. But recent advances in PEFT (LoRA, adapters, etc.) help bridge the gap, techniques like LoRA or prefix tuning allow large models to approach state-of-the-art accuracy with fewer updated weights. For example, small-parameter fine-tunings (MILE, VQA-Adapter) have achieved nearly the same performance as full fine-tuning at dramatically reduced cost. Moreover, the multi-task capability of Med-VLMs can reduce the number of models needs to be deployed, which the computation resources can be saved by another way, if number of down stream tasks is need to be used. Thus, in real-world clinical deployment, TSMs are easier to certify and run in resource-limited hospitals, whereas large generative models may require cloud services and careful monitoring but more powerful. In summary, current large medical VLMs is achieve high performance which are very close to state-of-the-art TSMs, but this comes with higher risk of hallucination, greater resource needs, and lower transparency. Also, the zero-shot VLMs still have the gap to fine-tuned methods. When focusing on absolute performance TSMs remain competitive (especially on closed questions) with much lighter footprint and more straightforward reasoning. The performance trends suggest that hybrid approaches, combining large vision-language models with task-specific models may offer the best balance for future Med-VQA systems.

7. Key challenges and open problems in Med-VQA

Modality Bias: In VQA domain, a significant challenge arises from models’ reliance on superficial linguistic correlations, commonly referred to as language priors, which can lead to inaccurate or misleading clinical answers. Zhan et al. (2023) discuss the possible of existence of language bias in current Med-VQA models by introducing a changing prior SLAKE-CP dataset, which is a restructured version of the SLAKE dataset with altered answer distributions between training and testing sets. This design challenges models to generalize beyond memorized biases, promoting a deeper understanding of visual and semantic relationships. Experimental results demonstrate that existing state-of-the-art Med-VQA models can not perform well on SLAKE-CP dataset. This suggests that the possibility of language biases, which can cause less reliable predictions. Thus, most current Med-VQA models still inadequately address the issue of language bias and lack robust evaluations under varying linguistic distributions. Consequently, future research should focus more on the sensitivity of models to language biases, aiming to develop Med-VQA systems with enhanced generalization capabilities and interpretability to ensure their reliability and effectiveness in real clinical environments.

Hallucination: Recent benchmarks reveal significant hallucination issues in state-of-the-art VLMs when performing Med-VQA. Wu et al. (2024) evaluate models under several conditions, including fake questions, “None of the Above” (NOTA) distractors, and image-question mismatches. Top models such as LLaVA-Med achieve only 0.33% overall accuracy, with particularly low rates 0.2% in fake and swap cases. Sepehri et al. (2025) employs visually distinct but embedding-similar radiology image pairs in a multiple-choice format. All evaluated models perform worse than random guessing, indicating systematic hallucination when visual grounding is required. Hallucinations may be driven by several factors. *Visual Embedding Overlap:* Contrastive image encoders (e.g., CLIP/BiomedCLIP) often compress clinically distinct images into overlapping embeddings, impairing fine-grained discrimination (Wu et al. , 2024). *Language Bias:* VLMs tend to rely on linguistic priors or generative patterns, favouring plausible textual predictions over image-grounded reasoning. *Absence of Cross-Modal Validation:* Current architectures lack mechanisms to verify consistency between image, question, and answer modalities, making them vulnerable to hallucination under mismatched conditions. *Fragmented Evaluation:* Existing benchmarks focus on specific hallucination scenarios or imaging modalities, but there is no unified, large-scale benchmark encompassing diverse tasks, modalities, and question formats. Thus, it is necessary to develop region-aware, high-resolution feature extractors that capture clinically relevant details and align more tightly with semantics during encoder training. In addition, introducing training objectives that penalize confident yet unsupported predictions, using perturbations such as fake questions, swaps, and NOTA; which can encourage grounding in visual evidence. Moreover, expanding the Med-VQA tasks from simple classification/generation to answers with grounding explanations, and involving radiologists in dataset curation and error review to flag hallucinations early. Hallucination remains a major barrier to safe deployment of VLMs in clinical Med-VQA settings. Systematic mitigation requires advances in visual grounding, modality consistency, instruction tuning, benchmark standardization, interpretability, and safety guardrails. Grounded, clinically aware VLMs will only emerge through holistic, multidisciplinary efforts.

Interpretability: A key strategy to mitigate hallucination in Med-VQA is enhancing model interpretability. Some approaches (e.g. Shui et al., 2025) improve interpretability by decomposing CT volumes into anatomical regions and aligning each region with its corresponding

(例如，VQA-Mix 在开放性问题上仅得了 13%)。总的来说，TSMs 方法在 Med-VQA 基准测试中大约达到 75-86% 的准确率（取决于数据集和方法）。唯一的例外是 Med-CoE（一个大型 VLM），它在 SLAKE 上达到 91.8 的准确率，这是所有方法中最高的（无论是 TSMs 还是 VLMs），并且大幅超过了那些 VLM 方法。

经过微调的医学视觉语言模型（VLMs）表现接近时间序列模型（TSMs）。表7显示，大多数微调的VLM系统在SLAKE上达到约85%，在VQA-RAD上达到约75%，这与表现最好的TSMs非常接近。例如，具有7B参数的PMC-VQA模型在SLAKE上取得了85.2%的成绩。然而，零样本/少样本的医学VLM（Med-VLMs）：使用冻结的大模型进行提示或检索通常表现不如微调的变体。例如，一个零样本的Med-VLM（GPT-4V）在VQA-RAD上的准确率仅为20%（而Q2ATransformer为80.5%）。其他零样本VLM同样落后于监督模型。目前的Med-VLMs通常生成自由文本答案，而TSMs使用固定答案集进行分类。TSMs更有可能与真实答案完全匹配，从而在精确度等严格指标下获得更高的评估分数。此外，这种比较并不公平，因为Med-VLMs可能生成语义上正确但词汇不同的答案（例如，“无异常”与“正常”），却会被指标惩罚。

性能数据揭示了TSM与大型VLM之间的明显权衡。泛化能力与专业化能力：基于大型VLM的系统往往在面对新的或开放性的问题时具有更好的泛化能力。相比之下，较小的融合型TSM通常对特定数据集或问题类型进行了精细调整，可能无法很好地迁移。然而，设计良好的TSM（例如通过注入领域对比学习的BiomedCLIP）在许多任务上可以与大型模型匹配。可解释性与可靠性：大型生成模型本质上是黑盒，这阻碍了可解释性。增加模型容量会以可解释性和计算效率为代价。在医疗环境中，这一点尤为关键，因为生成模型可能产生听上去合理但实际上错误的医疗信息。相比之下，TSM通常从固定词汇表或选定的图像区域生成答案，使其本质上更加可预测。为了减轻幻觉，最近的Med-VQA方法整合了知识

问题）具有更轻的占用和更直接的推理。性能趋势表明，混合方法，即将大型视觉-语言模型与特定任务模型结合，可能为未来的医疗视觉问答系统提供最佳平衡。

7. Med-VQA中的主要挑战和未解决的问题

模态偏差：在视觉问答 (VQA) 领域，一个显著的挑战来源于模型对表面语言相关性的依赖，这通常被称为语言先验，这可能导致临床答案不准确或具有误导性。Zhan 等人（2023）通过引入可变化先验的SLAKE-CP 数据集讨论了当前 Med-VQA 模型中语言偏差存在的可能性，该数据集是 SLAKE 数据集的重构版本，在训练集和测试集之间改变了答案分布。这一设计挑战模型在超越记忆偏差的情况下进行泛化，促进对视觉和语义关系的更深入理解。实验结果表明，现有的最先进 Med-VQA 模型在 SLAKE-CP 数据集上的表现不佳。这表明语言偏差的可能性，可能导致预测不够可靠。因此，大多数当前的 Med-VQA 模型仍未充分解决语言偏差问题，并且在不同语言条件下缺乏稳健的评估。

幻觉：最近的基准测试显示，最先进的视觉语言模型（VLM）在执行医学视觉问答（Med-VQA）时存在显著的幻觉问题。Wu 等人（2024）在多种条件下评估模型，包括虚假问题、“以上都不是”（NOTA）干扰项以及图像-问题不匹配。像 LLaVA-Med 这样的顶级模型总体准确率仅为 0.33%，在虚假和交换情况中的准确率尤其低，仅为 0.2%。Sepehri 等人（2025）在多项选择格式中使用视觉上不同但嵌入相似的放射学图像对。所有评估的模型表现都比随机猜测更差，表明在需要视觉锚定时存在系统性的幻觉。幻觉可能由多个因素驱动。视觉嵌入重叠：对比图像编码器（如 CLIP/BiomedCLIP）常常将临床上不同的图像压缩到重叠的嵌入中，损害了细粒度区分能力（Wu 等人，2024）。语言偏差：VLM 往往依赖语言先验或生成模式，偏向

可解释性：在医学视觉问答（Med-VQA）中，缓解虚构的关键策略是增强模型的可解释性。一些方法（例如，Shui 等，2025）通过将 CT 体积分解为解剖区域并将每个区域与其对应部分对齐来提高可解释性

radiology report segment via fine-grained contrastive pre-training. Such fine-grained, region-anchored strategies can directly counter hallucinations by enforcing that answers are supported by clearly localized visual features, not just heuristic or linguistic patterns. However, such approaches require large annotated datasets, which is labor-intensive and difficult to transfer to low-resource domains.

Generalization: A critical challenge in Med-VQA is generalization beyond chest X-ray datasets–most current VLM pre-training focuses narrowly on frontal chest images, limiting performance on other modalities (e.g., CT, MRI, ultrasound, pathology) or body regions. Studies such as CoSMIC (Liu et al., 2025b) leverage a large-scale multimodal dataset spanning four medical domains and evaluate it on fifteen downstream datasets across five domains: retinoscopy, radiography, ophthalmoscopy, dermoscopy, and an unseen histopathology setting, demonstrating strong zero-shot transfer and improved average performance. The advances suggest that broad, cross-modality pre-training can significantly enhance model robustness. Future directions include integrating diverse pre-training data spanning X-ray, CT, MRI, ultrasound, and pathology; developing unified architectures that balance uni-modal and fusion encoding; and rigorously evaluating transfer via cross-dataset benchmarks (e.g., chest-to-CT, X-ray-to-pathology) along with formal metrics for generalization performance. This will be essential to build trustworthy, widely applicable Med-VQA systems.

Strategic Considerations: When to Choose Task-Specific Models vs. Fine-Tuned vision-language models: An important practical question remains: should one invest in developing a TSM or fine-tune a large VLM? This decision hinges on several trade-offs related to performance, resources, and deployment constraints. While TSMs often require careful architectural design and task-specific engineering, such as custom fusion strategies, domain-informed training objectives, or handcrafted augmentations, they typically demand far fewer computational resources and deliver faster inference. This makes them especially well-suited for deployment in real-world clinical environments where low latency, explainability, and system certifiability are critical. However, these benefits come at the cost of limited flexibility: TSMs are usually specialized to a narrow task and require re-design or retraining when applied to new domains or modalities. On the other hand, large VLMs, particularly in their zero-shot or few-shot form–offer the appealing promise of general-purpose capability across diverse Med-VQA tasks. Yet in practice, current zero-shot VLMs often underperform compared to well-designed TSMs, especially on clinically specific or closed-ended questions. Moreover, their inference cost in terms of memory and latency is significantly higher, which limits real-time use in resource-constrained settings. Fine-tuned VLMs can match or even outperform TSMs in many tasks, as demonstrated by recent results on SLAKE and VQA-RAD. However, this performance comes at a steep cost: extensive labelled data, long training time, and significant GPU memory are required. Even with parameter-efficient tuning techniques like LoRA or adapters, the resource demands remain substantially higher than those of compact task-specific models. Ideally, a powerful zero-shot VLM capable of handling all sub-tasks with strong factuality, efficiency, and minimal tuning would be the best solution. However, when zero-shot performance is insufficient, the decision between building a TSM or fine-tuning a VLM becomes central. In low-resource settings or for well-bounded tasks, Task-Specific models remain a strong choice. In contrast, when multi-task generalization and flexibility are essential, and computational resources are available, fine-tuned VLMs offer superior scalability. Ultimately, future Med-VQA systems may benefit from hybrid architectures: using VLMs for broad, general-purpose reasoning while invoking smaller, specialized modules for high-stakes or domain-specific tasks. Such a design may offer the best balance between performance, reliability, and real-world deployability.

Toward Hybrid Med-VQA Systems: Unifying Task-Specific Precision with vision-language Model Flexibility: For Med-VQA, TSMs still outperform fine-tuned large Med-VLMs, and current zero-shot methods offer limited gains. However, the performance gap is narrowing: over the past two years, medical VLM methods such as MedVP-LLaVA and specialist VLMs have rapidly advanced, achieving results close to expert models. Currently task-specific models excel at narrow tasks, e.g., lesion grading, by capturing finely tuned features–often with fewer parameters–while fine-tuned medical VLMs provide adaptability across diverse modalities and question types. Thus, how to combine the strength of task-specific models and medical VLMs may be the next direction. Specialist frameworks like VILA-M3 (Nath et al., 2025) integrate domain-expert models (e.g. segmentation or classification) with VLM backbones, boosting accuracy by over prior state-of-the-art task-specific methods. Similarly, lightweight Med-MoE (Jiang et al., 2024) can match or exceed advanced Med-VQA models. This envisages hybrid systems that combine the best of both worlds: using medical VLMs for broad, flexible reasoning and experts for high-resolution tasks, with dynamic gating to minimize latency and resource use. Future research should explore:

- **Adaptive model routing**, where the system selects between medical VLM and specialist(s) at run time based on question type or image complexity.
- **Exemplar-based alignment**, refining response distributions so that zero-shot or few-shot medical VLM outputs are calibrated to mirror specialist outputs where appropriate.
- **Parameter-efficient orchestration**, using MoE or conditional computation to enhance accuracy with a fraction of model size.

This hybrid paradigm holds promise for future Med-VQA methods, combining broad adaptability with specialist precision, delivering both high performance and efficient deployment.

8. Conclusion

This survey presented a comprehensive comparison between task-specific models and large vision-language models in the context of Medical Visual Question Answering. Task-specific models exhibit strengths in computational efficiency and suitability for deployment in resource-constrained clinical environments, while large VLMs excel in generalizability and complex reasoning capabilities due to extensive pre-training on diverse datasets. Nonetheless, significant challenges persist, including addressing modality biases, interpretability, hallucinations, and the need for enhanced generalization across medical modalities and anatomical regions. Future research directions highlighted include the integration of multimodal knowledge sources, development of domain-specific pre-training strategies, and refining interpretability methods to ensure clinical trust and applicability. Addressing these challenges will be pivotal for realizing the full potential of Med-VQA in clinical practice.

CRediT authorship contribution statement

Huahu Xu: Investigation, Resources, Supervision; **Qishen Chen:** Conceptualization, Formal analysis, Investigation, Writing – original draft, Visualization; **Wenxuan He:** Writing – original draft, Visualization; **Xingyuan Chen:** Conceptualization, Investigation; **Honghao Gao:** Investigation, Resources, Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

通过细粒度对比预训练获得放射学报告片段。这种细粒度、区域锚定的策略可以通过确保答案由明确定位的视觉特征支持，而不仅仅依赖启发式或语言模式，直接对抗幻觉。然而，这种方法需要大量标注数据，这既劳动密集又难以迁移到低资源领域。

泛化能力：Med-VQA 中的一个关键挑战是超越胸部 X 光数据集的泛化能力——当前大多数 VLM 预训练主要集中在正面胸部图像上，这限制了其在其他模态（如 CT、MRI、超声、病理）或身体部位的表现。像 CoSMIC（Liu 等，2025b）这样的研究利用覆盖四个医学领域的大规模多模态数据集，并在五个领域的十五个下游数据集上进行评估：视网膜检查、放射摄影、眼底检查、皮肤镜检查，以及一个未见的组织病理学场景，展示了强大的零样本迁移能力和提高的平均性能。研究进展表明，广泛的跨模态预训练可以显著增强模型的稳健性。未来方向包括整合覆盖 X 光、CT、MRI、超声和病理学的多样化预训练数据；开发在单模态和融合编码之间取得平衡的统一架构；以及通过跨数据集基准（如胸部）严格评估迁移能力。

战略考量：何时选择任务特定模型与微调的视觉-语言模型：一个重要的实际问题仍然存在：应该投资开发任务特定模型（TSM）还是微调大型视觉-语言模型（VLM）？这一决策取决于与性能、资源和部署限制相关的若干权衡。虽然TSM通常需要精心的架构设计和任务特定的工程，例如自定义融合策略、基于领域的训练目标或手工设计的数据增强，但它们通常需要的计算资源远少于大型VLM，并提供更快的推理速度。这使得它们特别适合在低延迟、可解释性和系统可认证性至关重要的现实临床环境中部署。然而，这些优势是以灵活性有限为代价的：TSM通常专注于特定任务，并且在应用于新领域或新模态时需要重新设计或再训练。另一方面，大型VLM，特别是在

迈向混合医学问答系统：将任务特定的精度与视觉-语言模型的灵活性统一：对于医学问答，TSM仍然优于微调的大型医学视觉-语言模型，而当前的零样本方法提供的增益有限。然而，性能差距很小。

在过去两年中，医学视觉语言模型（VLM）方法如 MedVPLLaVA 和专科 VLM 已迅速发展，取得了接近专家模型的结果。目前，任务特定模型在狭窄的任务上表现出色，例如通过捕捉精细调优特征来进行病灶分级——通常参数更少——而经过微调的医学 VLM 在不同模态和问题类型上提供了适应性。因此，如何结合任务特定模型和医学 VLM 的优势可能是下一步方向。像 VILA-M3（Nath 等, 2025）这样的专科框架将领域专家模型（如分割或分类）与 VLM 主体集成，通过超过先前最先进的任务特定方法提高了准确性。同样，轻量化的 Med-MoE（Jiang 等, 2024）可以匹配或超过先进的 Med-VQA 模型。这设想了将两者优势结合的混合系统：利用医学 VLM 进行广泛、灵活的推理，而专家模型用于高分辨率任务，并通过动态门控来最小化

• 自适应模型路由，系统根据问题类型或图像复杂度，在运行时在医疗视觉语言模型（VLM）和专家之间进行选择。• 基于示例的对齐，通过优化响应分布，使零样本或少量样本的医疗VLM输出在适当情况下校准以反映专家输出。• 参数高效编排，使用MoE或条件计算以在仅使用部分模型规模的情况下提升准确性。

这种混合模式为未来的医学视觉问答（Med-VQA）方法带来了希望，它结合了广泛的适应性和专业的精确性，既提供高性能又实现高效部署。

8. 结论

本综述在医学视觉问答（Med-VQA）背景下，对任务特定模型与大型视觉-语言模型（VLMs）进行了全面比较。任务特定模型在计算效率和适合部署在资源有限的临床环境中表现出优势，而大型视觉-语言模型由于在多样化数据集上的广泛预训练，在泛化能力和复杂推理能力方面表现突出。然而，仍然存在显著挑战，包括解决模态偏差、可解释性、幻觉问题，以及在不同医疗模态和解剖区域中增强泛化能力的需求。未来研究方向包括多模态知识源的整合、开发领域特定的预训练策略，以及改进可解释性方法以确保临床信任度和适用性。应对这些挑战对于在临床实践中充分实现医学视觉问答的潜力至关重要。

CRediT 作者贡献声明

胡华徐：调查、资源、监督；陈启申：概念化、正式分析、调查、撰写 – 原始稿、可视化；何文轩：撰写 – 原始稿、可视化；陈星源：概念化、调查；高鸿浩：调查、资源、撰写 – 审阅与编辑、监督。

数据可用性

数据可根据请求提供。

利益冲突声明

作者声明他们没有已知的可能影响本文所报告工作的竞争性财务利益或个人关系。

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